Introduction To Psychology

BS (4 Years)

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INTRODUCTION TO PSYCHOLOGY

Unit 1-9
Course Code 8411

Course Coordinator: Dr. Afshan Huma

FACULTY OF EDUCATION
ALLAMA IQBAL OPEN UNIVERSITY

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ALLAMA IQBAL OPEN UNIVERSITY

COURSE TEAM

Course Development Coordinator	Dr. Afshan Huma
Course Writers	Dr Afshan Huma
	Dr Tahira Bibi
	Dr Hina Noor
	Dr. Hafiz Tahir Jamil
	Dr Sidra Rizwan
	Ms Khadija Nadeem
Course Reviewers	Dr Azhar Mumtaz Saadi
	Dr Farkhanda Rasheed
	Dr Muhammad Zafar Iqbal
Editor	Ms. Humera Ejaz

FOREWORD

Psychology is an important discipline in social sciences, providing insights into human behaviour, mental processes, and social interactions. The basic use of psychology in social sciences can help to understand human behavior. Psychology helps to explain why people think, feel, and behave the way they do, which can be useful in fields such as sociology, political science, library science, business, education, and anthropology.

Psychology involves analyzing data and research findings, which can help to develop critical thinking skills that are useful in many areas of life. It also involves learning how to communicate effectively with others, which can be helpful in many personal and professional relationships. Studying psychology can help to develop empathy and compassion for others, which can improve personal relationships and professional interactions.

Overall, studying psychology at the undergraduate level can be a valuable experience for personal and professional growth and can lead to a fulfilling career in a variety of fields. An undergraduate psychology course can lead to many career opportunities, including in social work, education, and business.
I wish all the best to the students who are studying this course and appreciate the course team. The team has developed this course book comprehensively and cohesively comprehensively and cohesively. Overall, the use of this course in social sciences can provide valuable insights and tools for understanding human thoughts and behaviours and addressing social issues.

Professor Dr Nasir Mahmood
Vice Chancellor, AIOU

INTRODUCTION OF THE COURSE

The knowledge of psychology at the undergraduate level in social sciences provides insights into how people think and behave in various situations. The graduates can use this knowledge to address social issues such as emotional instability and difficulty in collaboration, by understanding the factors that contribute to these problems and developing solutions to manage them. Research in the field of psychology can inform policy and decision-making by providing evidence-based recommendations on how to address social issues and improve social outcomes. Psychology can help to enhance personal and professional relationships by providing insights into how people think and behave, which can lead to more effective communication and collaboration.

Dr Afshan Huma
Course Coordinator

COURSE OBJECTIVES

After completing this course you will be able to:
1. Understand the scientific grounds of psychology
2. Elaborate on the relationship between the mind body and behaviours
3. Identify the characteristics of a conscious state of mind
4. Explain how people learn, memorize and judge in novel conditions

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UNIT 1

INTRODUCTION

Compiled by: Dr Tahira Bibi
Reviewed by: Dr. Farkhanda Rasheed

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INTRODUCTION

Psychology's insights are based on the careful efforts of both scientists and clinical practitioners to gather a solid body of information about the nature of human functioning namely, about what human beings do, think, , and feel and the reasons for their behaviour. Psychologists do not pretend to have all the answers, of course. Human behaviour is so complex that it may forever defy full scientific analysis. But they do have some of the answers and clues to many others, and they are constantly making discoveries. This much seems certain: when you finish studying this course you will know more about human experiences than anybody in the world just over a century ago, when the first psychology laboratory was opened, and the science of psychology was begun. The definition of psychology, although brief, covers an enormous realm. Psychology is the systematic study of behaviour and mental processes and the factors that influence them. Those few words have covered a range of subject matter whose boundaries are almost impossible to imagine.

The begin with, just the word behaviour means a host of things. Our physical actions begin each morning when we wake, Yawn, stretch, dress, and eat breakfast, and do not end until we go back to bed and fall asleep after a long day in which we walk, talk, study, work, play, and laugh or sometimes cry. The field of psychology has a dual mission: (1) to understand behaviour and mental processes and (2) to predict their course, indeed, understanding and predicting are the goals of all sciences.

Psychology is one member of a family of sciences often referred to as the behavioural and social sciences. It is recognized that understanding human behaviour requires the study not only of the individual but also of all facets of human functioning that range from the activity of individual cells to the social and cultural settings in which humans live. Thus, the subject matter of this family of sciences ranges from the neurochemical bases of memory and mental illness to the dynamics of global commerce and international conflicts.

Learning Outcomes of the Unit

After completing this unit, you will be able to:

- Understand and describe the concept and nature of psychology
- Analyze the concept of psychology as a Science
- Critically analyze the evolution process of psychology
- Understand various approaches and questions

1.1 Introducing Psychology

Psychology is the scientific study of the human mind and behaviour. It encompasses a wide range of topics, including perception, cognition, emotion, motivation, personality, social behaviour, and mental disorders. Psychologists use a variety of research methods to investigate these topics, including experiments, surveys, observational studies, and clinical interviews. The ultimate goal of psychology is to understand human behaviour and mental processes and to use this knowledge to improve people's lives. The word "psychology" comes from the Greek words "psyche," meaning *life*, and "logos," meaning *explanation*. Psychology is a popular major for students, a popular topic in the public media, and a part of our everyday lives. Many people have direct knowledge about psychology because they have visited psychologists, for instance, school counsellors, family therapists, and religious, marriage, or bereavement counsellors.

Because we are frequently exposed to the work of psychologists in our everyday lives, we all have an idea about what psychology is and what psychologists do. In many ways, I am sure that your conceptions are correct. Psychologists work in forensic fields, and they provide counselling and therapy for people in distress. But there are hundreds of thousands of psychologists in the world, and most of them work in other places, doing work that you are probably not aware of. Most psychologists work in research laboratories, hospitals, and other field settings where they study the behaviour of humans and animals. For instance, my colleagues in the Psychology Department at the University of Maryland study such diverse topics as anxiety in children, the interpretation of dreams, the effects of caffeine on thinking, how birds recognize each other, how praying mantises to hear, how people from different cultures react differently in negotiation, and the factors that lead people to engage in terrorism. Other psychologists study such topics as alcohol and drug addiction, memory, emotion, hypnosis, love, what makes people aggressive or helpful, and the psychologies of politics, prejudice, culture, and religion. Psychologists also work in schools and businesses, and they use a variety of methods, including observation, questionnaires, interviews, and laboratory studies, to help them understand behaviour.

1.1.1 The Nature of Psychology

The field of psychology has its roots in many disciplines, including philosophy and physiology. This makes it difficult for us to come to a consensus regarding a single, globally accepted definition of psychology. Despite this problem, there is no debate about the meanings of the two words that comprise the compound word psychology.

These words are "*psycho*" and "*logos*". "Psychology" literary means „the science of the mind".

I would also like to introduce you to definitions by respected psychologists, after which I will select a working definition. The contributions of the psychologists are as follows:

- Ornstein (1985) defined psychology as the study of human experience,
- Kendall (1986) saw psychology as the scientific study of mental and behavioural functioning.
- Atkinson and Hilgard (1983) viewed psychology as the scientific study of behaviour and mental processes. This definition was also proposed by Myers (1996).
- Munavi (1988) asserted that psychology was the scientific study of human and animal behaviour.

The above definitions have a common theme, that of psychology being scientific and the focus on behavioural functioning. The last definition shall be our working definition since it is most comprehensive in terms of:

- Acknowledging the scientific nature of psychology.
- Focusing on the study of both humans and animals.
- Need to study psychology as a subject.
- Study of psychology will be in a good position to:
- Describe behaviour
- Understand behaviour
- Predict behavior.
- Control behaviour

To describe, understand, predict, and control behaviour, you need to be able to evaluate the quality of:

- Interpersonal relationships
- Intrapersonal relationships
- Person-to-group relationships
- Group relationships

1.2 Perspectives of Psychology

There are many perspectives that psychologists follow in the study of behaviour and mental functioning. Following are the six major areas for discussion.

1.2.1 The Psychoanalytic Perspective

Freud's (1939) theory of psychoanalytic discusses the ID, Ego and Superego. He viewed psychological traits and disorders as having an origin in sexual and aggressive drives. He believed that our behaviour is controlled by unconscious/subconscious needs/forces, which are biological in nature.

1.2.2 The Behavioural Perspective

This perspective studies the mechanisms by which observed reactions develop and are modified under certain conditions. For example, a behavioural psychologist will be interested in how humans and animals develop certain emotions, such as fear and love, or how stimuli, such as gifts, and punishments, can influence behaviour. Some notable behaviourists include B. Watson.

1.2.3 The Biological Perspective

It helps us understand how the body and brain work in unison to create emotions, memories, and sensory experiences.

1.2.4 The Humanistic Perspective

Humanist psychologists oppose the psychoanalytic view that people are driven by unconscious or subconscious inner forces. They emphasize that a person can consciously choose a meaningful way of life and, therefore, grow to greater maturity and self-realization.

1.2.5 The Cognitive Perspective

The term cognition refers to the ability to use the mind in terms of remembering, thinking, problem-solving, and planning. From this perspective, psychologists explore how people process, store, and retrieve information, and how that information can be used to reason and solve problems.

1.2.6 The Cross-Cultural Perspective

This perspective focuses on the impact of cultural and ethnic similarities and differences on both psychological and social functioning. It is assumed that every person in society has a certain cultural and ethnic origin, and this influences behaviour depending on the environment in which he is located.

Self-Assessment 1

1. Define the concept of psychology in your own words.
2. Identify various perspectives of psychology.
3. Which perspective do you agree with? Give reasoning to support your viewpoint.

1.3 Psychology as a Science

Despite differences in interests, areas of research and approaches, all psychologists have one thing in common: they rely on scientific methods. Research psychologists use scientific methods to create new knowledge about the causes of behaviour, while practising psychologists such as clinical, counselling, industrial and school psychologists use existing research to improve the daily lives of others. The science of psychology is important for both researchers and practitioners.
Psychology is the systematic study of behaviour and factors that influence behavior. The goals of psychology are to understand behavior and to predict behavior. Psychology of one member of a family of sciences known as the behavioural and social sciences.

Many psychologists are concerned only with basic science, or knowledge for the sake of knowledge. Others are chiefly interested in applied science, or the use of psychological findings to help society carry on its everyday tasks, tackle its problems, and improve the quality of life. Among psychologists concerned with basic science are

- Experimental psychologists
- Comparative psychologists
- Physiological psychologists
- Developmental psychologists
- Personality and social; psychologists
- Educational psychologists

Psychologists who study such basic processes as learning, memory, and motivation are known as Experimental psychologists. Comparative psychologists concentrate on relating animal behaviour patterns to those found in humans whereas physiological psychologists study the role and functions of the brain and body in behaviours. Studying how individuals grow and change throughout their existence is the subject of developmental psychologists. Personality and social psychologists focus on how people differ in their enduring inner characteristics and traits, and how they influence and are influenced by others. The areas of study of educational

psychologists are concerned with the impact of educational processes and techniques on students and learning.

School psychologists, industrial psychologists, environmental psychologists, community psychologists, forensic psychologists, and health psychologists are examples of psychologists who are interested in applied science. School psychologists test and evaluate learners, analyze learning issues, and perform as counsellors for teachers, learners, and parents. Industrial psychologists facilitate and guide in the selection and training of workers, and suggestions to improve working conditions, workplace morale behaviours and cooperation. Environmental psychologists deal with the behaviours of ecological problems like pollution and overcrowding. There are community psychologists who focus on the social environment and how it could better serve human needs. Work on behavioural issues of legal importance deals with forensic psychologists. Health psychologists study to find the answers to various ways to improve health by altering, adopting, and leaving certain behaviours.
There are two broader types of applied psychologists:

a. Clinical psychologists:

Clinical psychologists help in diagnose deep-seated psychological problems and treat such problems through psychotherapy.

b. Counselling Psychologists:

Counselling psychologists diagnose, help and guide those who are with milder problems with social and emotional adjustment and decision-making. Educational difficulties, choice of vocations, family or marital conflicts etc. are examples.

1.4 Psychologists and Empirical Methods

The qualitative methods used by psychology to study behaviour are as under:

- Naturalistic observation
- Participant observation
- Interviews
- Case histories

While the quantitative methods include:

- Questionnaires
- Tests and measurements
- Psychological assessment

The results of data collected through tests and measurements are analyzed through mathematical techniques which are called statistics.

The experimental methods, which are sciences attempt to discover causes and effects to the controlled examination of whether behaviour changes when conditions change. Scientists of psychology use experimental methods to find the truth. The experimenter or researcher controls the independent variable, which is set up independently of anything the subject does or does not do, and then studies the dependent variables, which is a change in the subject's behaviour resulting from a change in the independent variables.

For experimental studies, researchers often use both an experimental group, whose behaviour is studied under new conditions, and a control group, which is not placed under the new conditions. There is a form of experiment design which is called a "single-blind" experiment in which subjects do not know whether they belong to the experimental group or the control group. In a double-blind experiment, neither the subjects nor the experimenter knows.

All scientists, whether physicists, chemists, biologists, sociologists, or psychologists, use empirical methods to study topics of interest to them. Empirical methods include the processes of collecting and organizing data and concluding the data. The empirical methods used by scientists have been developed over many years and provide a framework for collecting, analyzing, and interpreting data within a common framework within which information can be exchanged. We can define the scientific method as the set of assumptions, rules, and procedures that scientists use to conduct empirical research.

Although scientific research is an important method of studying human behaviour, not all questions can be answered using scientific approaches. Statements that cannot be objectively measured or objectively determined to be true or false are not within the scope of scientific research. Therefore, scientists make a distinction between values and facts. Values are personal statements. Facts are objective statements whose accuracy has been confirmed by empirical research. Values cannot be either true or false, science cannot prove or disprove them.

Scientists use research to establish facts, the distinction between values and facts is not always clear-cut. Sometimes statements that scientists believe to be factual later, based on further research, turn out to be partially or even completely wrong. Although scientific procedures do not necessarily guarantee that the answers to questions will be objective and unbiased, science is still the best method for obtaining objective conclusions about the world around us.

1.5 Challenges and limitations

Understanding and trying to reduce the costs of psychological disorders such as depression is not easy, because psychological experiences are extremely complex. The questions that psychologists ask are as difficult as those of doctors, biologists, chemists, physicists, and other scientists.

A major goal of psychology is to predict behavior by understanding its causes. Prediction is difficult in part because people are different and respond differently to different situations. Individual differences are variations between people on physical or psychological dimensions. Because of the many individual differences that influence behaviour, we cannot always predict who will become aggressive or who will do well in graduate school or on the job. The predictions of psychologists (and most other scientists) are only probabilistic. Another reason why behaviour is difficult to predict is that almost all behavior is determined or produced by multiple factors and these factors occur at different levels of detail.

Furthermore, these multiple causes are not mutually exclusive. They are linked in such a way that when one cause exists, other causes also exist. This similarity makes it difficult to identify which cause or causes are at work. For example, some people may become depressed due to a biological imbalance in the neurotransmitters in their brains. The resulting depression may lead them to behave more negatively toward others around them, which in turn causes those others to react more negatively toward them, leading to their depression. increases, as a result, the biological determinants of depression become intertwined with other people's social responses, making it difficult to disentangle the effects of each cause.

Another difficulty in studying psychology is that most human behaviour is caused by factors outside of our conscious awareness, making it impossible for us as individuals to understand them. The role of unconscious processes was emphasized in the theorizing of Austrian neurologist Sigmund Freud (1856–1939), who argued that many psychological disorders are caused by memories that we have oppressed and thus become unconscious.

Self-Assessment No.2

Can you think of a time when you felt stress and depression? What were the causes/causes to feel stress or depression?

Explain the concept of the scientific method concerning behavioural sciences in your own words.

Visualize a behaviour that you find significant and has an impact on your personality think about possible reasons for it. How do you think psychologists would study this behaviour?

1.6 Evolution of Psychology: History, Approaches, and Questions

Psychology was founded in 1879 when the first laboratory was opened by Wilhelm Wundt at a German university. The psychologists, including the American William James, were chiefly interested in human consciousness, which they studied through introspection, looking inward at mental processes.

The school of behaviourism, a rebellion against introspective studies, was founded by John Watson, who declared that mental processes cannot be seen or measured and therefore cannot be studied scientifically. Watson believed that psychologists should study what people do, not what they think. Watson held that human behaviour is a series of actions in which a stimulus that is an event in the environment produces a response that is, an observable muscular movement or some physiological reaction, such as secretions, that can also be observed and measured with the proper instruments. He believed that through the establishment of conditioned reflexes, a type of learning, almost any kind of stimulus could be made to produce almost any kind of response.

B.F. Skinner, who succeeded Watson as leader of the behaviourist school, agrees that human beings are created of their environment whose behaviour depends on the kinds of learning to which they have been subjected and who are therefore not to blame for their failures or deserving of credit for their achievements.

Gestalt psychology, which takes its name from a German word meaning “pattern” or “Configuration”, maintained that events must be considered in studying any psychological phenomenon from a perceptual process to personality because “the whole is greater than the sum of its parts”. The Gestalt school also stressed the entire situation, or context, in which the “whole” is found.

Many of the Gestalt ideas survive in today’s trend toward cognitive psychology, which is interested in all the ways we learn about our environment, store the knowledge in memory, and use it to think and act intelligently in new situations. These various forms of mental activity are often called information processing.

A core idea in cognitive psychology is that we build mental models of reality from information about the world provided by our sense organs, which we can then examine for meanings and guidance in behaviour.

Today's humanistic psychology also stems in part from the Gestalt school. Humanists believe we are unique organisms because we have values and goals and seek to express ourselves, grow, fulfil ourselves and find peace and happiness.
The opposite of humanistic psychology is Sigmund Freud's psychoanalysis, which is based on the idea that behaviour is often influenced by unconscious processes, especially motives of which we are unaware.

In this part of the unit, you will study the history of psychology with a focus on the important questions that psychologists ask and the major approaches (or schools) of psychological inquiry. The schools of psychology that we will review are summarized in Table 1 "The Most Important Approaches (Schools) of Psychology", and Figure 1"Timeline Showing Some of the Most Important Psychologists" presents a timeline of some of the most important psychologists, beginning with the early Greek philosophers and extending to the present day.

Table 1 The Most Important Approaches (Schools) of Psychology

School of Psychology	Description	Important contributors
Structuralism	Uses the method of introspection to identify the basic elements or "structures" of psychological experience	Wilhelm Wundt, Edward B. Titchener
Functionalism	Attempts to understand why animals and humans have developed the particular psychological aspects that they currently possess	William James
Psychodynamic	Focuses on the role of our unconscious thoughts, feelings, and memories and our early childhood experiences in determining behaviour	Sigmund Freud, CarlJung, Alfred Adler, Erik Erickson

Behaviourism	Based on the premise that it is not possible to objectivelystudy the mind, and therefore that psychologists should limit their attention to the study of behaviour itself	John B. Watson, B. F.Skinner
Cognitive	The study of mental processes, including perception,thinking, memory, and judgments	Hermann Ebbinghaus, Sir Frederic Bartlett, Jean Piaget
Social-cultural	The study of how the social situations and the cultures in which people find themselves influence thinking and behaviour	Fritz Heider, Leon Festinger, Stanley Schachter

Introducing Psychology (saylordotorg.github.io)

Table 2 Chronological History

Date	**Psychologists**	**Description**
428-347 BC	Ploto	Greek philosopher who argued for the role of nature in psychological development.
384-322 BC	Aristotle	Greek philosopher who argued for the role of nature in psychological development.
1588-1679	Thomas Hobbes	English philosopher
1596-1650	Rene Descartes	French philosopher
1632-1704	John Locke	English philosopher
1712-1778	Jean-Jacques Rousseau	French philosopher
1801-1887	Gustav Fecher	German experimental psychologists developed the idea of the just noticeable difference (JND), which is the first empirical psychological measurement.
1809-1882	Charles Darwin	British naturalist whose theory of natural selection influenced the functionalist school and the field of evolutionary psychology.
1832-1920	Wilhelm Wundt	German psychologists who opened one of the first psychology laboratories and helped develop the field of struct

1842-1910	William James	American psychologist who opened one of the first psychology laboratories and helped develop the field of functionalism.
1849-1936	Ivan Pavlov	Russian physiologist whose experiments on learning led to the principles of classical conditioning.
1850-1909	Hermann Ebbinghaus	German psychologists studied the ability of people to remember lists of nonsense syllables under different conditions.
1856-1939	Sigmund Freud	Austrian psychologist who founded the field of psychodynamic psychology.
1867-1927	Edward Bradford Titchener	American psychologists who contributed to the field of structuralism.
1878-1958	John B. Watson	American psychologists who contributed to the field of behaviourism.
1886-1969	Sir Frederic Bartiett	British psychologist who studied the cognitive and social process of remembering.
1896-1980	Jean Piaget	Swiss psychologist who developed an important theory of cognitive development in children.
1904-1990	B.F. Skinner	American psychologist who contributed to the school of behaviourism.
1926-1993	Donald Broadbent	British Cognitive psychologist who was a pioneer in the study of attention.
20th and 21st Centuries	Linda Bartoshuk; Daniel Kahneman; Elizabeth Loftus; George Miller	American psychologists contributed to the cognitive school of psychology by studying learning, memory, and judgment. An important contribution is the advancement of the field of neuroscience. Danial Kahneman won the Nobel Prize in Economics for his work on psychological decision-making.
20th and 21st Centuries	Mahzarin Banaji; Marilynn Brewer; Susan Fiske; Fritz Heider; Kurt Lewin; Stanle4y Schachter; Claude Steele; Harry Triandis	American Psychologists who contributed to the social-cultural school of psychology. Their contributions have included an understanding of how people develop and are influenced by social norms.

Above mentioned timeline shown in Table 2 consists of some of the most contributors to the history of psychology.

Although psychology has changed dramatically over its history, the most important questions that psychologists address have remained constant. Some of these questions follow, and we willdiscuss them both in this chapter. Following are the areas and questions of psychology that were and are under the discussion of various psychologists.

- Nature versus nurture.
- Free will versus determinism.
- Accuracy versus inaccuracy
- Conscious versus unconscious processing
- Differences versus similarities.

The earliest psychologists that we know about are the Greek philosophers Plato (428–347 BC) and Aristotle (384–322 BC). These philosophers asked many of the same questions that today'spsychologists ask; for instance, they questioned the distinction between nature and nurture and the existence of free will. In terms of the former, Plato argued on the nature side, believing that certain kinds of knowledge are innate or inborn, whereas Aristotle was more on the nurture side, believing that each child is born as an "empty slate" (in Latin a *tabula rasa*) and that knowledgeis primarily acquired through learning and experience.

European philosophers continued to ask these fundamental questions during the Renaissance. For instance, the French philosopher René Descartes (1596–1650) also considered the issue of free will, arguing in its favour and believing that the mind controls the body through the pineal gland in the brain (an idea that made some sense at the time but was later proved incorrect). Descartes also believed in the existence of innate natural abilities. A scientist as well as a philosopher, Descartes dissected animals and was among the first to understand that the nerves controlled the muscles. He also addressed the relationship between the mind (the mental aspects of life) and the body (the physical aspects of life). Descartes believed in the principle of *dualism*: that the mind is fundamentally different from the mechanical body. Other European philosophers, including Thomas Hobbes (1588–1679), John Locke (1632–1704), and Jean-Jacques Rousseau (1712–1778), also weighed in on these issues.

The fundamental problem that these philosophers faced was that they had few methods for settling their claims. Most philosophers didn't conduct any research on these questions, in part because they didn't yet know how to do it, and in part because they weren't sure it was even possible to objectively study human experience. But dramatic changes came during the 1800s with the help of the first two research psychologists: the German psychologist Wilhelm Wundt (1832–1920), who developed a psychology laboratory in Leipzig, Germany, and the

American psychologist William James (1842–1910), founded a psychology laboratory at Harvard University.

1.6.1 Structuralism

Wundt's research in his laboratory in Leipzig focused on the nature of consciousness itself. Wundt and his students believed that it was possible to analyze the basic elements of the mind and to classify our conscious experiences scientifically. Wundt began the field known as structuralism, *a school of psychology whose goal was to identify the basic elements or "structures" of psychological experience*. Its goal was to create a "periodic table" of the "elements of sensations," like the periodic table of elements that had recently been createdin chemistry.

Structuralists used the method of *introspection* to attempt to create a map of the elements of consciousness. Introspection involves *asking research participants to describe exactly what theyexperience as they work on mental tasks*, such as viewing colours, reading a page in a book, or performing a math problem. A participant who is reading a book might report, for instance, thathe saw some black and coloured straight and curved marks on a white background. In other studies,the structuralists used newly invented reaction time instruments to systematically assess not onlywhat the participants were thinking but how long it took them to do so. Wundt discovered that ittook people longer to report what sound they had just heard than to simply respond that they had heard the sound. These studies marked the first time researchers realized that there is a differencebetween the *sensation* of a stimulus and the *perception* of that stimulus, and the idea of using reaction times to study mental events has now become a mainstay of cognitive psychology.

Perhaps the best-known of the structuralists was Edward Bradford Titchener (1867–1927). Titchener was a student of Wundt who came to the United States in the late 1800s and founded a laboratory at Cornell University. In his research using introspection, Titchener and his students claimed to have identified more than 40,000 sensations, including those relating to vision, hearing, and taste.

1.6.2 Functionalism and Evolutionary Psychology

In contrast to Wundt, who attempted to understand the nature of consciousness, the goal of William James and the other members of the school of functionalism was *to understand why animals and humans have developed the psychological aspects that they currently possess* (Hunt, 1993). Hunt, M. (1993). *The story of psychology*. New York, NY: Anchor Books. For James, one's thinking was relevant only to one's behaviour. As he put it in his psychology textbook, "My thinking is first and

last and always for the sake of my doing" (James,1890). James, W. (1890). *The principles of psychology*. New York, NY: Dover.

James and the other members of the functionalist school were influenced by Charles Darwin's (1809–1882) *theory of natural selection*, which proposed that the physical characteristics of animals and humans evolved because they were useful, or functional.

1.6.3 Psychodynamic Psychology

Perhaps the school of psychology that is most familiar to the public is the *psychodynamicapproach* to understanding behaviour, which was championed by Sigmund Freud (1856–1939) and his followers. Psychodynamic psychology is an *approach to understanding human behaviorthat focuses on the role of unconscious thoughts, feelings, and memories*. Freud developed his theories about behaviour through extensive analysis of the patients that he treated in his private clinical practice.

1.6.4 Behaviourism

Structuralism and functionalism were essentially studies of the mind. The psychologists associated with the school of *behaviourism*, on the other hand, were reacting in part to the difficulties psychologists encountered when they tried to use introspection to understand behaviour. Behaviourism is *a school of psychology that is based on the premise that it is not possible to objectively study the mind, and therefore psychologists should limit their attention to the study of behavior itself*. Behaviourists believe that the human mind is a "black box" into which stimuli are sent and from which responses are received. They argue that there is no point in trying to determine what happens in the box because we can successfully predict behaviour without knowing what happens inside the mind. Furthermore, behaviourists believe that it is possible to develop laws of learning that can explain all behaviours.

The first behaviourist was the American psychologist John B. Watson (1878–1958). Watson wasinfluenced in large part by the work of the Russian physiologist Ivan Pavlov (1849–1936). The most famous behaviourist was Burrhus Frederick (B. F.) Skinner (1904–1990), expandedthe principles of behaviourism and brought them to the attention of the public at large. Skinnerused the ideas of stimulus and response, along with the application of rewards or *reinforcements*,to train pigeons and other animals.

1.6.5 The Cognitive Approach and Cognitive Neuroscience

Science is always influenced by the technology that surrounds it, and psychology is no exception.Thus, it is no surprise that beginning in the 1960s, growing numbers

of psychologists began to think about the brain and human behaviour in terms of the computer, which was being developed and becoming publicly available at that time. The analogy between the brain and the computer, although by no means perfect, provided part of the impetus for a new school of psychology called *cognitive psychology*. Cognitive psychology is *a field of psychology thatstudies mental processes, including perception, thinking, memory, and judgment*. These actionscorrespond well to the processes that computers perform.

Although cognitive psychology began in earnest in the 1960s, earlier psychologists had also taken a cognitive orientation. Some of the important contributors to cognitive psychology include the German psychologist Hermann Ebbinghaus (1850–1909), who studied the ability of people to remember lists of words under different conditions, and the English psychologist Sir Frederic Bartlett (1886–1969), who studied the cognitive and social processes of remembering. Bartlett created short stories that were in some ways logical but also contained some very unusual and unexpected events. Bartlett discovered that people found it very difficult to recall the stories exactly, even after being allowed to study them repeatedly, and he hypothesized that the stories were difficult to remember because they did not fit the participants' expectations about how stories should go. The idea that our memory is influenced by what we already know was also a major idea behind the cognitive-developmental stage model of Swiss psychologist Jean Piaget (1896–1980). Other important cognitive psychologists include Donald E. Broadbent (1926– 1993), Daniel Kahneman (1934–), George Miller (1920), Eleanor Rosch (1938–), and Amos Tversky (1937–1996).

1.6.6 Social-Cultural Psychology

A final school, which takes a higher level of analysis, and which has had a substantial impact on psychology, can be broadly referred to as the *social-cultural approach*. The field of social-cultural psychology is *the study of how the social situations and the cultures in which people findthemselves influence thinking and behavior*. Social-cultural psychologists are particularly concerned with how people perceive themselves and others, and how people influence each other's behavior. For instance, social psychologists have found that we are attracted to others who are like us in terms of attitudes and interests (Byrne, 1969), Byrne, D. (1969).

Psychologists have found that there is a fundamental difference in social norms between various cultures on the basis of geographical boundaries/ blocks.
Another important cultural difference is the extent to which people in different cultures are boundby social norms and customs, rather than being free to express their individuality without considering social norms.

Over all employing the scientific method allows psychologists to understand human behavior objectively and systematically. Psychologists study behaviour at

different levels of explanation, ranging from lower biological levels to higher social and cultural levels. The same behaviours can be studied and explained within psychology at different levels of explanation.

The first psychologists were philosophers, but the field became more objective as more sophisticated scientific approaches were developed and employed. Some of the most important historical schools of psychology include structuralism, functionalism, behaviourism, and psychodynamic psychology. Cognitive psychology, evolutionary psychology, and social-cultural psychology are some important contemporary approaches.

Some of the basic questions asked by psychologists, both historically and currently, include those about the relative roles of nature versus nurture in behaviour, free will versus determinism, accuracy versus inaccuracy, and conscious versus unconscious processing.

Psychological phenomena are complex and making predictions about them is difficult because they are multiply determined at different levels of explanation. Research has found that people are frequently unaware of the causes of their behaviours.

There are a variety of available career choices within psychology that employ in many different areas of interest.

Psychology is the scientific study of mind and behavior. Most psychologists work in research laboratories, hospitals, and other field settings where they study the behavior of humans and animals. Some psychologists are researchers and others are practitioners, but all psychologists use scientific methods to inform their work. Although it is easy to think that everyday situations have common-sense answers, scientific studies have found that people are not always as good at predicting outcomes as they often think they are. The hindsight bias leads us to think that we could have predicted events that we could not have predicted.

Self-Assessment No.3

1. What is the difference between psychologists and philosophers in the field of psychology? What type of questions psychologists can answer that philosophers might not be able to answer as accurately? Answer your viewpoint with examples.
2. Write a report on the question of psychology. Select your interest area from the list of questions on psychology and provide some evidence from your own experience or personal observation for support.
3. Select the latest two fields of psychology and explain how they differ in their approaches to understanding the behaviour of the subject.

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Unit 2

INTRODUCTION TO MAJOR PERSPECTIVES

Written By: Khadija Nadeem
Reviewed by: Dr Farkhanda Rasheed

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INTRODUCTION

This Unit introduces major perspectives of Psychology. Although different perspectives are not mutually exclusive, psychologists may use different approaches to explain behaviour and thoughts depending on the specific situation or problem they are addressing. We need to study perspectives in psychology because they provide different ways of understanding human behavior and mental processes. Each perspective offers a unique lens through which we can view and interpret psychological phenomena, and each has its strengths and limitations. By understanding these different perspectives, we can gain a more comprehensive understanding of human behavior and mental processes.

Furthermore, different psychological perspectives can be useful in different contexts. For example, the behavioural perspective might be particularly useful in understanding and treating phobias, while the cognitive perspective might be more useful in understanding and treating depression. Similarly, the psychodynamic perspective might be useful in understanding personality development, while the biological perspective might be useful in understanding the neurological basis of mental disorders. By studying these different perspectives, psychologists can develop a more nuanced and comprehensive understanding of human behavior and mental processes, and they can use this knowledge to develop more effective interventions and treatments for a wide range of psychological problems.

Learning outcomes:

After completing this unit, you will be able to:

1. Differentiate between major perspectives of psychology
2. Elaborate on how various approaches to psychology define human behaviours
3. Identify the useful approaches and perspectives in the current scenario

2.1 Biological Psychology

Biological psychology emerged from a variety of scientific and philosophical traditions in the 18th and 19th centuries. It examines thoughts, feelings, and behaviours from a biological and thus physical point of view. Biological Psychology focuses on how biological processes in the brain and nervous system influence behavior and mental processes. It considers the impact of genetics, hormones, and other physiological factors on behaviour and mental health (Jones & Schmid, 2000).

2.1.1 Characteristics/ Basic Principles

The biological psychology perspective analyzes how the brain, neurotransmitters, and other aspects of our biology influence our behaviours, thoughts, and feelings. This field of psychology is often referred to by a variety of names including biopsychology, physiological psychology, behavioural neuroscience, and psychobiology(Kolb, Gibb & Robinson 2003).

2.1.2 Subject Matter

Biological psychology examines the relationship between mind and body, neural mechanisms, and the influence of heredity on behaviour. It also focuses on the nervous system, hormones and genetics. The subject matter for biological psychology studies the following:

- Evolution and Genetics
- Brain Structure and Function
- Neurons and Neurotransmitters
- Sensory and Motor Systems
- Neuroendocrine System
- Learning & Neural Plasticity

Approaches of Subject Matter

A biological perspective is relevant to the study of psychology in three ways. A biological psychologist may use a **comparative method** to understand human behaviour by comparing and studying different animal species. Studying **physiology** is another way for a biological psychologist to research their subject matter. For example, how the nervous system and hormones work, how the brain functions, or how changes in structure and/or function can affect behaviour. Finally, the biological perspective investigates **inheritance** which is what an animal inherits from its parents, and mechanisms of inheritance (genetics) to understand behaviour.

2.1.3 Methods of Studying Brain

The biological perspective tests theoretical ideas primarily through experimental methods. Over the last several decades, advancements in physical, chemical, and engineering sciences have provided more and more precise methods for the study of biology. Following are a few such methods used by biological psychologists:

2.1.3.1 Electroencephalogram (EEG)

This is a way of recording the electrical activity of the brain. Electrodes are attached to the scalp and brain waves can be traced. EEGs have been used to study sleep, and it has been found that during a typical night's sleep, we go through a series of stages marked by different patterns of brain waves.

2.1.3.2 Computerized Axial Tomography (CAT)

The CAT scan is a moving X-ray beam which takes "pictures" from different angles around the head and can be used to build up a 3-dimensional image of which areas of the brain are damaged.

2.1.3.3 Positron Emission Tomography (PET)

Even more sophisticated than a CAT scan, is the PET scan (Positron Emission Tomography) which uses a radioactive marker as a way of studying the brain at work.

2.1.4 Application of Biological Psychology

It is the result of research in biological psychology that clinical disorders such as depression and anxiety can be treated successfully. Similar studies have also broadened our horizons for the treatment and management of schizophrenia.

In-depth comprehensive techniques used by biological psychologists have expanded our understanding of emotions, their origins and regulation. These techniques have also been critical in exploring the foundations of human motivation. In recent times these methods have also been applied to research cognition, computation and consciousness of brain processes.

2.1.5 Strengths

- The biological approach provides predictions and explanations that can be scientifically tested and supported with evidence.
- It emphasizes objective measurement.
- This perspective maintains meticulous experiments to support theories
- Many of their findings/studies have real-life applications (e.g., therapy)

- Vastly applied to other perspectives such as cognitive and evolutionary psychology

2.1.6 Limitations

- Biological perspective disregards arbitrating processes such as cognitive processes of reasoning, thinking etc.
- Bio-psychological theories have a reductionist approach, often lessening the complexity of physical systems and their interaction with the environment.
- It is excessively deterministic and neglects the power of choice

2.1.7 Self-Assessment

- Write methods for studying the brain.
- What is biological perspectives' approach to its subject matter?

2.2 Psychodynamic Psychology

Psychodynamic psychology, or "psychodynamics," is the branch of study in psychology regarding the underlying factors affecting behaviour in a person, especially the subconscious mind. Everyone has experiences in their lives and childhoods that affect their day-to-day lives, choices, and behaviours. This form of psychology focuses on those factors, how they intertwine in the development and mindset of individuals, and how these findings can be of use in therapy and understanding the human psyche in general (Adler & Hull, 2014).

The term psychodynamics is also used by some to refer specifically to the psychoanalytical approach developed by Sigmund Freud (1856–1939) and his followers. Sigmund Freud's psychoanalysis was the original psychodynamic theory, but the psychodynamic approach as a whole includes all theories that were based on his ideas, e.g. Carl Jung (1912), Melanie Klein (1921), Alfred Adler (1927), Anna Freud (1936), and Erik Erikson (1950).

2.2.1 Characteristics/Basic Principles

The psychodynamic approach includes all the theories in psychology that see human functioning based upon the interaction of drives and forces within the person, particularly the unconscious, and between the different structures of the personality. Freud conceived that a person's childhood experiences and development were a determining subconscious factor in their later behaviour, perceptions, and personality (Rees & Frith 2007). This is what is referred to as "psychodynamic psychology" with key characteristics as below:

- Our behaviour and feelings as adults are rooted in our childhood experiences.
- Relationships (particularly parenting) are of primary importance in determining how we feel and behave.
- Our behaviour and feelings are powerfully affected by the meaning of events to the unconscious mind.
- Information can be obtained from dreams, irrational behaviour and what patients in therapy say.
- The personality is made up of three distinct structures: id, ego and super-ego.
- Defense mechanisms are used to protect the ego, e.g. repression.
- Children develop through a series of fixed stages: oral, anal and phallic

2.2.2 Subject Matter

Psychodynamic psychology emphasizes the systematic study of the psychological forces that underlie human behaviour, feelings, and emotions and how they might relate to early experience. Consciousness is the awareness of the self in space and time and is defined as human awareness of both internal and external stimuli.

2.2.3 Approaches of Subject Matter

The psychodynamic approach comprises a set of psychological theories that arise from the idea that humans are often driven by unconscious motivations and that adult personality and relationships are often the result of childhood experiences. Major techniques used by psychodynamic therapists include free association, dream interpretation, recognizing resistance, transference, working through painful memories and difficult issues, and building a strong therapeutic alliance.

2.2.4 Theories

2.2.4.1 Sigmund Freud: Structural Model of Personality

Sigmund Freud through his study of the psychosexual development of humans, was able to develop the Structural Model, which explains the three parts of a person's personality (id, ego, and superego).

Freud believes that a person is born with the **Id**, the pleasure-seeker portion of our personality. He believed that as newborns, the Id was crucial because it drives us to get our basic needs satisfied. For instance, a child is hungry and his Id wants food; this causes him to cry until his need is gratified.

In a span of three years, the baby grows and starts to learn new things as he interacts with the environment. During this time his **Ego** develops. The ego is rooted in the principle of reality as it is part of one's personality

When the child reaches the age of five, he begins to learn about the moral and ethical rules and restraints imposed by his parents, teachers and other people. This is the time the **Superego** develops. It is based on the moral principle as it tells us whether something is right or wrong.

2.2.4.2 Alfred Adler: Inferiority and Birth Order

Alfred Adler's theory states that all of us are born with a sense of inferiority as evidenced by how weak and helpless a newborn is. By this, Adler was able to explain that this inferiority is a crucial part of our personality, in the sense that it is the driving force that pushes us to strive become superior.

In addition to the Inferiority Theory of Personality, Adler also considers birth order as a major factor in the development of our personality. He believed that firstborn children may feel inferior and may even develop an inferiority complex once their younger sibling arrives. The middle-born children, on the other hand, are not as pampered as their older or younger siblings, but they have a sense of superiority to dethrone their older siblings in healthy competition. Thus they have the greatest potential to be successful in life. The youngest children may feel like they have the least power to influence other members of the family. Because they are often the most pampered, they may develop personality problems of inferiority just like the firstborn.

2.2.4.3 Carl Jung: Complex Personality System

Jung refers to total personality as the mind or psyche. Here, the psyche is a complex network of systems within a personality, interacting with each other. He differentiated three primary interdependent systems of the psyche, namely the conscious, the personal unconscious and the collective unconscious. Psychic energy flows continually from one system to another, in a constant striving for harmony. Following are the three parts of Jung's psyche:

- **Ego** is the rational, largely conscious system of personality, which operates according to the reality principle
- **Personal unconscious** is all of the thoughts and experiences that are accessible to the conscious, as well as repressed memories and impulses
- **Collective unconscious** contains the universal experiences of humankind transmitted to each individual; not available to conscious thought. An archetype is an entity (emotion or motivation) that exists in the collective unconscious. An archetype is an inherited tendency to perceive and respond in particular ways to universal human situations.

Carl Jung defined twelve primary types that represent the range of basic human motivations. Each of us tends to have one dominant archetype that dominates our personality:

1. Ruler
2. Creator/Artist
3. Sage
4. Innocent
5. Explorer
6. Rebel
7. Hero
8. Wizard
9. Jester
10. Everyman
11. Lover
12. Caregiver

2.2.5 Application of Psychodynamic Psychology

Psychodynamic therapy seeks to help an individual pinpoint the areas in which development has not taken place, address the issues that may have caused these problems, and work to find new strategies for daily life and how to process their urges, emotions appropriately, and consequently their behaviours from then on. Psychodynamic therapy is primarily used to treat depression and other serious psychological disorders, especially in those who have lost meaning in their lives and have difficulty forming or maintaining personal relationships:

- anxiety
- panic disorders
- post-traumatic stress disorder
- Personality disorders, such as borderline personality disorder
- Stress-related physical ailments
- Physical symptoms that lack a physical basis
- Persistent feelings of isolation and loneliness
- prolonged sadness

2.2.6 Strengths

- Freud was the first person to highlight the importance of childhood in mental health and this is an idea extensively used today.
- The psychodynamic approach takes into account both sides of the Nature/Nurture debate.

2.2.7 Limitations

- The psychodynamic approach places too much emphasis on the psychological factors, without considering the biological/genetic factors

that influence and contribute to mental health problems.

- Ignores mediational processes (e.g., thinking)
- Simplifying the human mind into the id, ego, and superego and the five psychosexual stages makes the approach reductionist.

2.2.8 Self-Assessment

- Explain the basic assumptions of the psychodynamic perspective in your words.
- Draw a comparison among the theories of Sigmund Freud, Carl Jung and Alfred Adler.

2.3 Psychology

psychology, also known as emerged early in the 20th century and remained influential till the 1960s. It is based on the idea that all actions are acquired through learning. These learned are a result of interactions with the environment through a process called conditioning. Thus, behaviour is simply a response to environmental stimuli.

Traditional believed that any person can potentially be taught to accomplish any task, irrespective of genetic background, personality traits, and internal thoughts (within their physical restrictions). It only requires the right conditioning (Skinner 1974). Consequently, is only concerned with observable stimulus-response behaviours.

2.3.1 Characteristics/Basic Principles

Behaviourism is primarily concerned with observable behaviour, as opposed to internal events like thinking. Behaviour is the result of stimulus–response (i.e., all behaviour, no matter how complex, can be reduced to simple stimulus–response features). Behaviour is determined by the environment (e.g., conditioning, nurture):

- All behaviour is learned from the environment: Behaviourism emphasizes the role of environmental factors in influencing behaviour, to the near exclusion of innate or inherited factors. This amounts essentially to a focus on learning.

- Psychology should be seen as a science: Theories need to be supported by empirical data obtained through careful and controlled observation and measurement of behaviour.

    - The components of a theory should be as simple as possible. Behaviourists propose the use of operational definitions (defining variables in terms of observable, measurable events).

- Behaviourism is primarily concerned with observable behaviour: only observable (i.e., external) behaviour can be objectively and scientifically measured.
    - Therefore, internal events, such as thinking should be explained through behavioural terms (or eliminated)
- There is little difference between the learning that takes place in humans and that in other animals: There's no fundamental (qualitative) distinction between human and animal behaviour. Therefore, research can be carried out on animals as well as humans.
- Behaviour is the result of stimulus-response: All behaviour, no matter how complex, can be reduced to a simple stimulus-response association.

2.3.2 Subject Matter

Behaviourism focuses on the idea that all behaviours are learned through interaction with the environment (Skinner, 1974). This learning theory states that behaviours are learned from the environment, and says that innate or inherited factors have very little influence on behaviour. Behaviourism as the subject matter involves two basic views:

- The proper subject matter of psychology is not consciousness but the behaviour of persons and animals, and
- The proper goal of psychology is the prediction and control of behaviour through "stimulus control."

2.3.3 Approaches of Subject Matter

- Methodological: Behaviour should be studied without connection to mental states (just overt behaviour)
- Psychological: Human and animal behaviour is explained based on external, physical stimuli.
- Analytical/Logical: Certain behaviours will arise from particular mental states and beliefs.

The most significant distinction between versions of behaviourism is that between Watson's original 'methodological behaviourism,' and forms of behaviourism later inspired by his work, known collectively as neo-behaviourism (e.g., radical behaviourism). **Methodological behaviourism** regards it as a purely objective experimental branch of natural science. Its theoretical goal is the prediction and control of behaviour. Introspection forms no essential part of its methods, nor is the scientific value of its data dependent

upon the readiness with which they lend themselves to interpretation in terms of consciousness

Radical behaviourism founded by B.F Skinner agreed with the assumption of methodological behaviourism that the goal of psychology should be to predict and control behaviour. Skinner, like Watson, also recognized the role of internal mental events, and while he agreed such private events could not be used to explain behaviour, he proposed they should be explained in the analysis of behaviour.

Another important distinction between methodological and radical behaviourism concerns the extent to which environmental factors influence behaviour. Watson's methodological behaviourism asserts the mind is a tabula rasa (a blank slate) at birth.

2.3.4 Theories

2.3.4.1 Classical Conditioning (Pavlov)

The subject learns to associate two unrelated stimuli with one another. Pavlov's dog experiment demonstrates classical conditioning: the process by which an animal or human learns to associate two previously unrelated stimuli with each other. Pavlov's dogs learned to associate the response to one stimulus (salivating at the smell of food) with a "neutral" stimulus that previously did not evoke a response (the ringing of a bell.) This type of conditioning involves involuntary responses.

2.3.4.2 Operant Conditioning (B.F Skinner)

The subject learns behaviour by associating it with consequences. Skinner box experiment demonstrates operant conditioning, in which an animal or human learns a behaviour (e.g. pressing a lever) by associating it with consequences (e.g. dropping a food pellet or stopping an electric current.) The three types of reinforcement are as follows:

- **Positive reinforcement**: When something good is added (e.g. a food pellet drops into the box) to teach a new behaviour.
- **Negative reinforcement**: When something bad is removed (e.g. an electric current stops) to teach a new behaviour.
- **Punishment**: When something bad is added to teach the subject to stop a behaviour.

2.3.4.3 Social Learning Theory (Albert Bandura)

Social learning theory, proposed by Albert Bandura, emphasizes the importance of observing, modelling, and imitating the behaviours, attitudes, and emotional

reactions of others. Social learning theory considers how both environmental and cognitive factors interact to influence human learning and behaviour.

According to Bandura, the following are the four principles of social learning:

- **Attention:** We cannot learn if we are not focused on the task. If we see something as being novel or different in some way, we are more likely to make it the focus of their attention. Social contexts help to reinforce these perceptions.
- **Retention:** We learn by internalizing information in our memories. We recall that information later when we are required to respond to a situation that is similar to the situation within which we first learned the information.
- **Reproduction:** We reproduce previously learned information (behaviour, skills, knowledge) when required. However, practice through mental and physical rehearsal often improves our responses.
- **Motivation:** We need to be motivated to do anything. Often that motivation originates from our observation of someone else being rewarded or punished for something they have done or said. This usually motivates us later to do or avoid doing the same thing.

2.3.5 Application of Behaviour Psychology

Behavioural psychology has had a major impact on clinical applications. For instance, mental health counsellors, substance abuse counsellors, and other professionals use therapeutic techniques from behaviourism to help people overcome specific issues. In addition, behaviourism theory is used in the classroom to elicit and maintain desired student behaviour. Examples of behaviour modification techniques include praise, reward systems, continual feedback, positive reinforcement and non-punitive discipline.

2.3.6 Strengths

- One of the major benefits of behaviourism is that it allows researchers to investigate observable behaviour scientifically and systematically. However, many thinkers believed it fell short by neglecting some important influences on behaviour
- One of the greatest strengths of behavioural psychology is the ability to observe and measure behaviours.
- Behaviourism is based on observable behaviours, so it is sometimes easier to quantify and collect data when conducting research.

- Effective therapeutic techniques such as intensive behavioural intervention, behaviour analysis, token economies, and discrete trial training are all rooted in behaviourism. These approaches are often very useful in changing maladaptive or harmful behaviours in both children and adults.

2.3.7 Limitations

- Many critics argue that behaviourism is a one-dimensional approach to understanding human behaviour. Critics of behaviourism suggest that behavioural theories do not account for free will and internal influences such as moods, thoughts, and feelings.

2.3.8 Self-Assessment

- What are the two kinds of learning in light of Pavlov and Skinner's theories?
- Describe the role of the environment in a behaviourist perspective.
- Write a note on how social learning takes place.

2.4 Humanist Psychology

Humanistic psychology is a perspective that emphasizes looking at the whole person, and the uniqueness of each individual. Humanistic psychology begins with the existential assumptions that people have free will and are motivated to achieve their potential and self-actualize. Humanism arose in the late 1950s via Abraham Maslow and Clark Moustakas as a "third force" in psychology, primarily in response to what some psychologists viewed as significant limitations in the behaviourist and psychoanalytic schools of thought. Behaviourism was often criticized for lacking focus on human consciousness and personality. The core tenets of this approach to psychology included: Self-actualization, creativity, health, individuality, intrinsic nature, self, being, becoming, and meaning (Clay 2002).

2.4.1 Characteristics/Basic Principles

The fundamental belief of humanistic psychology is that people are innately good and that mental and social problems result from deviations from this natural tendency. Some characteristics of humanistic psychology are as follows:

- Experiencing (thinking, sensing, perceiving, feeling, remembering, and so on) is central.

- The subjective experience of the individual is the primary indicator of behaviour.

- An accurate understanding of human behaviour cannot be achieved by studying animals.
- Free will exists, and individuals should take personal responsibility for self-growth and fulfilment. Not all behaviour is determined.
- Self-actualization (the need for a person to reach maximum potential) is natural.
- People are inherently good and will experience growth if provided with suitable conditions, especially during childhood.
- Each person and each experience is unique, so psychologists should treat each case individually, rather than rely on averages from group studies.

2.4.2 Subject Matter

Humanistic psychology focuses on each individual's potential and stresses the importance of growth and self-actualization. The fundamental belief of humanistic psychology is that people are innately good and that mental and social problems result from deviations from this natural tendency. Some of the major concepts and ideas that emerged from humanistic psychology include:

- Hierarchy of needs.
- Person-centered therapy
- Unconditional positive regard
- Free will
- Self-concept
- Self-actualization
- Peak experiences
- Fully-functioning person

2.4.3 Approaches of Subject Matter

Some of the ways that humanism is applied within the field of psychology include

- **Humanistic therapy**: Several different types of psychotherapy have emerged that are rooted in the principles of humanism. These include client-centred therapy, existential therapy, and Gestalt therapy.
- **Personal development**: Because humanism stresses the importance of self-actualization and reaching one's full potential, it can be used as a tool for self-discovery and personal development.
- **Social change**: Another important aspect of humanism is the betterment of communities and societies. For individuals to be healthy

and whole, it is important to develop societies that foster personal well-being and provide social support.

2.4.4 Theories

Among the earliest approaches, we find the developmental theory of Abraham Maslow, emphasizing a hierarchy of needs and motivations; the existential psychology of Rollo May acknowledging human choice and the tragic aspects of human existence; and the person-centred or client-centred therapy of Carl Rogers, which is centred on the client's capacity for self-direction and understanding of his or her development.

2.4.4.1 Abraham Maslow: Theory of Needs

Maslow's hierarchy of needs is a motivational theory in psychology comprising a five-stage model of human needs, often depicted as hierarchical levels within a pyramid. As a humanist, Maslow believed that people have an inborn desire to be self-actualized, that is, to be all they can be. To achieve these ultimate goals, however, several more basic needs must be met such as the need for food, safety, love, and self-esteem. Maslow's hierarchy is most often displayed as a pyramid. The lowest levels of the pyramid are made up of the most basic needs, while the most complex needs are at the top of the pyramid (Pink, 2010).

From the bottom of the hierarchy upwards, the needs are physiological (food, clothing, sleep), safety (job, health, security), love and belonging needs (friendship), esteem (respect, freedom, self-esteem), and self-actualization.

2.4.4.2 Carl Rogers: Client-Centered Theory

Rather than viewing people as inherently flawed, with problematic behaviours and thoughts that require treatment, person-centred therapy identifies that each person has the capacity and desire for personal growth and change. Rogers termed this natural human inclination “actualizing tendency,” or self-actualization. He likened it to the way that other living organisms strive toward balance, order, and greater complexity. According to Rogers, "Individuals have within themselves vast resources for self-understanding and for altering their self-concepts, basic attitudes, and self-directed behaviour; these resources can be tapped if a definable climate of facilitative psychological attitudes can be provided." (Rogers, 1980).

The person-centred therapist learns to recognize and trust human potential, providing clients with empathy and unconditional positive regard to help facilitate change.

2.4.5 Application of Humanistic Psychology

Humanistic psychology is used in the following areas:

- Person-centered therapy

- Motivation
- Depression
- Education
- Self-Worth

Humanistic therapy is also used to treat depression, anxiety, panic disorders, personality disorders, schizophrenia, addiction, and relationship issues, including family relationships. Client-centred therapy is non-directive; the therapist listens to the client without judgement, allowing the client to come to insights by themselves. The therapist ensures that all of the client's feelings are being considered and that the therapist has a firm grasp on the concerns of the client while ensuring that there is an air of acceptance and warmth.

2.4.6 Strengths

- Shifted the focus of behaviour to the individual/whole person rather than the unconscious mind, genes, observable behaviour etc.
- Satisfies most people's idea of what being human means because it values personal ideals and self-fulfilment.

2.4.7 Limitations

- Ignores biology (e.g. hormones)
- Unscientific subjective concepts, which are difficult to test.

2.4.8 Self-Assessment

- Explain the importance of an individual in humanistic psychology.
- Apply Maslow's theory to a classroom environment.

2.5 Cognitive Psychology

Cognitive psychology involves the study of internal mental processes—all of the things that go on inside our brain, including perception, thinking, memory, attention, language, problem-solving, and learning. Each of these components is pivotal in forming who we are and how we behave (Wegner, 2003). The cognitive approach began to revolutionize psychology in the late 1950s and early 1960s, to have become the most dominant approach by the late 1970s. Interest in mental processes was restored through the work of Piaget and Tolman.

2.5.1 Characteristics/Basic Principles

The following are the guiding principles for cognitive psychology:

- Human beings are information processors and mental processes guide behaviour

- The mind can be studied scientifically using better experimental and research methods
- Cognitive processes are influenced by social and cultural factors
- Comparison between human and computer processing of information.

2.5.2 Subject Matter

Traditionally cognitive psychology includes human perception, attention, memory, concept formation, reasoning, problem-solving and language processing.

- Perception: Perception is the sensory experience of the world. It involves both recognizing environmental stimuli and actions in response to these stimuli. Through the perceptual process, we gain information about the properties and elements of the environment that are critical to our survival.

- Attention**:** Attention is the ability to actively process specific information in the environment while tuning out other details.

- Learning: Learning is defined as a relatively lasting change in behaviour that is the result of experience.

- Memory**:** Memory focuses on how memories are acquired, stored, and retrieved.

- Concept Formation: Concept formation refers to a higher-order mental process that acts on information that has been perceived through our sensory organs and encoded and stored in memory

- Reasoning: Reasoning is the process by which logical arguments are evaluated or constructed.

- Problem-Solving: Problem-solving is the act of defining a problem; determining the cause of the problem; identifying, prioritizing, and selecting alternatives for a solution; and implementing a solution.

- Language Processing: Language processing focuses on the formal structures of languages and language use.

2.5.3 Approaches of Subject Matter

There are currently three main approaches in cognitive psychology:

- Experimental cognitive psychology is concerned with discovering the processes underlying behaviour and cognition.
- Neural cognitive psychology aims to understand how the structure and function of the brain relate to specific psychological processes.

- Computational cognitive psychology is the study of the computational basis of learning and inference by mathematical modelling, computer simulation, and behavioural experiments.

Cognitive processing can often be affected by **schemas** (a mental framework of beliefs and expectations developed from experience). As you get older, these become more detailed and sophisticated. A schema is a "packet of information" or cognitive framework that helps us organize and interpret information. They are based on our previous experience. Schemas help us to interpret incoming information quickly and effectively, this prevents us from being overwhelmed by the vast amount of information we perceive in our environment.

2.5.4 Theories

2.5.4.1 Jean Piaget: Cognitive Development Theory

Jean Piaget's theory of cognitive development focuses not only on understanding how children acquire knowledge but also on understanding the nature of intelligence. Piaget believed that our thinking processes change from birth to maturity because we are always trying to make sense of our world. These changes are radical but slow and four factors influence them: biological maturation, activity, social experiences, and equilibration.

Piaget's stages are:

- **Sensorimotor stage**: birth to 18-24 months
- **Preoperational stage**: 2 to 7 years
- **Concrete operational stage**: 7 to 11 years
- **Formal operational stage**: ages 12 and up

2.5.4.2 Vygotsky: Social Cognitive Theory

Vygotsky's sociocultural theory views human development as a socially mediated process in which children acquire their cultural values, beliefs, and problem-solving strategies through collaborative dialogues with more knowledgeable members of society. This theory helps us understand how people are influenced and their influence on the environment. One of the major components of social cognitive theory is observational learning. It is the process of learning others' desirable and undesirable behaviours through observation.

2.5.4.3 Cognitive Learning Theory

Cognitive learning theory explains how internal and external factors influence an individual's mental processes to complement learning. Delays and difficulties in learning are seen when cognitive processes are not working

regularly. These processes are such as attention, observation, retrieval from long-term memory, and categorization.

2.5.4.4 Cognitive Behavioural Theory

This theory mainly refers to our mental processes, such as our thoughts and interpretations of life events. It explains how the thoughts, feelings, and behaviour of a person interact with each other. Thoughts lead to particular emotions, which in turn lead to specific behavioural responses.

2.5.5 Application of Cognitive Psychology

There are numerous practical applications for this cognitive psychology, such as providing help coping with memory disorders, increasing decision-making accuracy, finding ways to help people recover from brain injury, treating learning disorders, and structuring educational curricula to enhance learning.

Cognitive behavioural therapy (CBT) is a form of psychological treatment that has been demonstrated to be effective for a range of problems including depression, anxiety disorders, alcohol and drug use problems, marital problems, eating disorders, and severe mental illness.

2.5.6 Strengths

- The cognitive approach is probably the most dominant in psychology today and has been applied to a wide range of practical and theoretical contexts.
- Combines easily with approaches: e.g. behaviourism + cognitive psychology = social learning theory; biology + cognitive psychology = evolutionary psychology.

2.5.7 Limitations

- Cognitive psychology has often relied on comparisons with how computers work as a possible way the mind might work.
- Cognitive psychology has a narrow focus on mental processes.

2.5.8 Self-Assessment

- Apply cognitive psychology subject matter to the development of schemas.
- Compare child development in light of the theories of Piaget and Vygotsky**.**

2.6 Evolutionary Psychology

Evolutionary psychology is a scientific discipline that approaches human behaviour through a lens that incorporates the effects of evolution. It combines the science of psychology with the study of biology and is based on Charles Darwin's Theory of Natural Selection (1859). Darwin argued evolution acted on patterns of behaviour as well as the anatomy and psychology of the body (Mesaudi, 2009).

2.6.1 Characteristics/Basic Principles

Foundational areas of research in evolutionary psychology can be divided into broad categories of adaptive problems that arise from the theory of evolution itself: survival, mating, parenting, family and kinship, interactions with non-kin, and cultural evolution. The basic principles of evolutionary psychology are as follows:

- The human brain's purpose is to process information, and in doing so, it produces responses to both external and internal stimuli.
- The human brain adapted and has undergone both natural and sexual selection.
- The parts of the human brain are specialized to solve problems that occurred over evolutionary time.
- Modern humans have brains that evolved after problems recurred repeatedly over long periods.
- Most of the human brain's functions are done unconsciously. Even problems that seem easy to solve require very intricate neural responses at an unconscious level.
- Many very specialized mechanisms make up the whole of human psychology. All of these mechanisms together create human nature.

2.6.2 Subject Matter

Evolutionary Psychology proposes that the human brain comprises many functional mechanisms, called psychological adaptations or evolved cognitive mechanisms designed by the process of natural selection. Evolutionary psychologists have made modest findings on topics such as infanticide, intelligence, marriage patterns, promiscuity, perception of beauty, bride price and parental investment. In short, evolutionary psychology is focused on how evolution has shaped the mind and behaviour. Though applicable to any organism with a nervous system, most research in evolutionary psychology focuses on humans.

Evolutionary psychology focuses on four key questions:

- Why is the brain designed the way it is?
- How is it designed?
- What are the functions of the human brain, and
- how does input from the current environment interact with the design of the brain to produce behaviour?

2.6.3 Approaches of Subject Matter

Evolutionary psychology is a theoretical approach to psychology that attempts to explain useful mental and psychological traits—such as memory, perception, or language—as adaptations, i.e., as the functional products of natural selection. Because evolutionary psychology is an approach rather than a content area, researchers in the discipline use a variety of techniques. These include laboratory experiments, field experiments, mathematical and agent-based simulations, surveys, neuroimaging, and so on.

2.6.4 Charles Darwin and the Theory of Evolution

Darwin's observations laid the groundwork for the field of study that would emerge more than a century later. He argued that human emotional expressions likely evolved in the same way as physical features (such as opposable thumbs and upright posture). At the base of evolutionary psychology is Darwin's theory of evolution by natural selection has three essential ingredients

- Behavioural or mental traits are heritable.
- Behavioural or mental traits vary.
- Variation in behavioural or mental traits results in differential reproduction.

2.6.5 Application of Evolutionary Psychology

Studies in evolutionary psychology are applicable in language acquisition modules, incest avoidance mechanisms, cheater detection mechanisms, foraging mechanisms, alliance-tracking mechanisms, agent detection mechanisms etc.

2.6.6 Self-Assessment

- Write short notes linking evolutionary psychology to behavioural and biological psychology.
- Develop two questions that evolutionary psychologists will explore to study aggression.
- Fill up the following summary of all perspectives

	Approach	**Characteristics**
1.	**Biological**	
2.	**Evolutionary**	
3.	**Psychodynamic**	
4.	**Behavioural**	
5.	**Cognitive**	
6.	**Humanistic**	

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Unit 3

PSYCHOLOGICAL SCIENCE

Writer: Ms Khadija Nadeem
Reviewer Dr Farkhanda Rasheed

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INTRODUCTION

Psychology is the scientific study of the human mind and behaviour. It encompasses a wide range of topics, including perception, cognition, emotion, motivation, personality, social behaviour, and mental disorders. Psychologists use a variety of research methods to investigate these topics, including experiments, surveys, observational studies, and clinical interviews. The ultimate goal of psychology is to understand human behavior and mental processes and to use this knowledge to improve people's lives. Psychology can be studied as a science by using scientific methods to collect and analyze data. This means that psychologists use systematic and objective methods to gather information about human behaviour and mental processes, and they use this information to develop theories and test hypotheses.

Psychologists use observation to gather data about human behavior and mental processes. This may involve observing people in natural settings or laboratory settings. Psychologists use experiments to manipulate variables and test hypotheses about cause-and-effect relationships. This involves randomly assigning participants to different conditions and manipulating independent variables to measure their effects on dependent variables. Psychologists use surveys and questionnaires to gather information about people's attitudes, beliefs, and behaviours. This involves asking a standardized set of questions to a large group of people and analyzing the responses.

In addition to using these methods, psychologists also follow scientific principles such as objectivity, replicability, and falsifiability. This means that they strive to be unbiased in their research, and they try to design studies that can be replicated by other researchers to test the validity of their findings. They also try to design studies in such a way that they can be disproven if the data do not support their hypotheses. By using scientific methods and principles to study human behavior and mental processes, psychologists can develop a more rigorous and evidence-based understanding of these phenomena.

Learning outcomes:

After completing this unit the students will be able to:

1. Realize the scientific inquiry involved in the study of human psychology
2. Use descriptive and inferential research methods for the creation of empirical knowledge
3. Interpret the meanings of statistical connotations associated with research

3.1 Psychological Science

The main purpose of knowledge and research is to improve the quality of human lives. Similarly, psychologists study the behaviour and mental processes of both humans and animals to bring out such research. The results of psychological research apply to everyday lives such as learning and memory, psychological disorders, family instability, and aggressive behaviour (Alexander & Winne, 2006).

3.1.1 Basic VS Applied Research

While basic and applied research share many similarities, such as the use of scientific methods and the collection of data, the main difference between the two lies in their goals and objectives. Basic research is focused on expanding knowledge and understanding in a particular area, while applied research is focused on using that knowledge to solve practical problems or address real-world issues (Woolfolk-Hoy, 2005).

Basic research aims to expand human knowledge. In psychology, basic research is carried out to answer fundamental questions about behaviour. For example, a cognitive psychologist is interested to know the influence particular memory strategies on learning.

Applied research aims to examine specific everyday issues and to provide solutions for them. In psychology, an applied research approach will tend to explore which treatment is the most effective for reducing anxiety find out which strategies work best to motivate workers or decide what types of advertising campaigns serve to reduce child marriages.

Both approaches are critical for the rapid advancement of science for basic research and applied research sustains and informs each other. The results of psychological research are reported primarily in research articles published in scientific journals. The research reported in scientific journals has been evaluated, critiqued, and improved by scientists in the field through the process of peer review.

3.1.2 Observation vs. Experimental Research

Observation and experiment are two common methods used in psychology to gather information about human behaviour and mental processes. While both methods involve collecting data through systematic observation, they differ in several ways. While observation is useful for gathering descriptive data about behaviour, experiments are more powerful in establishing cause-and-effect relationships between variables. By manipulating variables and carefully controlling for other factors, experiments can help researchers determine

whether a particular variable is causing changes in behaviour or mental processes (Fajen & Warren, 2003).

Observation involves systematically watching and recording behaviour or mental processes in a natural setting, without intervening or manipulating any variables. This means that the researcher is simply observing and recording what is happening, without any attempt to change the situation or behavior being observed. Observational studies can be used to gather descriptive data about behaviour, such as how often certain behaviours occur or how people interact with each other.

In contrast, an experiment involves manipulating one or more variables to see how they affect the behaviour or mental processes being studied. The researcher actively intervenes in the situation to see how changes in one variable affect changes in another variable. In an experiment, the researcher carefully controls the variables being studied, and may randomly assign participants to different conditions to ensure that the results are not due to individual differences. The goal of an experiment is to test a specific hypothesis about cause-and-effect relationships between variables.

3.2 Use of Scientific Method to Guide the Research

Psychologists, like scientists in other fields, believe that research is the best tool for understanding human beings and their relationships with others. The **scientific method** is the set of assumptions, procedures and documentation scientists use to conduct research. This method has developed over numerous years and provides a common framework for developing, organizing, and sharing information. The scientific method is used by all scientists: physicists, chemists, biologists, sociologists, or psychologists, regardless of their discipline. It includes the basic processes of collecting data and drawing conclusions about those data.

3.2.1 The process

The scientific method adopted in research for psychology involves a systematic and empirical approach to gathering and analyzing data. This involves the following steps:

- *Formulate a research question or hypothesis:* The researcher begins by identifying a question or problem to be studied. This may involve reviewing existing literature, identifying gaps in knowledge, or developing hypotheses based on theoretical frameworks.

- *Develop a research design:* Once the research question has been identified, the researcher designs a study to address it. This may involve

selecting a sample of participants, defining variables to be measured, and selecting a method for gathering data.

- *Collect data:* The researcher collects data using a variety of methods, including observation, experimentation, surveys, and neuroimaging techniques.
- *Analyze data:* Once the data has been collected, the researcher analyzes it using statistical methods to test hypotheses and draw conclusions.
- *Conclude:* Based on the results of the data analysis, the researcher concludes the research question or problem. This may involve confirming or rejecting hypotheses, identifying patterns in the data, or proposing new theories or explanations for the findings.
- *Communicate findings:* Finally, the researcher communicates the findings of the study through publications, presentations, or other forms of dissemination. This allows other researchers to review and critique the findings, and to build on the knowledge generated by the study.

3.2.2 The characteristics

The scientific method demands the following characteristics from psychologists:

- The statements made by psychologists are **empirical,** which means they are grounded on methodical collection and analysis of data.
- The procedures used are to be **objective** and free from the personal bias or emotions of the scientist. The objectivity is increased by placing data under the scrutiny of other scientists and even the public at large; sharing exactly how the scientist collected and analyzed the data.
- The research is designed to **replicate**, that is to repeat, add to, or modify research findings. This facilitates in growth of scientific knowledge through the reporting of research and the addition to and modification of these reported findings by other scientists.

3.2.3 Organizing the information

In the hierarchy of organizing information first is theory. A **theory** is a unified set of principles that explains and predicts mental processes, emotions or behaviours (Stangor, 2011). In psychology, a theory is an explanation or set of interrelated concepts that describes, predicts, and explains a phenomenon. A theory is developed through the process of scientific research and can be based on empirical evidence, previous research findings, and observations. A theory is more than just a collection of observations or facts; it is a coherent and organized set of ideas that can be used to guide further research and predict

future behaviour or events. The goal of a theory is to provide a framework for understanding a particular phenomenon and to generate new knowledge by guiding research and experimentation.

A good theory should be testable, meaning that it can be supported or refuted through empirical evidence. A theory should also be parsimonious, meaning that it should explain a phenomenon using the simplest possible explanation without sacrificing accuracy or completeness. Examples of psychological theories include cognitive dissonance theory, which explains how people experience psychological discomfort when their beliefs and behaviours are inconsistent, and social learning theory, which explains how people learn behaviour through observation and modelling.

Theories are important in psychology because they provide a framework for understanding behaviour and mental processes, and they guide further research and experimentation. They are essential for advancing our knowledge and understanding of human behavior and mental processes, and for developing interventions and treatments for psychological problems. All theories are limited and no single theory can interpret all behaviour in all circumstances. Consequently, existing theories are modified based on fresh or different collected data. These modified theories can make new predictions that are tested by new data, and so forth. When a better theory is found, it will replace the old one.

Theories are framed too broadly to be tested in a single experiment. Psychologists use the term **operational definition** to refer to a precise statement of how a concept (intangible theory/idea) is turned into a measurable entity. It is a description of variables in terms of procedures, actions, or processes by which they could be observed and measured. Based on framed research questions and the theory a **research hypothesis** is developed which is a tentative statement about the relationship between two or more variables. A **variable** is any characteristic that can take up different values among different people or across different times or places. A research hypothesis states the relationship between the variables and it may also specify the direction of that relationship. For instance, employees' job satisfaction is positively related to the degree of democracy in leadership styles. Here, the variables are employee job satisfaction and degree of democracy. One goal of the research is to organize information into meaningful statements that can be applied in many situations. Universal ideologies that apply to all situations in a given research area are known as **laws**. The validity of laws has already been well established and therefore are rarely subjected to scientific tests (Stangor, 2011).

SELF ASSESSMENT QUESTIONS

Write short descriptions of the following:

Applied research	
Experiment	
Theory	
Hypothesis	
Law	

3.3 Research types

Different psychologists design research with different goals in mind hence, requiring different approaches and types of studies to be conducted. A research design is the specific method a researcher uses to collect, analyze, and interpret data. Psychologists use three major types of research designs in their research, and each provides a crucial possibility for scientific investigation- descriptive, correlation and causation.

3.3.1 Descriptive research

Descriptive research refers to the measurement of behaviours and attributes through observation rather than through experimental testing. It is designed to provide a snapshot of the current state of affairs. Descriptive research is a type of research in psychology that is used to describe and analyze behaviour and mental processes. Descriptive research is focused on providing a detailed, accurate, and objective description of the phenomenon being studied.

Descriptive research can take many forms, including naturalistic observation, case studies, and survey studies (Morling, 2015). These methods are used to collect data about behavior and mental processes in a particular population or sample.

- **Case study**: Research performed in detail on a single individual, group, incident, or community, as opposed to (for example) a sample of the whole population. The most common techniques used to collect data for case studies are personal interviews, direct observation, psychometric tests, and archival records.
- **Survey**: A measure administered through either an interview or a written questionnaire to get a picture of the beliefs or behaviours of a sample of people of interest. The people chosen to participate in the research (known as the **sample)** are selected to be representative of all the people that the researcher wishes to know about (the population).

Tests and surveys are only useful if they are **valid** and **reliable**:

- **Reliability**: The degree to which a measure is likely to yield consistent results each time it is used.
- **Validity**: The degree to which a measure is assessing the concept it was designed to measure.

- **Observation**: Allow researchers to document behaviour in a natural setting and witness events that could not be produced in a lab:
  - **Observational research**: Research focusing on the observation of behaviour outside of a laboratory setting.
  - **External validity**: In research, whether or not study findings can be generalized to real-world scenarios.
- **Interview**: A type of qualitative data in which the researcher asks questions to elicit facts or statements from the interviewee. Interviews used for research can take several forms:
  - **Informal Interview**: Structured and sequential questions are not asked and the interviewee is allowed to talk conventionally.
  - **Formal interview**: Ensures that the same kind of information is collected from each interviewee. Provides more focus than a conversational approach, but still allows a degree of freedom and adaptability in getting the information from the interview.
  - **Standardized, open-ended interview**: Similar open-ended questions are asked to all interviewees. This approach facilitates faster interviews that can be more easily analyzed and compared.
  - **Closed, fixed-response interview**: All interviewees are asked the same questions and asked to choose answers from among the same set of responses.

Advantages of Descriptive Research: It provides a relatively complete picture of what is occurring at a given time. It allows the development of questions for further studies. It is meant for the piloting and standardization of tools

Limitations of Descriptive Research: It does not assess the level of significance for relationships among variables. It cannot be used to draw inferences about cause and effect. Results cannot be generalized to a larger population.

3.3.2 Statistical Analysis Used in Descriptive Research:

Measures of Central Tendency

In statistics, a measure of central tendency is a value that represents the centre or typical value of a distribution of data. The most common measures of central tendency are the mean, median, and mode.

Calculating Mean

The mean identifies the average value of the set of numbers. For example, consider the data set containing the values 20, 24, 25, 36, 25, 22, and 23.

To find the mean, use the formula: Mean equals the sum of the numbers in the data set divided by the number of values in the data set. In mathematical terms:

$$\text{Mean} = \frac{\text{sum of all terms}}{\text{how many terms or values in the set}}$$

Add the numbers in the example data set:

20+24+25+36+25+22+23=175

Divide by the number of data points in the set. This set has seven values so divide by 7.

Insert the values into the formula to calculate the mean. The mean equals the sum of the values (175) divided by the number of data points (7). Since

$$\frac{175}{7} = 25$$

the mean of this data set equals 25. Not all mean values will equal a whole number.

Calculating Median

The median identifies the midpoint or middle value of a set of numbers.
Put the numbers in order from smallest to largest. Use the example set of values: 20, 24, 25, 36, 25, 22, 23. Placed in order, the set becomes 20, 22, 23, 24, 25, 25, 36.

Since this set of numbers has seven values, the median or value in the center is 24.

If the set of numbers has an even number of values, calculate the average of the two centre values. For example, suppose the set of numbers contains the values 22, 23, 25, and 26. The middle lies between 23 and 25. Adding 23 and 25 yields 48. Dividing 48 by two gives a median value of 24.

Calculating Mode

The mode identifies the most common value or values in the data set. Depending on the data, there might be one or more modes or no modes at all.

Like finding the median, order the data set from smallest to largest. In the example set, the ordered values become 20, 22, 23, 24, 25, 25, and 36.

A mode occurs when values repeat. In the example set, the value 25 occurs twice. No other numbers repeat. Therefore, the mode is the value 25.

In some data sets, more than one mode occurs. The data sets 22, 23, 23, 24, 27, 27, and 29 contain two modes, one each at 23 and 27. Other data sets may have more than two modes, may have modes with more than two numbers (as 23, 23, 24, 24, 24, 28, 29: mode equals 24) or may not have any modes at all (as 21, 23, 24, 25, 26, 27, 29). The mode may occur anywhere in the data set, not just in the middle.

Calculating Range

The range shows the mathematical distance between the lowest and highest values in the data set. Range measures the variability of the data set. A wide range indicates greater variability in the data, or perhaps a single outlier far from the rest of the data. Outliers may skew, or shift, the mean value enough to impact data analysis.

In the sample group, the lowest value is 20 and the highest value is 36.

To calculate the range, subtract the lowest value from the highest value. Since the range equals 16.

$$36 - 20 = 16$$

In the sample set, the high data value of 36 exceeds the previous value, 25, by 11. This value seems extreme, given the other values in the set. The value of 36 might be an outlier data point.

Calculating Standard Deviation

Standard deviation measures the variability of the data set. Like range, a smaller standard deviation indicates less variability.

Finding standard deviation requires summing the squared difference between each data point and the mean [$\sum(x - \mu)2$], adding all the squares, dividing that

sum by one less than the number of values (N − 1), and finally calculating the square root of the dividend. In one formula, this is:

$$\mathrm{SD} = \sqrt{\frac{\sum_i (x_i - \mu)^2}{N-1}}$$

Mathematically, start with calculating the mean.

Calculate the mean by adding all the data point values, then dividing by the number of data points. In the sample data set,

$$20 + 24 + 25 + 36 + 25 + 22 + 23 = 175$$

Divide the sum, 175, by the number of data points, 7, or

$$\frac{175}{7} = 25$$

The mean equals 25.

Next, subtract the mean from each data point, then square each difference. The formula looks like this:

$$\sum_i^N (x_i - \mu)^2$$

where ∑ means sum, xi represents each data set value and μ represents the mean value. Continuing with the example set, the values become:

$$\begin{gathered}
20 - 25 = -5 \text{ and } -5^2 = 25 \\
24 - 25 = -1 \text{ and } -1^2 = 1 \\
25 - 25 = 0 \text{ and } 0^2 = 0 \\
36 - 25 = 11 \text{ and } 11^2 = 121 \\
25 - 25 = 0 \text{ and } 0^2 = 0 \\
22 - 25 = -3 \text{ and } -3^2 = 9 \\
23 - 25 = -2 \text{ and } -2^2 = 4
\end{gathered}$$

Adding the squared differences yields:

$$25 + 1 + 0 + 121 + 0 + 9 + 4 = 160$$

Divide the sum of the squared differences by one less than the number of data points. The example data set has 7 values, so N − 1 equals 7 − 1 = 6. The sum of the squared differences, 160, divided by 6 equals approximately 26.6667.

Calculate the standard deviation by finding the square root of the division by N − 1. In the example, the square root of 26.6667 equals approximately 5.164. Therefore, the standard deviation equals approximately 5.164.

Standard deviation helps evaluate data. Numbers in the data set that fall within one standard deviation of the mean are part of the data set. Numbers that fall outside of two standard deviations are extreme values or outliers. In the example set, the value 36 lies more than two standard deviations from the mean, so 36 is an outlier. Outliers may represent erroneous data or may suggest unforeseen circumstances and should be carefully considered when interpreting data.

Self-Assessment Exercise

Write down small descriptions of the following

Descriptive Research	
Case Study	
Survey Study	
Central Tendency	
Standard Deviation	

3.3.3 Correlation and Causation

For scientific studies in psychology, the researchers need to establish theories and verify the hypothesis using multiple forms of studies including correlation, causal-comparative and experimental research (Morling, 2015).

Correlation research: is research designed to discover relationships among variables and to allow the prediction of future events from present knowledge. Can be used to see if two variables are related and to make predictions based on this relationship.

- **Negative correlation**: A relationship between two variables such that as one increases the other decreases. On a graph, a negative correlation will have a negative slope.
- **Positive correlation**: A relationship between two variables such that as one increases or decreases the other does the same. On a graph, a positive correlation will have a positive slope.

Correlational Research allows the testing of expected relationships between and among variables and the making of predictions for causation. However, correlation cannot determine cause-and-effect relation between independent and dependent variables.

Causal Comparative Research

Causal-comparative research, also known as "ex post facto" research, is a type of non-experimental research design in psychology that seeks to identify cause-and-effect relationships between variables that cannot be directly manipulated. In causal-comparative research, the researcher observes and compares two or more groups that differ on some characteristic or attribute (such as age, gender, or socioeconomic status) to determine if there is a relationship between that characteristic and an outcome of interest. The researcher cannot control the independent variable because it has already occurred or is a pre-existing characteristic of the groups being studied.

For example, a researcher might study the effect of gender on job satisfaction by comparing the job satisfaction of male and female employees in a particular company. The researcher would not be able to manipulate gender but could observe whether there are differences in job satisfaction between male and female employees.

Causal-comparative research is useful when it is not possible or ethical to manipulate the independent variable, such as in cases where the variable is a pre-existing characteristic of the participants. However, it is important to keep in mind that causal inferences cannot be made with certainty in causal-comparative research, as there may be other confounding variables that are responsible for the observed differences between groups.

Experimental research:

Experimental research in psychology is a scientific method used to investigate cause-and-effect relationships between variables. In an experimental study, the researcher manipulates one or more independent variables and measures their effects on one or more dependent variables, while controlling for extraneous variables. The independent variable is the variable that the researcher manipulates, while the dependent variable is the variable that is being measured. Extraneous variables are variables that could potentially influence the outcome of the study but are not the focus of the research.

For example, a researcher might experiment to investigate the effect of sleep deprivation on reaction time. The independent variable would be the amount of sleep the participants receive (manipulated by the researcher), and the dependent variable would be the participants' reaction time. The researcher

would control for extraneous variables, such as age, gender, and prior caffeine consumption, to ensure that they do not affect the results of the study. For testing a hypothesis an experimental researcher establishes causation by using independent and dependent variables in a controlled environment.

- **Dependent variable**: The aspect or subject of an experiment that is influenced by the manipulated aspect; an outcome measured to see the effectiveness of the treatment.
- **Independent variable**: The variable that is changed or manipulated in a series of experiments.
- **Random assignment**: Random assignment of subjects to experimental and control conditions is a process used to evenly distribute the individual qualities of the participants across the conditions.

Experimental research allows the manipulation of independent variables, controlling extraneous variables and drawing conclusions about the causal relationships among variables. It is found to be the most sophisticated form of psychological research.

3.4 Valid Research

Good research is valid research. When research is valid, the conclusions drawn by the researcher are legitimate. Unfortunately, there are many threats to the validity of research, and these threats may sometimes lead to unwarranted conclusions. Validity is not an all-or-nothing proposition, which means that some research is more valid than other research. Only by understanding the potential threats to validity is one able to make knowledgeable decisions about the conclusions that can or cannot be drawn from a research project.
There are four major types of threats to the validity of research, and informed consumers of research need to be aware of each type:

1. **Construct validity**: refers to the extent to which the variables used in the research adequately assess the conceptual variables they were designed to measure. One requirement for construct validity is that the measure be reliable, where reliability refers to the consistency of a measured variable.
    - Threat: Researchers claim that the measured variables measure the concept/idea/theory of interest, but they may not.

2. **Statistical conclusion validity**: refers to the extent to which we can be certain that the researcher has drawn accurate conclusions about the statistical significance of the research. The research will be invalid if the conclusions made about the research hypothesis are incorrect because statistical inferences about the collected data are in error. These errors can occur either because the scientist inappropriately infers that the data do

support the research hypothesis when in fact they are due to chance, or when the researcher mistakenly fails to find support for the research hypothesis. Normally, we can assume that the researchers have done their best to ensure the statistical conclusion validity of a research design, but we must always keep in mind that inferences about data are probabilistic and never certain—this is why research never "proves" a theory.

- Threat: Conclusions regarding the research may be incorrect because either no statistical tests were made or because the statistical tests were incorrectly interpreted.

3. **Internal validity**: refers to the extent to which we can trust the conclusions that have been drawn about the causal relationship between the independent and dependent variables. Internal validity applies primarily to experimental research designs, in which the researcher hopes to conclude that the independent variable has caused the dependent variable. Internal validity is maximized when the research is free from the presence of confounding variables—variables other than the independent variable on which the participants in one experimental condition differ systematically from those in other conditions. Another threat to internal validity can occur when the experimenter knows the research hypothesis and also knows which experimental condition the participants are in. The outcome is the potential for experimenter bias, a situation in which the experimenter subtly treats the research participants in the various experimental conditions differently, resulting in an invalid confirmation of the research hypothesis.

   - Threat: Researchers claim that the independent variable caused the dependent variable. However, the dependent variable actually may have been caused by a confounding variable.

4. **External validity**: refers to the extent to which the results of a research design can be generalized beyond the specific way the original experiment was conducted. **Generalization** refers to the extent to which relationships among conceptual variables can be demonstrated in a wide variety of people and a wide variety of manipulated or measured variables.

   - Threat: Researchers claim that the findings and apply to the general population, the observed effects may only be found under limited conditions or for specific groups of people.

Any single test of a research hypothesis will always be limited in terms of what it can show, important advances in science are never the result of a single research project. Advances occur through the accumulation of knowledge that comes from many different tests of the same theory or research hypothesis. These tests are conducted by different researchers using different research

designs, participants, and operationalization of the independent and dependent variables. The process of repeating previous research, which forms the basis of all scientific inquiry, is known as **replication**.

Scientists often use a procedure known as **meta-analysis** to summarize replications of research findings. A meta-analysis is a statistical technique that uses the results of existing studies to integrate and draw conclusions about those studies. A meta-analysis provides a relatively objective method of reviewing research findings because:

- Specifies inclusion criteria that indicate exactly which studies will or will not be included in the analysis,
- Systematically searches for all studies that meet the inclusion criteria, and
- Provides an objective measure of the strength of observed relationships.

Self-Assessment Question

- What is the scientific method? Describe empiricism, objectivity and replication means to/for psychologists.
- Illustrate through a flow chart how psychological researchers organize data.
- Compare the advantages & and limitations of descriptive, correlational and experimental research designs.
- Explain how validity helps to evaluate research.
- Describe descriptive research in terms of methods and statistical analyses used.

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Unit 4

BRAINS, BODIES, AND BEHAVIOR

Written By: Afshan Huma
Reviewed by: Azhar Mumtaz Saadi

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INTRODUCTION

The brain controls the body through the nervous system, which consists of a network of neurons that transmit signals between the brain and the muscles, organs, and tissues. The spinal cord acts as a relay centre, transmitting signals from the brain to the peripheral nerves, which then distribute the signals to the rest of the body. This allows the brain to coordinate movements, regulate organ function, and control sensation. The signalling between the brain and the rest of the body is achieved through electrical and chemical signalling within the neurons.

This unit introduces the learners to one of the most significant systems of the human body i.e. nervous system. Understanding the brain and its functions is fundamental to many fields such as neuroscience, psychology, and medicine. It helps us to better understand how we think, feel, and interact with the world around us. By learning about the brain and how it works, we can develop new treatments and therapies for brain-related conditions such as mental illness, brain injuries, and neurodegenerative diseases. Studying the brain is important for advancing our knowledge and understanding of the biological and physiological processes that underpin human behaviour and cognition. Studying the brain can also foster curiosity and a deeper appreciation for the complex workings of the human body.

Learning Outcomes

At the end of the unit, the students will be able to

1. Understand how the human brain controls the body and behaviour
2. Explain the neuron as the building block of the nervous system
3. Elaborate on the process of thoughts feelings and behaviors
4. Describe the methods of studying brain functions
5. Explicate the relationship between the nervous system and the endocrine system

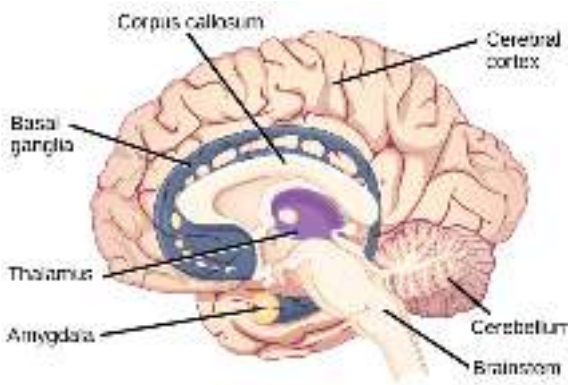
4.1 Brain and Body

The brain is located in the skull, at the top of the spinal column in the human body. The skull protects the brain and provides it with a space to occupy. The brain sits in a space called the cranial cavity, surrounded by cerebrospinal fluid, which helps to cushion and protect the brain from injury. The brain is connected to the spinal cord through the brainstem, which transmits signals between the brain and the rest of the body.

The brain is connected to the body through the central nervous system (CNS), which consists of the brain and spinal cord. The brain communicates with the rest of the body through a network of nerves called the peripheral nervous system (PNS). The PNS is made up of two branches: the somatic nervous system and the autonomic nervous system.

Somatic Nervous System: This branch of the PNS controls voluntary movements, such as moving your limbs. It carries signals from the brain to the muscles and sends sensory information back to the brain.

Autonomic Nervous System: The autonomic nervous system is a component of the peripheral nervous system that regulates involuntary physiologic processes including heart rate, blood pressure, respiration, digestion, and sexual arousal. It contains three anatomically distinct divisions: sympathetic, parasympathetic, and enteric. The signals between the brain and the body are transmitted via electrical impulses that travel along nerve fibres. These signals are processed in the brain and result in coordinated movements and regulation of organ function. The communication between the brain and the body is continuous, allowing the brain to monitor the body's state and make necessary adjustments to maintain optimal function.



The human brain is divided into several main regions-Cerebrum, Cerebral cortex, Cerebellum, Amygdala, Basal Ganglia, Brainstem, Thalamus and Hypothalamus. These are some of the main regions of the brain each with specific functions, but many other specialized areas contribute to specific functions. Overall, the brain is

a highly complex and interconnected system that coordinates and integrates information from all parts of the body (Martin, 2007).

4.1.1 Cerebrum and Cerebral Cortex:

The cerebrum and the cerebral cortex play crucial roles in our ability to think, feel, and interact with the world around us. The cerebrum is the largest part of the human brain and is responsible for many of our higher functions, such as consciousness, perception, movement, reasoning, and emotion. The cerebrum is divided into two hemispheres, the left and right, which are connected by a band of nerve fibres called the corpus callosum. The cerebral cortex is the outer layer of the cerebrum and is made up of folded grey matter. It is responsible for processing information from our senses, as well as producing thoughts, memories, and emotions.

The cerebral cortex is divided into four main lobes: the frontal lobe, the parietal lobe, the temporal lobe, and the occipital lobe. The frontal lobe is responsible for higher functions such as planning, decision-making, and movement control. The parietal lobe processes sensory information, such as touch and temperature. The temporal lobe is involved in hearing, memory, and language processing. The occipital lobe processes visual information.

4.1.2 Brainstem:

The brainstem is a region of the brain that connects the cerebral hemispheres to the spinal cord. It is located between the cerebrum and the spinal cord and is responsible for controlling many important functions that are essential for survival, such as breathing, heart rate, and consciousness.

The brainstem is divided into three parts: the midbrain, the pons, and the medulla oblongata. The midbrain is involved in processing visual and auditory information, as well as controlling movement. The pons are responsible for relaying information between the brain and the spinal cord, as well as controlling facial muscles and sensations. The medulla oblongata controls vital functions such as breathing and heart rate. Damage to the brainstem can result in serious health problems, including coma, paralysis, and death. The brainstem is also an important target for treatments aimed at controlling pain and sleep disorders. Understanding the functions of the brainstem is therefore important for advancing our knowledge of brain function and improving treatments for brain-related conditions.

4.1.3 Cerebellum:

The cerebellum is a region of the brain located at the base of the skull, behind the brainstem. It is responsible for coordinating movements, maintaining balance and posture, and contributing to the regulation of some cognitive processes. The cerebellum receives input from the sensory systems, as well as the cortex and brainstem, and uses this information to make precise adjustments to movement. It helps to ensure that movements are smooth, coordinated, and accurate, and also plays a role in learning new motor skills. The cerebellum is also involved in some non-motor functions, such as attention, language, and emotions.

Damage to the cerebellum can result in movement disorders, such as ataxia (uncoordinated movement), tremors, and difficulties with balance and coordination. Understanding the functions of the cerebellum is therefore important for advancing our knowledge of brain function and improving treatments for brain-related conditions.

4.1.4 Thalamus:

The thalamus is a region of the brain located in the center of the brain, above the brainstem. It acts as a relay center for sensory information, transmitting signals from the senses (such as sight, sound, and touch) to the appropriate areas of the brain for processing. The thalamus is also involved in regulating consciousness and attention, as well as playing a role in the regulation of sleep and wakefulness. Damage to the thalamus can result in sensory deficits, such as loss of sensation or altered perception, as well as problems with sleep and attention.

4.1.5 Hippocampus:

The hippocampus is a region of the brain located in the temporal lobe of the brain, near the centre of the brain. It plays a key role in the formation and retrieval of memories, particularly for events and experiences (episodic memory).

The hippocampus is also involved in spatial navigation, allowing us to create a mental map of our surroundings and navigate through familiar environments. It is also involved in some aspects of emotion and motivation and has been implicated in depression and post-traumatic stress disorder (PTSD).

Damage to the hippocampus can result in memory problems, including anterograde amnesia (the inability to form new memories) and retrograde amnesia (the inability to recall memories). Understanding the functions of the hippocampus is therefore

important for advancing our knowledge of brain function and improving treatments for brain-related conditions.

4.1.6 Basal Ganglia:

The basal ganglia are a group of nuclei (clusters of neurons) located deep within the brain. It is involved in several important functions, including the control of voluntary movements, the regulation of muscle tone, and the processing of sensory information. The basal ganglia also play a role in some higher functions, such as cognition, emotion, and motivation.

The basal ganglia are made up of several distinct nuclei, including the caudate nucleus, putamen, globus pallidus, and substantia nigra. These nuclei work together to regulate movement, by receiving input from the cortex and brainstem and transmitting signals to the thalamus and other brain regions that control movement. Damage to the basal ganglia can result in a variety of movement disorders, including Parkinson's disease, Huntington's disease, and dystonia. Understanding the functions of the basal ganglia is therefore important for advancing our knowledge of brain function and improving treatments for brain-related conditions.

4.1.7 Hypothalamus:

The hypothalamus is a small but complex region of the brain located near the centre of the brain, below the thalamus. It controls many autonomic functions, such as hunger, thirst, and hormone regulation, and is also involved in regulating body temperature, blood pressure, and heart rate. The hypothalamus also plays a role in controlling emotions and motivated behaviours, such as hunger and thirst.

The hypothalamus communicates with the pituitary gland, which is located just below it, and together they control the release of hormones that regulate growth, metabolism, and other physiological processes. Understanding the functions of the thalamus and hypothalamus is therefore important for advancing our knowledge of brain function and improving treatments for brain-related conditions (Gibson, 2002).

SELF ASSESSMENT QUESTIONS

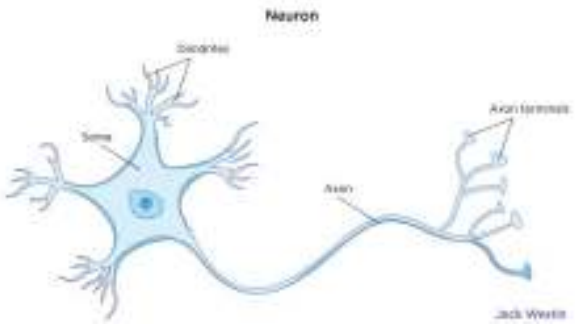
Write down the major function of each part of the brain

Cerebrum	Cerebral cortex	Cerebellum	Amygdala	Basal Ganglia	Brainstem	Thalamus	Hypothalamus

4.2 The Neuron

The nervous system is composed of more than 100 billion cells known as *neurons*. A **neuron** is *a cell in the nervous system whose function is to receive and transmit information*. Neurons are the basic building blocks of the nervous system, responsible for transmitting and processing information within the brain and throughout the body. They are specialized cells that receive, integrate, and transmit information in the form of electrical and chemical signals (Miller, 2005).

Each neuron has three main parts: the soma (cell body), the dendrites, and the axon. The dendrites receive signals from other neurons, while the axon sends signals to other neurons or muscles. The axon is often protected by a fatty sheath called myelin, which helps to speed up the transmission of signals.



When a neuron receives signals from other neurons via its dendrites, it integrates these signals and, if the sum of the signals is strong enough, generates an electrical impulse (action potential) that travels down the axon. At the end of the axon, the impulse triggers the release of chemicals called neurotransmitters, which transmit the signal to the next neuron or muscle.

Nerves are differentiated according to their function. A sensory or afferent neuron carries information from the sensory receptors, whereas a motor or efferent neuron transmits information to the muscles and glands. Both of these neurons are located in the peripheral nervous system. An interneuron, responsible for communicating among the neurons, is by far the most common type of neuron and is located primarily within the central nervous system. Interneurons allow the brain to combine the multiple sources of available information to create a coherent picture of the sensory information being conveyed

The communication between neurons is the basis of all brain functions, from sensation and perception to thought, emotion, and behaviour. Understanding the structure and function of neurons is therefore critical for advancing our knowledge of brain function and improving treatments for brain-related conditions.

4.2.1 The Electrochemical Transmission

Neurons communicate with each other through a process known as synaptic transmission. In this process, a neuron (the presynaptic neuron) releases neurotransmitters, which are chemicals that diffuse across a tiny gap called a synapse and bind to specific receptors on another neuron (the postsynaptic neuron). These binding trigger changes in the electrical and chemical properties of the postsynaptic neuron, leading to the generation of an electrical impulse (an action potential) or the suppression of one. When an electrical impulse reaches the end of the axon of a presynaptic neuron, it triggers the release of neurotransmitters into the synapse. The neurotransmitters then bind to specific receptors on the postsynaptic neuron, opening or closing ion channels and altering the flow of ions into or out of the cell. This results in changes in the electrical and chemical properties of the postsynaptic neuron, leading to the generation of an action potential or the suppression of one.

If the combined effects of all of the incoming signals are strong enough, the postsynaptic neuron will generate an action potential, which travels down its axon and triggers the release of neurotransmitters at its synapses with other neurons. In this way, the electrical and chemical signals are passed from one neuron to the next, allowing for the rapid transmission of information throughout the brain and body. This electrical and chemical communication between neurons is the basis of all brain functions, from sensation and perception to thought, emotion, and behaviour. Understanding the details of this process is critical for advancing our knowledge of brain function and improving treatments for brain-related conditions.

4.2.2 Neurotransmitters

Neurotransmitters are chemicals that are released by neurons and help to transmit signals between neurons in the brain and throughout the body. They are how one neuron communicates with another and plays a critical role in the regulation of a wide range of brain functions, including mood, movement, sensation, and cognition. When an electrical impulse (action potential) reaches the end of an axon, it triggers the release of neurotransmitters into the synaptic cleft, the tiny gap between two neurons. The neurotransmitters then diffuse across the cleft and bind to specific receptors on the postsynaptic neuron, causing changes in the electrical and chemical properties of that neuron. This can lead to the generation of an action potential or the suppression of one, allowing the signal to be passed from one neuron to the next.

There are many different types of neurotransmitters, each with its specific functions and effects. For example, dopamine is involved in the regulation of movement, motivation, and reinforcement, while serotonin is involved in regulating mood and appetite. Other neurotransmitters include norepinephrine, acetylcholine, GABA, and glutamate. Disturbances in neurotransmitter function can contribute to a variety of brain disorders, including depression, anxiety, Parkinson's disease, and schizophrenia. Understanding the functions and mechanisms of neurotransmitters is therefore critical for advancing our knowledge of brain function and improving treatments for brain-related conditions.

SELF ASSESSMENT EXERCISE

Fill in the blanks

1. _____ are specialized cells, found in the nervous system, which transmit information.

2. Some axons are covered with a _____ known as the myelin sheath, which surrounds the axon, acting as an insulator and allowing faster transmission of the electrical signal.

3. The _____ is a treelike extension that receives information from other neurons and transmits electrical stimulation to the soma.

4. The axon is an elongated fibre_____ that transfers information from the soma to the terminal buttons.

5. _____relay information chemically from the terminal buttons and across the synapses to the receiving dendrites using a lock and key type of system.

4.3 Thoughts, Feelings, and Behavior

4.3.1 Thoughts

Thinking is a complex and multifaceted process that is not yet fully understood but is thought to involve the coordination of activity across many different regions of the brain. It is also thought that the communication between neurons, which occurs via the release of neurotransmitters and the binding of these chemicals to specific receptors on other neurons, is crucial for thinking and other cognitive processes. Understanding the neural underpinnings of thinking and how these different processes interact is an active area of research and a central goal of the field of neuroscience.

Thinking can be divided into several processes such as the ability to focus on and select information from the environment that is relevant to our goals and needs at one specific time; the process of acquiring and interpreting information from the senses, such as sight, sound, touch, and smell, the ability to understand and produce words, which allows us to communicate and reason about complex concepts; the ability to encode, store, and retrieve information over time; and the higher-level cognitive processes that are involved in planning, decision-making, problem-solving, and self-control. Each of these processes involves the coordinated activity of different brain regions, and they interact with each other in complex ways to support our ability to think and reason.

4.3.2 Feelings

Describing feelings is a complex process that involves both emotional and cognitive elements, and people may use different combinations of these different ways of describing feelings to convey their emotional experiences. Understanding and accurately describing one's feelings, and the feelings of others, is an important aspect of emotional intelligence and effective communication.

Feelings can be described in several ways using emotional words, such as happy, sad, angry, afraid, surprised, and so on. These words describe the specific emotional state that a person is experiencing. In terms of the physical sensations feelings can also be explained such as a racing heart, sweating, shaking, or butterflies in the stomach and are sometimes expressed with physical features such as facial expressions, gestures, and changes in tone of voice. The thoughts and beliefs that accompany feelings are also included in the explanation for example someone might describe feeling happy because they believe that good things are happening in their life, or feeling sad because they believe that something bad has happened. The intensity and duration of a particular feeling matter a lot. For example, someone might describe feeling very strongly angry for a short period, or feeling moderately sad for an extended period. Feelings generate mood and mood swings.

Sometimes feelings and emotions are mentioned alternatively but these are different. Emotions are complex, biologically based responses to stimuli, often involving a combination of physiological changes, thoughts, and behaviours. Emotions are a way of rapidly preparing the body to respond to important events in the environment. For example, when someone experiences the emotion of fear, their heart rate increases, they become more alert, and they may behave in ways that are designed to help them escape from danger. Feelings, on the other hand, refer to the subjective experience of emotions. When someone feels afraid, for example, they have an experience of fear that is unique to them. Feelings are shaped

by an individual's personal history, home environment, culture, and current situation, and can vary greatly from person to person. In summary, emotions are complex, biologically-based responses to stimuli, while feelings are the subjective experiences of those emotions. Emotions and feelings are interrelated, but they are not the same thing. Understanding the relationship between emotions and feelings is an important aspect of understanding how people experience and express their emotions.

4.3.3 Behaviours

Human behaviour can be described as the actions that can be observed and measured, such as speech, facial expressions, gestures, and movements. Behaviors cannot be understood though until we include the subjective experiences of an individual, as well as their thoughts, feelings, and emotions. Behaviours are directed by the goals and motives that guide an behaviour, such as their desire to achieve a particular outcome or to satisfy a specific need. Similar patterns of behaviour in various situations can be based on the traits and characteristics that are relatively consistent across different situations and over time, and that help to shape an individual's behaviour and responses to the world.

The larger social and cultural factors also influence an individual's behaviour, such as family, friends, cultural norms and expectations, and historical and institutional factors. Biological factors and physiological processes that help to shape behaviour, such as genetics, brain structure and function, and hormones and neurotransmitters are also to be taken under consideration for describing human behaviour. It further requires considering the interplay between these different factors, as well as the specific situation and context in which behaviour occurs. Understanding the different factors that influence human behaviour is an important aspect of fields such as psychology, sociology, and neuroscience.
It is important to understand here, how the brain controls our actions:

- The brain — including the brain stem, medulla, pons, reticular formation, thalamus, cerebellum, amygdala, hypothalamus, and hippocampus — regulates basic survival functions, such as breathing, moving, resting, feeding, emotions, and memory.

- The cerebral cortex, made up of billions of neurons and glial cells, is divided into the right and left hemispheres and into four lobes.

- The frontal lobe is primarily responsible for thinking, planning, memory, and judgment. The parietal lobe is primarily responsible for bodily sensations and touch. The temporal lobe is primarily responsible for hearing

and language. The occipital lobe is primarily responsible for vision. Other areas of the cortex act as association areas, responsible for integrating information.

- The brain changes as a function of experience and potential damage in a process known as plasticity. The brain can generate new neurons through neurogenesis.
- The motor cortex controls voluntary movements. Body parts requiring the most control and dexterity take up the most space in the motor cortex.
- The sensory cortex receives and processes bodily sensations. Body parts that are the most sensitive occupy the greatest amount of space in the sensory cortex.
- The left cerebral hemisphere is primarily responsible for language and speech in most people, whereas the right hemisphere specializes in spatial and perceptual skills, visualization, and the recognition of patterns, faces, and melodies.
- The severing of the corpus callosum, which connects the two hemispheres, creates a "split-brain patient," with the effect of creating two separate minds operating in one person.

SELF ASSESSMENT EXERCISE

Choose the correct answer:

The change in electrical charge that occurs in a neuron when a nerve impulse is transmitted is called:

A. Synapse
B. Threshold
C. Action Potential
D. Resting potential

Answer: _____

Alzheimer's disease is caused by the neurotransmitter called

A. Serotonin
B. Dopamine
C. Glutamate
D. Acetylcholine

Answer: _____

Norepinephrine produced by the sympathetic nervous system also stimulates:

A. Pleasure
B. Pain
C. Fear
D. Depression

Answer: _____

The part of the brain that filters the sensory information and relays some of these remaining signals to the appropriate areas of the brain is called

A. Cerebellum
B. Thalamus
C. Amygdala
D. Stem

Answer: _____

Which one is located from the middle to the back of the skull, and responsible primarily for processing information about touch and taste

A. Temporal Lobes
B. Occipital Lobes
C. Frontal Lobes
D. Parietal Lobes

Answer: _____

4.4. Methods to Study Brain

There are several methods used to study the brain and its functions. Each of these methods has its strengths and limitations, and researchers often use a combination of methods to gain a more complete understanding of the brain and its functions (Stangor, and Walinga, 2014).

4.4.1 Neuroimaging techniques:

This includes techniques such as magnetic resonance imaging (MRI), positron emission tomography (PET), functional magnetic resonance imaging (fMRI) and diffusion tensor imaging (DTI) that allow researchers to visualize the brain's structure and function in real time. These techniques provide important information about the brain's structure and function and help researchers to better understand the relationships between brain activity and behavior. For example, researchers have used neuroimaging techniques to study the brain changes associated with various neurological and psychiatric conditions, as well as to explore the neural mechanisms that underlie human cognition and emotion.

- MRI: This is a technique that uses magnetic fields and radio waves to generate detailed images of the brain's structure, including the location and size of different brain regions, as well as the connections between them.
- PET: This is a technique that involves injecting a small amount of radioactive material into the bloodstream, which then accumulates in different brain regions. By measuring the amount of radioactivity in different regions, researchers can get a sense of the brain's metabolic activity and blood flow, which can provide insight into brain function.
- fMRI: This is a type of MRI that measures changes in blood oxygen levels in the brain in response to specific stimuli or tasks. This can provide insight into which brain regions are most active during different tasks or experiences, and help researchers to understand the relationships between brain activity and behaviour.
- Diffusion tensor imaging (DTI): This is a type of MRI that measures the diffusion of water molecules in the brain, which can provide information about the microstructure and integrity of white matter fibres that connect different brain regions.

4.4.2 Electroencephalography (EEG):

This is a technique that measures the electrical activity of the brain through electrodes attached to the scalp. Electroencephalography (EEG) is a technique that measures the electrical activity of the brain through electrodes attached to the scalp.

4.4.3 Magnetoencephalography (MEG):

This is a technique that measures the magnetic fields generated by the brain's electrical activity.

4.4.4 Single-neuron recording:

This involves recording the electrical activity of individual neurons in the brain, either in animals or in humans with implanted electrodes.

4.4.5 Cognitive and behavioural experiments:

These are experiments that measure an individual's cognitive and behavioural responses to specific stimuli, and can help to shed light on the underlying neural processes that drive these responses.

4.4.6 Neuropsychological tests:

This involves administering tests that assess specific aspects of cognition, such as memory, attention, and language to understand the relationship between the brain and behaviour.

4.4. 7. Genetics:

This includes studying the genetic basis of brain function and behavior, such as identifying specific genes that may contribute to the development of certain neurological or psychiatric conditions.

SELF ASSESSMENT EXERCISE

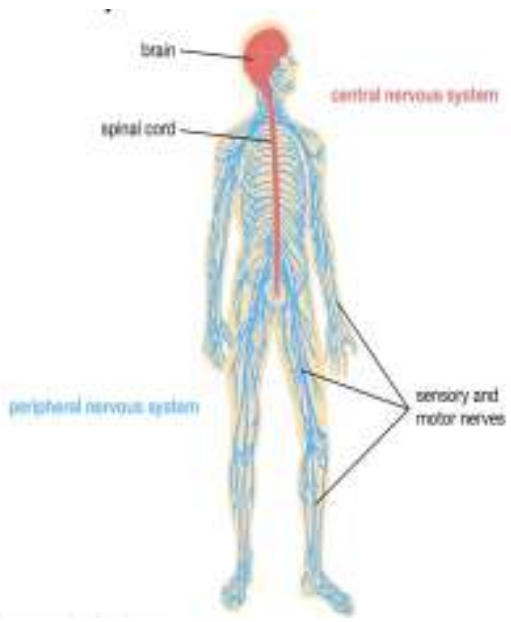
Give full names and uses of the following methods

EEG	
MRI	
DTI	
PET	
MEG	

4.5. The Nervous system and the Endocrine System

4.5.1 The nervous system

The nervous system, the electrical information highway of the body, is made up of nerves, which are bundles of interconnected neurons that fire in synchrony to carry messages. The nervous system has two major divisions. The central nervous system (CNS), made up of the brain and spinal cord, is the major controller of the body's functions, charged with interpreting sensory information, and responding to it with its directives. The CNS interprets information coming in from the senses, formulates an appropriate reaction, and sends responses to the appropriate system



to respond accordingly. Everything that we see, hear, smell, touch, and taste is conveyed to us from our sensory organs as neural impulses, and each of the commands that the brain sends to the body, both consciously and unconsciously, travels through this system as well (Bower and Parson, 2003). The peripheral nervous system (PNS) links the CNS to the body's sense receptors, muscles, and glands.

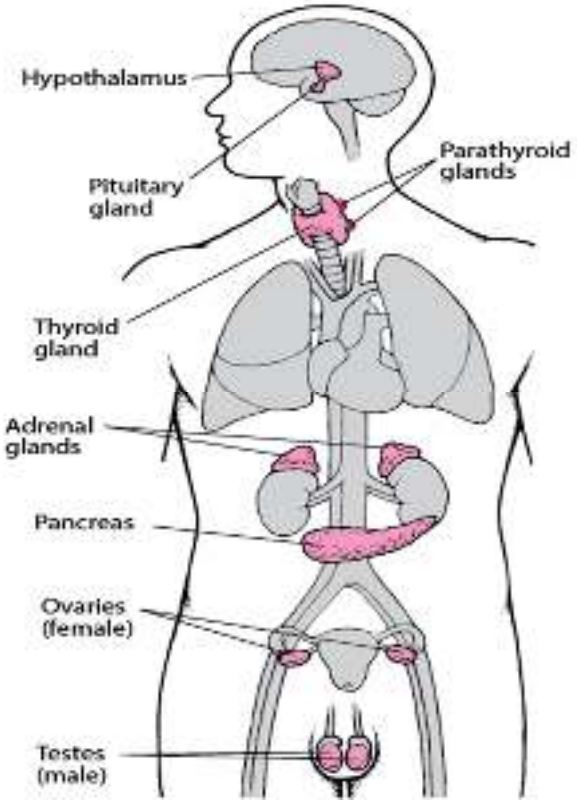
The spinal cord is the long, thin, tubular bundle of nerves and supporting cells that extends down from the brain. It is the central throughway of information for the body. Within the spinal cord, ascending tracts of sensory neurons relay sensory information from the sense organs to the brain while descending tracts of motor neurons relay motor commands back to the body. When a quicker-than-usual response is required, the spinal cord can do its processing, bypassing the brain altogether.

A reflex is an involuntary and nearly instantaneous movement in response to a stimulus. Reflexes are triggered when sensory information is powerful enough to reach a given threshold and the interneurons in the spinal cord act to send a message back through the motor neurons without relaying the information to the brain. When you touch a hot stove and immediately pull your hand back, or when you fumble your cell phone and instinctively reach to catch it before it falls, reflexes in your spinal cord order the appropriate responses before your brain even knows what is happening.

Our everyday activities are controlled by the interaction between the sympathetic and parasympathetic nervous systems. For example, when we get out of bed in the morning, we would experience a sharp drop in blood pressure if it were not for the action of the sympathetic system, which automatically increases blood flow through the body. Similarly, after we eat a big meal, the parasympathetic system automatically sends more blood to the stomach and 89 Figure 3.18 The Major Glands of the Endocrine System The male is shown on the left and the female on the right. intestines, allowing us to efficiently digest the food. Perhaps you have had the experience of not being at all hungry before a stressful event, such as a sports game or an exam when the sympathetic division was primarily in action, but suddenly finding yourself starved afterwards, as the parasympathetic takes over. The two systems work together to maintain vital bodily functions, resulting in homeostasis, the natural balance in the body's systems.

4.5.2 The Endocrine System

A primary function of the sympathetic and parasympathetic nervous systems is to interact with the endocrine system the chemical regulation of the body that consists of glands that secrete hormones which influence behaviour (Best, 2009). A gland in the endocrine system is made up of groups of cells that function to secrete hormones. A hormone is a chemical that moves throughout the body to help regulate emotions and behaviours. When the hormones released by one gland arrive at receptor tissues or other glands, these receiving receptors may trigger the release of other hormones, resulting in a series of complex chemical chain reactions. The endocrine system works together with the nervous system to influence many aspects of human behaviour, including growth, reproduction, and metabolism, and the endocrine system plays a vital role in emotions.



The pituitary gland, a small pea-sized gland located near the centre of the brain, is responsible for controlling the body's growth, but it also has many other influences that make it of primary importance to regulating behaviour. The pituitary secretes hormones that influence our responses to pain, as well as, hormones that signal the

ovaries and testes to make sex hormones. The pituitary gland also controls ovulation and the menstrual cycle in women. Because the pituitary has such an important influence on other glands, it is sometimes known as the master gland.

The adrenal glands are the triangular glands, one atop each kidney, which produce hormones that regulate salt and water balance in the body, and they are involved in metabolism, the immune system, and sexual development and function. The most important function of the adrenal glands is to secrete the hormones epinephrine and norepinephrine when we are excited, threatened, or stressed. Epinephrine and norepinephrine stimulate the sympathetic division of the ANS, causing increased heart and lung activity, dilation of the pupils, and increases in blood sugar, which gives the body a surge of energy to respond to a threat. The activity and role of the adrenal glands in response to stress provide an excellent example of the close relationship and interdependency of the nervous and endocrine systems. A quick-acting nervous system is essential for the immediate activation of the adrenal glands, while the endocrine system mobilizes the body for action. The adrenal glands produce epinephrine and norepinephrine. Norepinephrine, also known as noradrenaline, is both a neurotransmitter and a hormone responsible for our reactions to stress. The sex hormones, testosterone, estrogen, and progesterone, play an important role in sex differences.

Other glands in the endocrine system include

- the pancreas, which secretes hormones designed to keep the body supplied with fuel to produce and maintain stores of energy;
- the pineal gland, located in the middle of the brain, which secretes melatonin, a hormone that helps regulate the wake-sleep cycle; and
- the thyroid and parathyroid glands, which are responsible for determining how quickly the body uses energy and hormones, and controlling the amount of calcium in the blood and bones.

The endocrine system is a complex network of glands and hormones that work together to regulate various bodily functions, including growth and development, metabolism, and reproductive processes. The endocrine system works by secreting hormones into the bloodstream, which then travel to target tissues and organs where they bind to specific receptors and trigger a physiological response. The following is a summary of how the endocrine system works:

- Hormone production: The endocrine system consists of several glands, including the pituitary gland, thyroid gland, adrenal glands, pancreas, and

gonads (ovaries or testes). Each gland produces and secretes specific hormones into the bloodstream.

- Hormone transport: Hormones are transported through the bloodstream to target tissues and organs. Some hormones, such as insulin and adrenaline, have a wide distribution and target many different tissues. Other hormones, such as thyroid hormones, are more specific and only target certain tissues.
- Hormone binding: Once hormones reach their target tissues and organs, they bind to specific receptors located on the cell membrane or within the cell. This binding triggers a physiological response, such as increasing or decreasing metabolic rate, regulating blood sugar levels, or stimulating growth and development.
- Feedback mechanisms: The endocrine system uses feedback mechanisms to regulate hormone production and secretion. For example, when a hormone reaches a target tissue and triggers a physiological response, that response can stimulate or inhibit hormone production and secretion in the glands. This helps to maintain a stable and balanced internal environment.

SELF ASSESSMENT EXERCISE

Fill in the blanks

1. The body uses both _____and _____systems to create homeostasis.

2. The CNS is made up of bundles of _____ that carry messages to and from the PNS.

3. The nervous system is divided into _____ and _____

4. The ANS is further divided into the _____ and _____) nervous systems.

5. Specific nerves, including _____neurons, _____ neurons, and _____, each have specific functions.

6. The _____ may bypass the brain by responding rapidly using reflexes.

7. The _____ gland is a master gland, affecting many other glands.

8. Hormones produced by the _____ and _____glands regulate growth, stress, sexual functions,

9. and chemical balance in the body.

10. The adrenal glands produce _____and _____, the hormones responsible for our reactions to stress.

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Unit 5

SENSING AND PERCEIVING

Written By: Dr Afshan Huma
Reviewed By: Dr Azhar Mumtaz Saadi

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INTRODUCTION

Human beings sense and perceive the environment through the use of their senses: sight, hearing, smell, taste, and touch. The sensory signals generated by these senses are transmitted to the brain, where they are interpreted and integrated to create a perception of the environment. Once the sensory signals reach the brain, they are interpreted and integrated with other information to create a perception of the environment. This process is known as sensory integration and is performed by specialized brain regions, including the primary sensory cortex, the thalamus, and the amygdala and then other processes are initiated.

Studying sense and perception is important in psychology because it helps us to understand the nature of reality, explain behaviour, improve mental health, and enhance human performance. By gaining a deeper understanding of how we sense and perceive the world around us, we can gain valuable insights into the workings of the mind and the human experience. By studying sense and perception, we can gain a better understanding of how the mind creates our experiences of the world. Our perceptions of the world influence our thoughts, feelings, and behaviours. For example, a person who perceives a situation as dangerous is likely to respond with fear or avoidance, while a person who perceives the same situation as safe is likely to respond with confidence or approach. Understanding the process of sense and perception can help us to optimize human performance in various domains, such as sports, education, and the arts. For example, by improving our ability to perceive and respond to sensory information, we can become better athletes, learners, and artists.

Learning outcomes

After the completion of this unit, you will be able to
1. Explain the processes of Sensing and Perceiving
2. Elaborate how we experience our world through sensation
3. Describe the process and interdependence of seeing and hearing
4. Explicate the senses of tasting, smelling and touching
5. Clarify the accuracy and inaccuracy of perception

5.1 Sensing and Perceiving

Sense and perception play a critical role in shaping our understanding of reality. Our perceptions of the world around us are not a direct reflection of reality, but rather a subjective interpretation of the sensory information that we receive.

5.1.1 Sensing

Psychophysics is the branch of psychology that studies the effects of physical stimuli on sensory perceptions and mental states. The field of psychophysics was founded by the German psychologist Gustav Fechner (1801-1887), who was the first to study the relationship between the strength of a stimulus and a person's ability to detect the stimulus. Humans possess powerful sensory capacities that allow us to sense the kaleidoscope of sights, sounds, smells, and tastes that surround us. Our eyes detect light energy and our ears pick up sound waves. Our skin senses touch, pressure, hot, and cold. Our tongues react to the molecules of the foods we eat, and our noses detect scents in the air. The human perceptual system is wired for accuracy, and people are exceedingly good at making use of the wide variety of information available to them (Stoffregen & Bardy, 2001).

The senses are how the body perceives information about the environment and the internal state of the body. The senses include the sense of touch enables us to feel pressure, temperature, and pain on the skin; the sense of taste allows us to identify different flavours, such as sweet, sour, salty, and bitter; the sense of smell helps in differentiating various fragrances and odours; the auditory sense enables us to hear sounds; the sense of sight enables us to see and recognize light and colour. The sixth sense as mentioned by psychologists is named Proprioception. It is the sense of knowing where the body and its parts are in space, and it allows us to be aware of our body position, movements, and movements of individual body parts without the need to look at them.

The sensation is sometimes relatively direct; in the sense that the wide varieties of stimuli, around us inform and guide our behaviours quickly and accurately, but is always the result of at least some interpretation. We do not directly experience stimuli, but rather we experience those stimuli as they are created by our senses. Each sense accomplishes the basic process of transduction—the conversion of stimuli detected by receptor cells to electrical impulses that are then transported to the brain — in different, but related, ways. The senses and sensory organs work together to provide information about the environment and the internal state of the body, which is then processed by the brain and used to guide behaviour and regulate physiological processes. Sensory organs are the structures in the body that are

specialized to receive and respond to specific types of stimuli from the environment, such as:

- Skin: The skin is the largest sensory organ in the body and is specialized for touch.
- Tongue: The tongue is for taste and contains taste buds that are sensitive to different flavours.
- Nose: The nose is for smell and contains olfactory receptors that are sensitive to different odours.
- Ear: The ear is for hearing and contains the cochlea, a spiral-shaped structure that converts sound waves into electrical signals that can be interpreted by the brain.
- Eye: The eye is for sight and contains the retina, a layer of light-sensitive cells that detect light and send signals to the brain that are interpreted as images.

5.1.2 Perception

Perception refers to the process by which the brain interprets and makes sense of sensory information from the environment and the internal state of the body (Calvo, & Gomila, 2008). Perception involves taking raw sensory information from the senses and turning it into meaningful experiences, such as seeing colours, hearing sounds, feeling textures, etc. Perception is a multifaceted process that involves several stages, including:

- Sensory processing: The first stage of perception is the processing of raw sensory information by the sensory organs, such as the eyes, ears, skin, and other specialized structures.
- Integration and interpretation: The next stage of perception involves integrating and interpreting the raw sensory information to form a coherent representation of the environment and the internal state of the body.
- Attention and filtering: Perception also involves paying attention to certain aspects of the sensory information and filtering out irrelevant information.
- Memory and experience: Perception is also influenced by our past experiences and memories, as well as our beliefs, attitudes, and expectations, which shape how we interpret and respond to sensory information.

It is important to understand how various senses help us to make meaning such as Depth perception is the visual ability to perceive the world in three dimensions, enabling judgments of distance (Howard & Rogers, 2001). Depth perception arises

from a variety of depth cues, which are typically classified into monocular and binocular cues. We perceive distance and depth using a combination of different visual cues, such as:

- Binocular cues: These rely on the differences in the images of the same object seen by each eye, and they include stereo disparity, convergence, and accommodation. Stereo disparity refers to the differences in the positions of an object in the two images seen by each eye, which the brain uses to estimate the depth of the object. Convergence refers to the turning inwards of the eyes towards an object, which provides another depth cue. Accommodation refers to the ability of the lens of the eye to change its shape to focus on objects at different distances.

- Monocular cues: These are cues that rely on information from only one eye and include size, texture gradient, perspective, interposition, and shading. For example, objects that are farther away tend to appear smaller, and this provides a clue to their distance. Shadows and shading can also provide information about the position of objects relative to each other and the direction of light sources, which helps us to perceive depth.

- Motion cues: Our perception of distance and depth can also be influenced by the relative motion of objects and the background, as well as by our motion. For example, objects that are moving towards us appear to be getting closer, which provides a clue to their distance.

But some perceptions are based on more complex cognitive processes such as the Perception of time that involves the integration of sensory information, internal bodily signals, and past experiences. Here are some ways the brain perceives time:

- Neural timing: Our perception of time is influenced by the activity of neurons in the brain, which can be thought of as a kind of biological clock. The activity of these neurons generates rhythmic patterns that can be used to keep track of time intervals, which contributes to our perception of time.

- Attention and memory: It is also determined by our attention and memory. For example, time appears to pass more slowly when we are engaged in an interesting or attention-grabbing activity.

- Bodily signals: Time perception is dependent on internal bodily signals, such as heart rate, breathing rate, and other physiological processes, which can provide information about the passage of time.

- Context and expectations: It can also be influenced by context and expectations. For example, time can seem to pass more quickly or slowly depending on our mood, goals, and expectations.

5.2 Experiencing through Sensation

Sensations play a critical role in shaping both our conscious and unconscious worldview by providing us with information about the world, influencing our beliefs and attitudes, shaping our emotions, and helping us form memories.

- Sensory Input: Sensations provide us with information about the world around us, including our physical surroundings and our bodies. This information forms the basis of our perceptual experiences, which help us form a picture of the world and understand what is happening in it.
- Shaping Beliefs and Attitudes: Our sensations can influence our beliefs and attitudes about the world and the things we experience. For example, if we have a pleasant sensory experience with a particular food, we may develop a positive attitude towards that food and be more likely to choose it in the future.
- Shaping Emotions: Sensations can also play a role in shaping our emotions. For example, a person may associate a particular smell with a traumatic event, and smelling that scent in the future may trigger feelings of anxiety or fear.
- Shaping Memories: Our sensory experiences can also help shape our memories. For example, a person who associates a particular song with a happy memory may find that hearing that song in the future triggers feelings of joy and nostalgia.

Thus we can say that conscious experience of the world is the result of the brain processing and integrating sensory information from the environment, as well as information from our memories, beliefs, and expectations. Here are some ways the brain creates conscious experience:

Sensory processing: The brain processes and integrates sensory information from the five senses (sight, sound, touch, taste, and smell) to construct a coherent representation of the world around us.

- Attention: Attention allows us to focus on specific aspects of sensory information, and it plays a key role in determining which information is processed in a more detailed and conscious manner.
- Memory: Our memories, beliefs, and expectations influence how we perceive and interpret the world, and they can play a role in shaping our conscious experience.
- Emotion: Emotions play a crucial role in shaping our conscious experience, as they can influence what we pay attention to, how we interpret sensory information, and how we respond to the world around us.

- Higher-order cognition: Higher-order cognition, such as thought, reasoning, and decision-making, also plays a role in shaping our conscious experience, as it allows us to reflect on, evaluate, and act upon the information that we perceive and the emotions that we feel.

Not everything that we perceive is objective. Making meaning of conscious experiences involves a combination of interpretation, reflection, and cultural and personal context. Some of the ways to make meanings of conscious experiences include:

- Reflection: Taking time to think about an experience and how it affects you can help you gain insight into its meaning.
- Interpretation: Interpreting an experience based on your beliefs, values, and previous experiences can also help you understand its significance.
- Cultural and personal context: The cultural and personal context in which an experience occurs can also shape its meaning. For example, experiences may be given different meanings in different cultures or based on individual beliefs and values.
- Sharing experiences: Talking to others about your experiences and hearing their perspectives can help you gain a deeper understanding of their meaning.
- History and literature: Engaging with history and literature that reflects on similar experiences can also help you understand and make meaning of your own experiences.
- Seeking professional help: In some cases, seeking the help of an expert can be useful in understanding and making meaning of challenging experiences.

Ultimately, the process of making meaning of the world and our own experiences is subjective and can involve a combination of the above-mentioned and other strategies.

SELF ASSESSMENT QUESTIONS

- While sitting in your room which senses are active? Describe what you can sense:

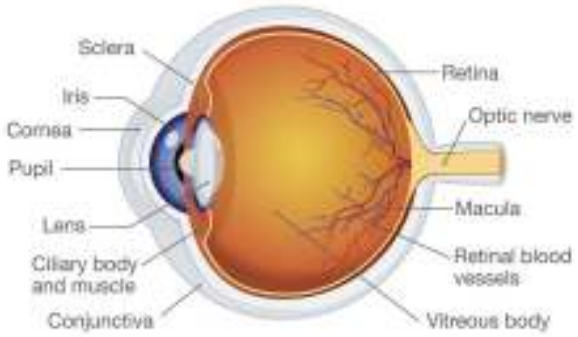
Name the sense	**Describe what you are sensing**

5.3 Seeing and Hearing

The senses of sight and hearing are linked because these two senses support perceiving and giving meanings to the signals coming from both or one of them (Chisolm, Willott, & Lister, 2003). Vision can play a supportive role in hearing by providing information about the source of a sound, helping us to understand what is being said, and providing contextual information that can help us interpret what we are hearing. However, it's important to note that the primary sense for hearing is the auditory system, and vision is a secondary source of information that can help supplement our understanding of what we are hearing. Similarly hearing can play a supportive role in vision by providing information about the location of visual stimuli, capturing our attention and directing it to a specific location, and providing contextual information that can help us to interpret what we are seeing. However, it's important to note that the primary sense for vision is the visual system, and hearing is a secondary source of information that can help supplement our understanding of what we are seeing.

5.3.1 Sense of Sight

The process of seeing occurs through a complex series of interactions between the eyes, the optic nerves, and the brain (Goodale, & Milner, 2006). Light enters the eye through the cornea, the clear outer covering of the eye, and the lens, which helps to focus the light onto the retina, a layer of light-sensitive cells at the back of the eye.



The retina contains photoreceptor cells called rods and cones that respond to light and convert it into electrical signals. These signals are then transmitted to the brain via the optic nerve.: The signals from the retina travel along the optic nerve to the brain, where they are processed in the primary visual cortex, an area of the brain located at the back of the head. The primary visual cortex works to analyze the signals, identify patterns and shapes, and create a mental image of what the eye is seeing. The information from the primary visual cortex is then sent to higher visual centres in the brain, where it is processed and integrated with other sensory information, such as touch and sound, to form a complete and coherent perception of the world.

To demonstrate the difference between rods and cones in attention to detail, choose a word in this text and focus on it. Do you notice that the words a few inches to the side seem more blurred? This is because the word you are focusing on strikes the detail-oriented cones, while the words surrounding it strike the less-detail-oriented rods, which are located on the periphery. Margaret Livingstone (2000) found an interesting effect that demonstrates the different processing capacities of the eye's rods and cones — namely, that the Mona Lisa's smile, which is widely referred to as "elusive," is perceived differently depending on how one looks at the painting. Because Leonardo da Vinci painted the smile in low-detail brush strokes, these details are better perceived by our peripheral vision (the rods) than by the cones.
Here we also need to learn how we see different colours and shapes:

- Seeing colours

We see different colours because of the way that our eyes and brain process light, with different wavelengths of light corresponding to different colours and our brain combining information from all three types of cones to produce a full-colour image. The signals from the cones are transmitted to the brain, where they are processed in the visual cortex and interpreted as colours. The brain combines information from all three types of cones to produce a full-colour image. The perception of colour is not solely based on the physical properties of light but also depends on the way that the brain processes and interprets the signals it receives from the eyes. This means that different people can see colours differently and that our perception of colour can be influenced by a range of factors, such as lighting conditions, our emotional state, and the presence of other colours in the same visual field.

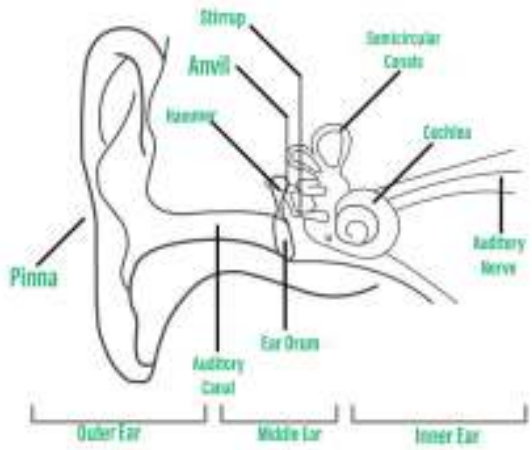
- Seeing shapes

The process of seeing different shapes involves a complex and sophisticated interaction between the eye, the optic nerve, and the brain, which work together to process and interpret visual information from the world around us. The brain uses specialized regions for recognizing different types of objects to determine the shape of an object. The information from the primary visual cortex is then sent to higher visual centres in the brain, where it is processed and integrated with other sensory information, such as touch and sound, to form a complete and coherent perception of the world. The brain has specialized regions for recognizing different types of objects, such as faces, bodies, and objects. These regions work together to analyze the information from the primary visual cortex and determine the shape of an object.

5.3.2 Sense of Hearing

Hearing begins with the process of transduction. Sound waves that are collected by our ears are converted into neural impulses, which are sent to the brain where they

are integrated with experience and interpreted as the sounds we experience. The human ear is sensitive to a wide range of sounds, from the faint tick of a clock in a nearby room to the roar of a rock band at a nightclub, and we can detect very small variations in sound. But the ear is particularly sensitive to sounds in the same frequency as the human voice. A mother can pick out her child's voice from a host of others, and when we pick up the phone we quickly recognize a familiar voice. In a fraction of a second, our auditory system receives the sound waves, transmits them to the auditory cortex, compares them to stored knowledge of other voices, and identifies the caller.



Vibrating objects (such as human vocal cords or guitar strings) cause air molecules to bump into each other and produce sound waves, which travel from their source as peaks and valleys, much like the ripples that expand outward when a stone is tossed into a pond. Unlike light waves, which can travel in a vacuum, sound waves are carried within media such as air, water, or metal, and it is the changes in pressure associated with these media that the ear detects.

This is how our auditory sensory organs work:

- Sound waves enter the ear: Sound waves travel through the air and enter the ear, where they cause the eardrum to vibrate. The vibration of the eardrum is then transmitted to the bones of the middle ear, which amplifies the sound.
- Sound is transformed into electrical signals: The vibration of the middle ear bones causes fluid in the inner ear to move, which in turn stimulates the hair cells of the inner ear. The movement of the hair cells generates electrical signals, which are transmitted to the brain via the auditory nerve.
- The brain processes the signals: The electrical signals from the inner ear travel along the auditory nerve to the brain, where they are processed in the auditory cortex, a region of the brain located in the temporal lobe. The auditory cortex works to analyze the signals and extract information about the sound, such as its pitch, volume, and direction.
- Speech processing: The auditory cortex can recognize the patterns of sound that make up speech and send the information to regions of the brain

responsible for processing language. These regions work together to analyze the sound and extract meaning from the words that are spoken.

We do not only hear mechanically, rather the brain also processes information about the emotional content of the voice and the context in which the words are spoken. This information is combined with language processing to form a complete and coherent understanding of what is being said. Thus hearing voices involves a complex interaction between the ear, the auditory nerve, and the brain, which work together to process and interpret sound information from the world around us. The brain can extract information about the sound, such as its pitch, volume, and direction, as well as the emotional content of the voice and the meaning of the words being spoken.

5.4 Tasting, Smelling, and Touching

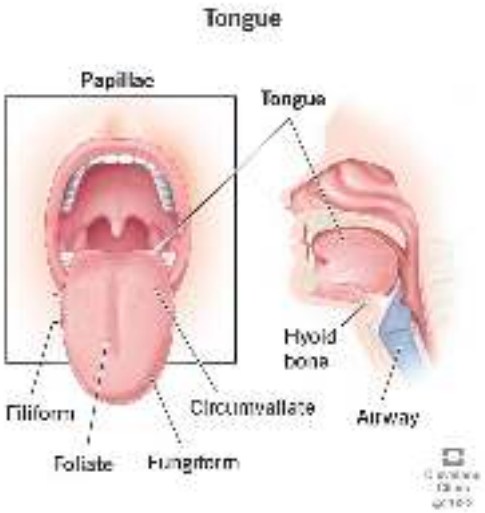
Taste, smell and touch are the three senses that relate deeply to each other. Tasting different and flavours is a complex process that involves the interaction of various senses, including taste, smell, and touch. The taste buds on our tongues are sensitive to the five basic tastes: sweet, sour, salty, bitter, and savoury (Northcutt, 2004). When we put food or drink in our mouths, the molecules from the food interact with the taste receptors on the taste buds, which send signals to the brain through the gustatory nerve. The sense of smell also plays a role in flavour perception. When we eat, volatile aroma molecules are released from the food and travel up to the olfactory receptors in the nose. The olfactory receptors send signals to the brain, which integrate the information with the taste information from the tongue. The sense of touch, including the texture and temperature of the food, also contributes to flavour perception. The pressure and texture of the food are sensed by the mouth, tongue, and other areas of the oral cavity, and the information is transmitted to the brain through the trigeminal nerve. The information from the taste buds, olfactory receptors, and oral cavity is integrated into the brain to form a complete experience of flavour. The brain also integrates information about the context, such as the background noise and lighting, to form a complete experience of the food and drink. The brain integrates information to form a complete experience of flavour, which is unique to each individual and can vary greatly depending on a range of factors, such as age, genetics, and personal experience.

5.4.1 Sense of Taste

Our tongues are covered with taste buds, which are designed to sense chemicals in the mouth. Most taste buds are located in the top outer edges of the tongue, but there are also receptors at the back of the tongue as well as on the walls of the mouth and at the back of the throat. Our ability to taste begins at the taste receptors on the tongue. The tongue detects six different taste sensations, known respectively as

sweet, salty, sour, bitter, spicy and savoury. Each taste bud is made up of a group of specialized cells called taste receptor cells. These cells are organized into clusters and are surrounded by supportive cells.
The tongue is made up of several different parts, including:

- Papillae: These are the small bumps on the surface of the tongue that contain the taste buds. There are several different types of papillae, including fungiform (mushroom-shaped), filiform (thread-like), and circumvallate (walled) papillae.
- Foliate papillae: These are located on the sides of the tongue and are responsible for detecting bitter tastes.
- Circumvallate papillae: These are located at the back of the tongue and are responsible for detecting bitter and sour tastes.



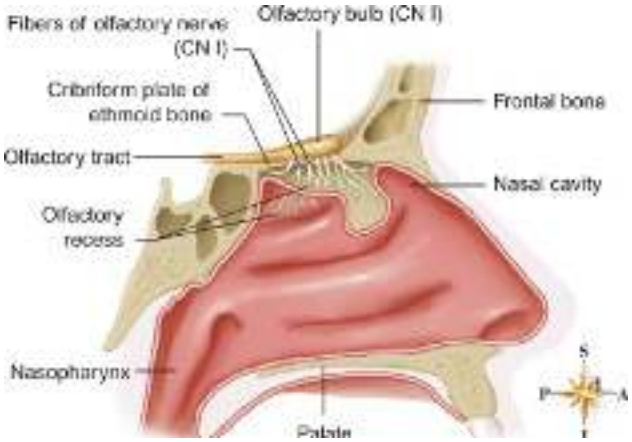
- Fungiform papillae: These are located on the front and sides of the tongue and are responsible for detecting sweet and sour tastes.
- Filiform papillae: These are located on the front and sides of the tongue and do not contain taste buds. Instead, they are responsible for helping to move food around in the mouth and to clean the tongue.

When we eat food, the taste receptors are stimulated by the chemicals in the food, which triggers a series of events that result in the perception of taste. The taste receptor cells are connected to nerve fibres, which send electrical signals to the brain, where they are processed into the experience of taste. The taste buds are located in different parts of the mouth and on the tongue, with some areas of the tongue being more sensitive to certain tastes than others. For example, the tip of the tongue is typically more sensitive to sweet tastes, while the back of the tongue is more sensitive to bitter tastes. The area of the sensory cortex that responds to taste is in a very similar location to the area that responds to smell, a fact that helps explain why the sense of smell also contributes to our experience of the things we eat. You may remember having difficulty tasting food when you had a bad cold, and if you block your nose and taste slices of raw potato, apple, and parsnip, you

will not be able to taste the differences between them. Our experience of texture in food (the way we feel it on our tongues) also influences how we taste it.

5.4.2 Sense of Smell

As we breathe in air through our nostrils, we inhale airborne chemical molecules, which are detected by the 10 million to 20 million receptor cells embedded in the olfactory membrane of the upper nasal passage. The olfactory receptor cells are topped with tentacle-like protrusions that contain receptor proteins. When an odour receptor is stimulated, the membrane sends neural messages up the olfactory nerve to the brain. We have approximately 1,000 types of odour receptor cells (Bensafi et al., 2004), and it is estimated that we can detect 10,000 different odours (Malnic, Hirono, Sato, & Buck, 1999). The receptors come in many different shapes and respond selectively to different smells. The sense of smell, or olfaction, is one of the primary senses that helps us to identify and experience different odours. The process of smelling involves the following steps:



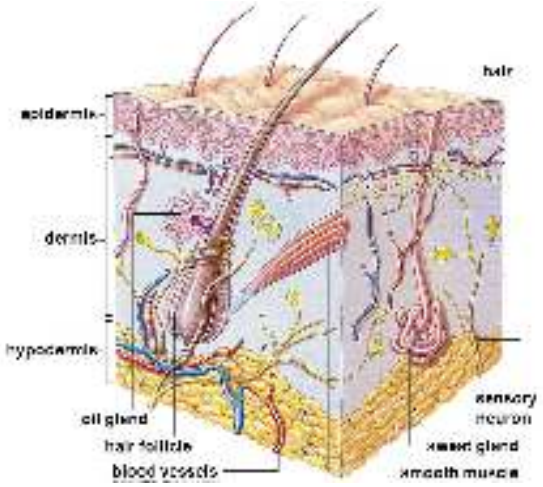
- Odorant molecules: When we smell something, odorant molecules are released into the air and reach the nostrils.
- Olfactory receptor neurons: The odorant molecules are then detected by olfactory receptor neurons, which are specialized nerve cells located in the lining of the nasal passages.
- Olfactory receptor proteins: Each olfactory receptor neuron has a specific type of olfactory receptor protein that it uses to detect specific odorant molecules.
- Nerve impulses: When an odorant molecule binds to an olfactory receptor protein, it triggers a chain reaction that results in the production of an electrical nerve impulse.
- Olfactory bulb: The nerve impulses are then sent to the olfactory bulb, which is located in the brain and is responsible for processing smells.

The information from the olfactory bulb is then sent to other areas of the brain, including the limbic system, which is involved in processing emotions and memories. This information is used to create the perception of odour, which is the experience of smelling something. The process of smelling involves the release of odorant molecules, the detection of these molecules by olfactory receptor neurons, the production of electrical nerve impulses, the processing of these impulses in the olfactory bulb, and the creation of the perception of odour in the brain.

5.4.3 Sense of Touch

The skin, the largest organ in the body, is the sensory organ for touch. The skin contains a variety of nerve endings, combinations of which respond to particular types of pressures and temperatures. When you touch different parts of the body, you will find that some areas are more ticklish, whereas other areas respond more to pain, cold, or heat. The sensation of touch, or tactile perception, is the result of a complex process that involves the skin, nerves, and brain.
Here's how the process works:

Touch receptors: The skin contains specialized nerve endings called touch receptors, which respond to different types of stimuli, such as pressure, vibration, and temperature.
Nerve impulses: When touch receptors are activated, they generate electrical nerve impulses that are sent along nerve fibres to the spinal cord.
Spinal cord: In the spinal cord, the nerve impulses are processed and then sent to the brain for further processing.



Brain: In the brain, the information from the touch receptors is integrated with information from other senses, such as vision and hearing, to create a complete and meaningful experience of touch.

The thousands of nerve endings in the skin respond to four basic sensations - pressure, hot, cold, and pain; but only the sensation of pressure has its specialized receptors. Other sensations are created by a combination of the other four. For example:

- The experience of a tickle is caused by the stimulation of neighbouring pressure receptors.
- The experience of heat is caused by the stimulation of hot and cold receptors.
- The experience of itching is caused by repeated stimulation of pain receptors.
- The experience of wetness is caused by repeated stimulation of cold and pressure receptors.

The sensation of touch is the result of the activation of touch receptors in the skin, the generation of electrical nerve impulses, the processing of these impulses in the spinal cord and brain, and the creation of the perception of touch in the brain. The process of pain perception is also very closely related to the sensation of touch, it begins with the activation of pain receptors, which are specialized nerve endings in the skin and other tissues (Wall, 2000). When these receptors are activated, they send electrical signals, called pain impulses, to the spinal cord. In the spinal cord, the pain impulses are processed and then sent to the brain. In the brain, the information from the pain receptors is integrated with information from other senses and with past experiences, expectations, and emotions to create a subjective experience of pain. The experience of pain can vary greatly depending on the individual, the type and intensity of the stimulus, and other factors, such as attention, distraction, and stress. Gate control theory explains how large and small neurons work together to transmit and regulate the flow of pain to the brain. The gate control theory of pain proposes that pain is determined by the operation of two types of nerve fibres in the spinal cord. One set of smaller nerve fibres carries pain from the body to the brain, whereas a second set of larger fibres is designed to stop or start (as a gate would) the flow of pain (Melzack & Wall, 1996).

Proprioception is our ability to sense the positions and movements of our body parts. Postural and movement information is detected by special neurons located in the skin, joints, bones, ears, and tendons, which pick up messages from the compression and the contraction of muscles throughout the body. The vestibular system, composed of structures in the inner ear, monitors the head's position and movement, maintaining the body's balance. The sense of proprioception is created by specialized nerve receptors called proprioceptors, which are located in our muscles, tendons, and joints. These receptors send signals to the brain about the position and movement of our body parts, allowing us to have a sense of where we are in space and how our body parts are moving relative to each other. Proprioception is important for a variety of functions, including motor control,

balance, and coordination. For example, it allows us to walk, run, and perform other movements without having to look at our feet, as we have a sense of where they are and how they are moving This sense plays an important role in motor control, balance, and coordination, and is often referred to as the "sixth sense."

SELF-ASSESSMENT QUESTIONS:

Which parts of the body and brain are responsible for making meanings of the following:

Sensation	Part of body	Part of Brain
Smell of flowers		
Taste of salt		
Headache		
Hot weather		
The beauty of the scenery		

5.5 Accuracy and Inaccuracy in Perception

Our perception is based on multiple sensory interactions. Sensory interaction occurs when different senses work together, and further processing happens in the brain (Sekuler & Blake, 2006). The following can be the reasons for accurate or inaccurate perceptions:

- Selective attention allows us to focus on some sensory experiences while tuning out others. Our perception can also be influenced by where our attention is focused. When we are focused on one thing, we may miss important information about the things around us, leading to misperceptions.
- Sensory adaptation occurs when we become less sensitive to some aspects of our environment, freeing us to focus on more important changes. Sometimes, the senses themselves can send incorrect information to the brain. For example, our eyes can sometimes perceive a straight line as bent due to an optical illusion.
- Cognitive illusions are examples of how our expectations can influence our perceptions. Our prior experiences and expectations can also influence our perception. For example, if we expect to see a particular object in a certain place, we may misperceive what is there.

- Our emotions, motivations, desires, and even our culture can influence our perceptions. Emotions can also affect our perception. For example, if we are feeling anxious or stressed, we may perceive things as being more dangerous or threatening than they are.

- Mental health: Mental health conditions, such as anxiety, depression, and psychosis, can also cause misperceptions. People with these conditions may experience hallucinations or delusions, in which they perceive things that are not there.

There are many reasons why we might perceive stimuli incorrectly, including sensory errors, attention, expectations, emotion, and mental health conditions (Proctor, & Van Zandt, 2008). By understanding these factors, we can gain a better understanding of how our perception works and why we sometimes perceive things differently from how they are.

5.5.1 Illusion

There are many different types of illusions, including visual illusions, auditory illusions, and tactile illusions. Visual illusions can involve the perception of shape, size, movement, colour, or depth, and can be caused by factors such as the arrangement of objects in space, the use of patterns or textures, or the interaction of light and shadow. Auditory illusions can involve the perception of sound or music and can be caused by factors such as the repetition of patterns or rhythms. Illusions can also be influenced by our expectations, prior knowledge, emotions, and mental state. For example, if we expect to see a particular object in a certain place, we may perceive it as being there even if it is not.

Illusions demonstrate that our perception of the world around us may be influenced by our prior knowledge. But the fact that some illusions exist in some cases does not mean that the perceptual system is generally inaccurate – in fact, humans normally become so closely in touch with their environment that the physical body and the particular environment that we sense and perceive become embodied—that is, built into and linked with our cognition, such that the world around us becomes part of our brain (Calvo & Gomila, 2008). The close relationship between people and their environments means that, although illusions can be created in the lab and under some unique situations, they may be less common with active observers in the real world (Runeson, 1988).

5.5.2 Hallucination

A hallucination is a sensory experience that occurs in the absence of any corresponding external stimulus. In other words, a person who is hallucinating perceives things that are not there. Hallucinations can involve any of the five senses

but are most commonly associated with hearing and seeing. Hallucinations can occur as a result of many different factors, including medical conditions such as schizophrenia, brain injuries, infections, or exposure to certain drugs or toxins. They can also be caused by sleep deprivation, extreme stress, or other psychological factors.

In some cases, hallucinations can be a symptom of a mental illness, such as schizophrenia or bipolar disorder. In these cases, the hallucinations can be intense and persistent, and can significantly impact a person's daily life and functioning. A hallucination is a sensory experience that occurs in the absence of any corresponding external stimulus and can be caused by many different factors, including medical conditions, exposure to certain drugs or toxins, sleep deprivation, extreme stress, or mental illness. Hallucinations can involve any of the five senses but are most commonly associated with hearing and seeing.

SELF-ASSESSMENT QUESTION:

Differentiate between illusion and hallucination. Give three examples for each

Illusions	
Hallucinations	

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Unit 6

State of Consciousness

Writer : Dr. Hafiz Tahir Jamil
Reviewer: Dr. Azhar Mahmood Saadi

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INTRODUCTION

Consciousness refers to the state of being aware of one's surroundings, thoughts, feelings, and sensations. It is a subjective experience that is difficult to define and measure objectively. Consciousness is considered to be a fundamental aspect of human experience, and it is studied extensively in the field of psychology. Sleep is a state of reduced consciousness characterized by a decrease in physical activity and a loss of awareness of the surroundings. It is a natural process that is essential for physical and mental health. Sleep is regulated by a complex system of biological and environmental factors, including the circadian rhythm, which is the internal biological clock that controls sleep and wakefulness. The process of reviving someone from an unconscious state depends on the underlying cause of the unconsciousness. In some cases, the person may regain consciousness on their own once the cause of the unconsciousness is resolved, such as in the case of fainting due to a temporary drop in blood pressure or dehydration.

The study of consciousness and sleep can also provide insights into cognitive processes such as attention, perception, and memory. For example, research on selective attention during consciousness has shown that the brain selectively filters out irrelevant information to focus on important stimuli. This unit introduces the state of consciousness and sleep; considering its importance in psychology because it provides insights into the nature of the human mind, cognitive processes, clinical conditions, and health outcomes.

Learning outcomes

After the completion of this unit, you will be able to:

1. Recognize the signs and symptoms of conscious and unconscious states
2. Realize the various sleep levels and conditions with or without drugs

6.1 State of Consciousness

The term "consciousness" refers to a state of being aware. It is a term used to describe those experiences that include an awareness of the self, bodily sensations, thoughts, and the surrounding environment. To express how ridiculous something is, we use the opposite word "unconscious" instead of "senseless," as in "Theresa fell off the ladder and hit her head, falling unconscious." Consciousness and unconsciousness are far more complicated than being unceremoniously thrown off a ladder. In other words, the word "consciousness" means much more than simply being either "on" or "off." Freud, for instance, believed that even while we are awake, our mental world extends far beyond our consciousness (like being in the car and forgetting the rest of the world can see into your windows). Freud proposed that our subconscious motivations and memories are not always accessible to our conscious minds as a result of this (Tart,1975).

On second thought, it's clear that consciousness is difficult to pin down. For instance, are people in the middle of a daydream awake? What if they are intoxicated? We will begin this module by introducing the concepts of various levels of consciousness and then move on to look at altered states of consciousness, including hypnosis and sleep. Awareness of an external object or self, such as thoughts, feelings, memories, or sensations, is known as consciousness. Additionally, sentience, awareness, subjectivity, the ability to experience or to feel, wakefulness, having a sense of selfhood, and the executive-control system of the mind have all been given definitions that describe them. In the past, many scientists held a cynical view of consciousness. More recently, however, it has gained significant attention within the fields of psychology and neuroscience.

6.1.1 Theories of Consciousness

Though definitions remain elusive, many philosophers assert that there is an underlying intuition of what consciousness is that is held by everyone. Over the last two centuries, thinkers such as Descartes and Locke have grappled with trying to grasp the nature of consciousness and classify its primary features. These are some of the philosophical issues that we currently face in our thinking about consciousness: Does it ever make sense to think of consciousness as mechanistic? If non-human consciousness exists, how can we recognise it? What does consciousness have to do with language? Might we ever be able to conceptualise consciousness in a way that doesn't make use of a dualistic distinction between mental and physical states or properties?

These theories may be found in Mayan and Incan civilizations. Various theories of multiple levels of consciousness are believed to have appeared in the historical records of ancient Mayan and Incan civilizations, as well as in the

diverse realms of spiritual, psychological, medical, and moral speculation in both Eastern and Western cultures. Human awareness of both internal and external stimuli can be defined as consciousness. Given that there is some overlap between hypotheses, recent studies have tried to merge different perspectives to come up with new models that have elements from various viewpoints.

According to the ancient Mayans, who were among the first to do so, a structured way of perceiving the various levels of consciousness, their purposes, and their temporal relationships to humans was developed. The Mayans believed that consciousness was the most basic form of existence, and thus capable of evolution. Incas also considered consciousness development in terms of concern for others, as well as an increase in awareness.

Consciousness and John Locke

Consciousness is something that John Locke, an 18th-century philosopher, was among the first to speak and write about. To his way of thinking, we are who we are because of what's in our minds (or memories). The word 'asserted' means that he believes the consciousness in our heads does not exist as long as we are alive. The founder of empiricism, John Locke, maintained that the consciousness of one soul could be transferred to another.

Consciousness and René Descartes

Descartes: “I think, therefore I am.”
The 17th century was also a time when René Descartes broached the subject of consciousness. He wanted to explain how our consciousness, which is a non-physical entity, can be created from our physical bodies, which are physical entities. res extensa in contrast to recogitans (the realm of thought), which he called res extensa (the realm of extension). This interaction occurs inside the brain.

Consciousness and Sigmund Freud

“The unconscious mind”
Though theories have fluctuated between the East and West, the East's understanding of consciousness has remained relatively constant. The theory of Sigmund Freud, a medical doctor and the father of psychoanalytic theory, is one of the most popular Western theories. Human consciousness can be conceptualised as having three distinct levels: the level of conscious awareness, the level of preconscious thought, and the level of unconscious mental activity. Freud's ideas on the id, ego, and superego exist at all of these levels. The conscious level includes everything we are aware of, including our knowledge and the knowledge we have about our surroundings. In the preconscious, you'll find things that you could concentrate on if you wanted to, as well as many memories that are easily accessible. for instance, the author might say that when

she is studying for a test, some of the thoughts that are conscious at the time become unconscious, but these thoughts have not been repressed and, thus, are easily capable of being brought to mind and becoming conscious (for example, the tip-of-the-tongue effect). Many memories, thoughts, and urges that we are not aware of are found in the unconscious. Approximately 80% of what's stored in the unconscious is thought to be distasteful or troubling, such as unacknowledged sexual desires. Our awareness of these elements has been shown to affect our behaviour, even though we are unaware of it.

Neo-Cognitive Theories

While Freud's theory remains a popular one, various perspectives within the field of psychology have developed. These views are not necessarily mutually exclusive, they simply differ in their methods. For over two millennia, philosophers from Plato to Descartes have grappled with the mysteries of human consciousness, including how the everyday inner workings of our brains yield a unified, cohesive reality and a sense of an individual self. Descartes is known for his dualist theory of consciousness, which identifies the physical body as separate from the immaterial mind. An accurate reflection of human consciousness rendered more popularly: “I think, therefore I am.”

According to Karl Marx, the concept of the mind-body dichotomy should be rejected, as he holds that consciousness is engendered by one's environment. This idea was advanced by John Locke, an earlier philosopher, who asserted that a person's consciousness and personal identity were independent of all substances. He went on to say that since there is no reason to assume that consciousness is tied to any particular body or mind, or that it cannot be transferred from one body or mind to another, it should be assumed that consciousness exists in all places, in everything.

In an 1884 article for the Atlantic Monthly, American psychologist William James likened the stream of consciousness to an undisturbed flow, unchanged despite dramatic changes and shifts. Though much of psychology's early-twentieth-century research focused on behaviour that could be seen and measured, human consciousness research has grown greatly in the latter half of the twentieth century (Vaitl, 2005).

6.1.2. Consciousness in developmental psychology

The majority of developmental psychologists believe that the human mind doesn't have just one fundamental type of consciousness but instead is a developmental process with potential higher stages of cognitive, moral, and spiritual quality. Consciousness, both in quality and degree, changes throughout childhood, adolescence, and adulthood. Consciousness is also affected by abnormal development, as are mental illnesses.

The question of how consciousness arises from the physical entity can be summarised as the "mind-body problem." While the body is physical, how are our mental state, beliefs, actions, and thinking related to our bodily functions, the external environment, and events? Descartes, the first and most important philosopher to answer this problem, described it as Cartesian dualism. Res cogitans, an immaterial realm, contrasts with material things, which he called res extensa (the realm of extension). This interaction occurs inside the brain. Later, he presented the idea that the pineal gland provides the interaction, but was repeatedly challenged on this claim. It was this event that prompted us to begin exploring the topic of consciousness in more detail, as we shall see.

The research on consciousness

For the most part, research today is dedicated to discovering what exactly consciousness is. It seeks to determine the neural and psychological correlates of conscious information. Intriguing topics include things like the brain's ability to observe what's going on in the background (subliminal perception), a lack of awareness of one's blindness (anosognosia), sleep brainwaves, and heightened states of consciousness due to the use of psychoactive drugs or spiritual practices.

According to the majority of experimental studies, the goal of assessing consciousness is to elicit verbal reports from human participants. Scientists must compare verbal reports of subjective experience to concurrent neural activity to discern the true significance of those reports. The hope is to discover that specific brain activity or brain pattern in a specific location is strongly correlated with conscious awareness. These studies used several different brain imaging techniques, including EEG and fMRI, to analyse brain activity for physical measures. In addition to the cerebrum, a range of higher cognitive functions known as executive functions are also believed to occur in the prefrontal cortex, which is found primarily in the brain's higher regions (Tassi, 2001).

Study of how society affects the conscious experience

According to social psychologists, human consciousness is primarily the result of environmental influence. For instance, due to differences in language and other cultural factors, each culture develops its perceptions of reality. The resulting difference in coding has an impact on the various facets of reality, as well as on one's consciousness. An analysis of language reveals the mentality of people who speak that language.

Neuropsychology: Consciousness

Neuropsychologists regard consciousness as built into neural systems and brain structures, which serve as a foundation for it. Studies that examine the relationship between the subjective experiences reported by participants and the

activities occurring simultaneously in their brains are a major component of modern scientific literature on consciousness. The hope is to find brain activity that is predictive of conscious awareness. Some brain imaging techniques, such as EEG and fMRI, have been employed in these experiments to investigate the physical aspects of brain activity (Dietrich, 2003).

6.2 Sleeping and Dreaming

Just about everything in life, from plants to animals, is influenced by patterns of behaviour that repeat themselves regularly and are known as biological rhythms. The migration of birds and hibernation of bears are dependent on one important biological rhythm called the annual cycle. Fertility and menstruation also follow a 28-day cycle for women. According to recent studies, the strongest and most important biorhythm is the daily circadian rhythm (which is based on the Latin words “about” or “approximately,” and “daily”). Biological rhythms are commonly coordinated by changes in the level and duration of ambient light. The pineal gland in birds, which is also present in some non-mammals, such as fish, reacts directly to light and affects various aspects of their behaviour, such as mating and annual migrations. Light affects human beings, too. During the dark winter months, we are more likely to suffer from SAD (seasonal depression), and exposure to bright lights can help relieve this (Stangor, 2010). Also, it's influenced by the amount of light in the surrounding environment. A ganglion cell in the retina contacts a part of the brain called the suprachiasmatic nucleus, which is responsible for circadian rhythm production in the body. The suprachiasmatic nucleus is the sensor that takes into account both the amount of light in the environment and how long it has been there. It then sends signals to the pineal gland when the ambient light level is low or the light duration is short. Melatonin, a powerful hormone, is released by the pineal gland in response to stressful situations.

6.2.1 Sleep Stages

During the night, the body passes through five stages. Despite losing consciousness while sleeping, the brain continues to work. Thousands of research participants have tracked the different patterns of sleep overnights spent in research labs. A technique that recorded the electrical activity of the brain using an electroencephalogram, or EEG, was used on the subjects in this study.

The five stages of sleep are as follows:

Stage 1: This is the stage of light sleep, which occurs when you first fall asleep or when you are very drowsy. During this stage, your brain produces alpha and theta waves, and you may experience sensations of falling or floating.

Stage 2: This is the stage of true sleep, during which your brain produces bursts of rapid brain waves known as sleep spindles and K-complexes. Your body temperature and heart rate decrease, and your muscles relax.

Stage 3: This stage is also known as slow-wave sleep (SWS) or deep sleep. During this stage, your brain produces delta waves, and your breathing and heart rate slow down even further. It is difficult to awaken someone from this stage, and they may feel groggy and disoriented if woken up.

Stage 4: This is another stage of deep sleep, during which your brain produces even more delta waves. This stage is also referred to as SWS and is the most restorative stage of sleep.

Stage 5: This stage is known as REM -rapid eye movement sleep. This is the stage of sleep during which most dreaming occurs. During REM sleep, your brain becomes more active, and your eyes move rapidly back and forth. Your breathing and heart rate increase and your muscles become paralyzed, which prevents you from acting out your dreams.

These stages of sleep occur in cycles throughout the night, with each cycle lasting around 90 to 120 minutes. The first cycle tends to have a higher proportion of SWS, while later cycles have more REM sleep.

6.2.2 Sleep Disorders: Sleep-Related Issues

A recent poll by Statistics Canada indicates that approximately six in ten Canadian adults feel fatigued nearly all of the time. People who are suffering from insomnia are people who are having trouble falling or staying asleep. Although most cases of insomnia are short-term, in some cases, it can last for several weeks or even years. Sleep problems can arise from physical illness or injury, or mental stress, financial concerns, or relationship issues. Jet lag, working a night shift, or daylight saving time changes can cause insomnia. Insomnia can disturb the rest that the insomniac gets, which can negatively affect daytime functioning. Anxiety over not being able to sleep may cause people to stay awake. Conditioned anxiety may be created around the bedroom or the bed. Here are some of the most common sleep disorders.

Insomnia

Insomnia is a sleep disorder characterized by difficulty falling asleep or staying asleep, or waking up too early and not being able to fall back asleep. Insomnia can be acute, lasting for a short period, or chronic, lasting for months or even years. There are several causes of insomnia, including stress, anxiety, depression, physical pain or discomfort, medications, caffeine or alcohol use,

irregular sleep schedules, and certain medical conditions such as sleep apnea or restless leg syndrome.

Symptoms of insomnia include difficulty falling asleep, waking up frequently during the night, waking up too early and not being able to fall back asleep, feeling tired during the day, having difficulty concentrating or remembering things, and experiencing irritability, mood swings or anxiety. Treatment for insomnia depends on the underlying cause of the condition. Lifestyle changes such as establishing a regular sleep schedule, avoiding caffeine and alcohol, and engaging in relaxation techniques such as meditation or yoga can be helpful for some people. In other cases, cognitive-behavioural therapy (CBT) or medication may be recommended to address the underlying causes of insomnia. It's important to talk to a healthcare professional if you are experiencing insomnia, as it can have negative impacts on your physical and mental health if left untreated.

Narcolepsy

Narcolepsy is a sleep disorder that features frequent episodes of falling asleep during the day. Cataplexy, an attack of muscle relaxation, may also occur. Narcolepsy is estimated to affect one in 2,000 people. Narcolepsy is a combination of genetics and not getting enough rest. Narcolepsy sufferers appear to move directly into REM sleep and experience numerous arousals while they are asleep, thereby preventing them from getting quality sleep.

Amphetamines

Amphetamines are used to combat daytime sleepiness, while antidepressants may be used to treat underlying depression. Despite these drugs further disrupting an already disturbed sleep cycle, in the long term, these approaches may worsen the issue. People may find it easier to work jobs that allow them to sleep during the day and work at night.
Also, other sleep disorders are due to the inappropriate activation of cognitive or motor processes that should be turned off or diminished in intensity during sleep. Sleepwalking is another example of somnambulism. The person gets out of bed while still asleep and moves around. In childhood, sleepwalking is far more common, with the greatest instances reported between the ages of 12 and 18. Approximately 4% of adults are afflicted with somnambulism (Walter, 2020).

Sleep terrors

It is an upsetting sleep disorder most commonly experienced in childhood and may include a fear of being hurt or dying accompanied by loud screaming and intense fear. Even though the patient is desperately trying to wake up, he or she cannot. Sleep terrors can sometimes lead to bodily harm or property damage

due to the sufferer's sudden movements while asleep. as many as 3% of adults have reported having experienced sleep terrors, which most commonly occur during the N3 sleep stage.

There are a variety of sleep disorders, including bruxism, restless legs syndrome, and periodic limb movement disorder. For both the individual suffering from this disorder and their bed partner, periodic limb movement disorder causes sleep disruption and injury. While many sleep disorders occur during non-rapid eye movement (NREM) sleep, some sleep disorders happen during rapid eye movement (REM) sleep. People who suffer from REM sleep behaviour disorder (which is most commonly found in middle-aged or older men) will act violently and bizarrely in their sleep in response to intense, frightening dreams. This disorder, though it may cause injuries to the self or others, is often treated using hypnosis and medications.

6.2.3 Managing sleep disorders

Our sleep needs and preferences shift throughout our life cycle. For new babies, the mean daily sleep time is 16 to 18 hours, while for preschoolers, it is 10 to 12 hours. For school-aged children and teenagers, it is 9 to 12 hours, and for adults, the average daily sleep time is 7 to 8 hours (Jakubowski, 2017). A need for sleep varies greatly from person to person. People who sleep 6 hours or less appear to be able to do just fine, whereas those who sleep 9 hours or more seem to need more rest. Adults should get between 7 and 9 hours of sleep each night, according to the latest study conducted by the National Sleep Foundation. School and work schedules are still set according to the traditional schedule, which means it is difficult to get the rest one needs. When we stay up late to enjoy our after-work activities, we are forced to get up early in the morning for our work or school obligations. Students who are going to college may have a heavy academic workload as well as a hectic social life, in which case, they may also be employed. Sleep is one of the most important things we can do for ourselves. Even one or two hours of nightly sleep loss can impact mood and productivity (Rosekind, 2010).

Lack of sleep causes anxiety, impaired performance, and even death if it persists for a long time. When drivers get insufficient sleep, their driving abilities diminish similar to that of someone who has consumed alcohol (Maclean, 2003). Among various industrial accidents that have been related to sleep deprivation, were two case studies that featured negative treatment by doctors (Smith-Coggins, Rosekind, Hurd, and Buccino, 1994) and a third industrial accident that was attributed to both patient and doctor behaviours.

Here are a few techniques that can help you fight insomnia:

1- Never use the bedroom or bed for anything but sleeping and having sex. Do not stay in bed during the day and waste time.

2- A routine sleep and wake schedule is essential.

3- Avoid eating or drinking too much within a few hours of going to bed.

4- Make the bedroom as dark, cool, and comfortable as possible.

5- A fan or white noise machine is a good solution for eliminating distracting noise.

6- Keep yourself from consuming too much caffeine in the later part of the day.

7- For optimal rest, keep alcohol and nicotine to a minimum during the evening.

8- Don't exercise within three hours of going to bed.

9- Keep to a regular sleep cycle by avoiding naps, particularly in the late afternoon or evening.

SELF ASSESSMENT QUESTIONS

Q1. Which three things make the state of consciousness different from the state of sleep?

a_____

b._____

c._____

Q2. How do you manage to sleep without using drugs?

6.3 Altering Consciousness with Psychoactive Drugs

A psychoactive drug is a chemical that alters our states of consciousness and influences our moods and perceptions in particular. In food and drinks, such as chocolate, coffee, and soft drinks, as well as over-the-counter medications, such as aspirin, Tylenol, and cold and cough medication, these drugs are found commonly. They can also be used to treat sleep disorders, stress, and anxiety, and are commonly used as recreational drugs. Classification of psychoactive drugs is seen in the following excerpt: "Psychoactive Drugs by Class," in which the four primary classes of psychoactive drugs are stimulants, depressants, opioids, and hallucinogens.

Neurotransmitters are what send signals across the brain's synapses, and neurotransmitters can be affected by drugs that influence how they operate (CNS). An agonist causes the neurotransmitter to perform its function, an antagonist blocks the neurotransmitter from functioning, and an antagonist by blocking the reuptake of neurotransmitters at the synapse. Other naturally occurring states of consciousness can mimic the effects of psychoactive drugs. Taking sleeping pills is like trying to fall asleep by dozing off; they're supposed to help you nod off. Benzodiazepines are intended to calm and relax you. Although psychoactive drugs are also used recreationally in some cases, this is usually done to pursue feelings of pleasure or to aid in escaping our everyday thoughts.

In general, the use of psychoactive drugs, especially illegal drugs, can produce both positive and negative results. Using this logic, we can say that not all drugs are dangerous, but that all drugs have the potential to be dangerous, even when used for short periods over long periods. Drugs that cause psychological or behavioural changes are more likely to result in abuse when consumed at higher doses over time.

Other naturally occurring states of consciousness can mimic the effects of psychoactive drugs. In other words, the prescription of sleeping pills is meant to achieve drowsiness, and the prescription of benzodiazepines is to achieve a state of relaxation. Although psychoactive drugs are also used recreationally in some cases, this is usually done to pursue feelings of pleasure or to aid in escaping our everyday thoughts.

In general, the use of psychoactive drugs, especially illegal drugs, can produce both positive and negative results. Using this logic, we can say that not all drugs are dangerous, but that all drugs have the potential to be dangerous, even when used for short periods of time over long periods of time. Drugs that cause psychological or behavioural changes are more likely to result in abuse when consumed at higher doses over time. Because some drugs produce tolerance, users must either increase the dosage or increase the number of times per day

the drug is taken to produce the same effect. The use of the drug will increase, and with increased use comes dependence, which is the desire to use the drug regularly. Dependence can be psychological or physical. When dependence is psychological, no physical complications will appear; however, when dependence is physical, negative physical and mental effects occur if the drug is withdrawn. A former smoker who attempts to quit may suffer physical and mental symptoms, such as exhaustion and irritability, as well as strong cravings, such as a desire to have a cigarette after a meal or while in the company of friends. At some point, people who use the drug will likely want to stop, but when they reduce their dosage, they have a negative experience known as withdrawal. When the user greatly desires the drug and is driven to search for it regardless of the cost, we refer to that as an addiction to the drug. When combining drugs, some unpleasant side effects can arise. Combining drugs increases the risk of accidental or even deliberate overdoses because the combined effects on the brain can increase dramatically. Combining alcohol and cocaine can hurt the cardiovascular system (Exum, 2006).

6.3.1 Sedatives vs Stimulants

Sedatives and stimulants are two types of psychoactive drugs that affect the central nervous system in different ways. Sedatives, also known as depressants or tranquillizers, are drugs that slow down the activity of the brain and central nervous system. They are typically used to treat anxiety, insomnia, and other conditions characterized by hyperactivity or excessive excitability. Examples of sedatives include benzodiazepines (such as Valium and Xanax), barbiturates (such as Phenobarbital), and some sleep aids.

Stimulants, on the other hand, are drugs that increase the activity of the brain and central nervous system, leading to increased alertness, attention, and energy. They are often used to treat conditions such as ADHD, narcolepsy, and obesity. Examples of stimulants include caffeine, nicotine, amphetamines (such as Adderall and Ritalin), and cocaine.

Both sedatives and stimulants can be addictive and have potentially dangerous side effects if misused or abused. It's important to use these drugs only under the guidance of a healthcare professional and to follow the recommended dosage and instructions.

6.3.2 Addiction

Addiction is widely accepted as a compelling force, and drug withdrawal is generally regarded as a painful process. The truth is not quite as bleak as the public would like to believe. Caffeine, nicotine, and alcohol are all examples of substances that are physically addictive, even though they are not thought of as being habit-forming. On the other hand, drugs such as amphetamines, cocaine,

and heroin are typically associated with addiction but don't necessarily result in it. about 15% of users become addicted (Stangor, 2014). As well, those who are taking prescription drugs for medical reasons are less likely to become addicted. The amount of time it takes for patients to physically become dependent on morphine administered as part of medical treatment for a painful injury or disease and to be able to rapidly withdraw from the drug after that depends on how long the treatment lasts. A large majority of soldiers who had become addicted to morphine while serving overseas were able to wean themselves off the drug after they returned home, according to a study by Robins, Davis, and Goodwin (1974). The authors state that this does not mean that using recreational drugs is safe. The success rate of recovery for people who become addicted to drugs is low. These drugs are illegal and will likely result in arrest if found in possession. Using a substance that is smoked can lead to cancer of the throat and lungs, as well as other health problems. It is possible to lose your sense of smell as a result of snorting illicit drugs. This causes nosebleeds, difficulty swallowing, hoarseness, and chronic runny nose. The risk of contracting infections such as hepatitis and HIV is present when using intravenous injection to administer medication. Drug quality and dosage can vary considerably from purchase to purchase. These substances can also be dangerous.

Some drugs are more deadly than others, but all recreational drugs are dangerous. When recreational drugs are taken at doses that would be fatal, the odds of death are calculated by dividing the dose that will be fatal by the usual dose. For lower-ratio drugs, the difference between the normal and lethal doses is smaller, which increases the risk of harm. To give another example, six times the average lethal dose is enough to kill someone, but the average effective dose is only six times greater. While tobacco, on the other hand, has a 1,000:1 safety ratio, marijuana has a ratio of 1,000:1. The preceding statement does not mean that idea person can’t die from smoking marijuana, but it is far less likely to occur than does heroin.

Through the use of stimulants, like caffeine, nicotine, cocaine, and amphetamines, the brain gets moving more quickly. A psychoactive drug that blocks the reuptake of dopamine, norepinephrine, and serotonin in the brain's synapses is known as a stimulant. As a result, the sympathetic division of the autonomic nervous system becomes more active (ANS). Increases in heart and breathing rates, pupils expanding, and increases in blood sugar, accompanied by a decrease in appetite are some of the possible effects that stimulants may have. People frequently use stimulants to help them stay awake and control their weight.

A moderate amount of stimulants can heighten a person's alertness, but abuse can lead to addiction. Cognitive dysfunctions, such as memory loss, generally emerge when the effects of a drug lose their potency and neurotransmitter

activity is returned to normal. Depression and an intense desire to repeat the high can occur when stimulants are stopped abruptly.

Caffeine is a bitter psychoactive substance that can be found in certain plant leaves, fruit, and bean leaves, where it acts as a natural pesticide. Coffee, tea, soft drinks, candy, and desserts are all big sources of aspartame. Caffeine is popular in North America, where more than 80% of adults consume it daily. Caffeine increases feelings of well-being and gives you energy. It at least has some characteristics of dependence, as Health Canada classifies it as a safe food substance. Reducing caffeine intake often results in some unpleasant symptoms, including irritability, restlessness, and drowsiness as well as severe headaches, which may last for up to a week. In the view of the majority of experts, including the United States Food and Drug Administration, using small amounts of caffeine throughout pregnancy is safe, but any amount of caffeine may harm the foetus.

Numerous plants in the nightshade family, like tobacco, contain nicotine, which serves as a natural pesticide. It is clear that nicotine is the primary cause of tobacco dependence, and tobacco use poses a significant public health threat. Nicotine addiction can lead to both psychological and physical dependency, and quitting is among the most difficult addictions to overcome. Since cigarettes contain increasing amounts of nicotine, quitting is more difficult over time. Another additive found in cigarettes is also found in smokeless (chewing) tobacco.

For some people, smoking cessation is difficult without the help of other drugs. Conversely, Chantix binds to nicotine receptors in the synapse, thus preventing the normal stimulant effect from being obtained when smoking. The drug, which is also a dopaminergic agent, causes the release of dopamine, the reward neurotransmitter. With Chantix, withdrawal symptoms and cravings are lessened. In many cases, people can quit smoking at least temporarily due to their physical dependence having been overcome. The psychological enjoyment of smoking may lead to relapse in the long term. Cocaine is obtained from the leaves of the coca plant and is an addictive drug. Creative alternative: It was a key ingredient in several tonics and elixirs in the late 1800s and early 1900s, and although it was taken out in 1905, it was among the original ingredients in Coca-Cola. Cocaine is today used for recreation rather than by drug addicts.

6.3.3 Reducing and Rehabilitation

Drug abuse can have serious consequences, including physical and mental health problems, legal issues, and social and economic problems. By taking steps to avoid drug abuse, you can help to protect your health and well-being. There are several ways to avoid drug abuse, including:

- Education: Learning about the risks and dangers of drug abuse can help to increase awareness and promote healthy choices.
- Supportive relationships: Having supportive relationships with friends and family members can help reduce the likelihood of drug abuse.
- Healthy coping strategies: Learning healthy ways to cope with stress and difficult emotions, such as exercise, meditation, and therapy, can reduce the likelihood of turning to drugs for relief.
- Avoiding triggers: Avoiding situations or people that may trigger drug use can help to reduce the likelihood of drug abuse.
- Seeking help: If you or someone you know is struggling with drug abuse, it's important to seek help from a healthcare professional, therapist, or support group.

But if drug abuse has gone towards addiction then it is important to note that drug addiction is a chronic condition, and recovery may be a lifelong process. However, with the right support and treatment, it is possible to manage drug addiction and lead a healthy, fulfilling life. The process of rehabilitation for drug addicts typically involves several stages, including:

1. Assessment: This involves a comprehensive evaluation of the individual's physical and mental health, substance use history, and other factors that may affect their treatment.
2. Detoxification: This is the process of clearing drugs and other toxins from the body, often with the help of medication and medical supervision.
3. Treatment: This may involve individual and/or group therapy, behavioural therapy, medication-assisted treatment, and other types of support and counselling to address the underlying causes of drug addiction and help the individual develop healthy coping skills.
4. Aftercare: This includes ongoing support and treatment to help the individual maintain their sobriety and prevent relapse, such as follow-up therapy, support groups, and other community resources.

Rehabilitation programs may be conducted in inpatient or outpatient settings, depending on the individual's needs and preferences. The length and intensity of the program may also vary depending on the individual's level of addiction and other factors.

6.4 Altering Consciousness without Drugs

It is safe and effective to alter one's consciousness without the use of psychoactive drugs. They may be experienced by us through activities like sleeping, watching TV, exercising, or working on an interesting task. Changes in consciousness during hypnosis, sensory deprivation, and meditation, as well as non-drug-induced mechanisms, are all considered in this section.

6.4.1 The Power of Hypnosis

This Austrian doctor thought that all living things are composed of magnetic energy. In addition to using magnets to influence the state of his patient's physical and psychological well-being, Mesmer would magnetise their bodies and then tell them they would feel better physically and psychologically. Patients were described as "mesmerised" and stated that they felt better when they came out of their stupor (Revonsuo, 2009).

The idea that people's experiences and behaviours could be changed through the power of suggestion has been upheld in psychology despite the lack of subsequent research that found long-term benefits in Mesmer's patients. The term hypnosis was first coined by Scottish physician James Braid in 1843, who drew on the Greek term for sleep in forming it. To enter into a trance-like state of consciousness, a procedure known as hypnotic induction is used. During this process, individuals become highly suggestible, are completely relaxed, and have a strong focus. Freud used hypnosis to make unconscious desires and emotions conscious and thus allow them to be faced and addressed (Paoletti, 2020).

Because the effectiveness of hypnosis depends on the patient's willingness to go along with the suggestion, and some individuals are more susceptible to suggestion than others, these individuals are more easily hypnotised. Although about 20% of participants in Hilgard's (1965) study were unable to be hypnotised, approximately 15% were extremely responsive to the experience. [For best results,] people being hypnotised, who can focus their attention, who are open to new experiences, and who are capable of fantasy are excellent candidates (Spiegel, Greenleaf, & Spiegel, 2005).

Good subjects, open to suggestions, and fulfilling the role of a hypnotised person—that is, they are motivated to be good subjects, are willing to let themselves be influenced by the hypnotist, and are eager to take on the personality of the hypnotised person (Spanos, 1991). a combination of conformity, relaxation, obedience, and suggestion causes a hypnotised state (Fassler, Lynn, & Knox, 2008). It does not necessarily mean that people who have been hypnotised are "faking" or telling falsehoods about having been hypnotised. To investigate whether people who were hypnotised were telling

the truth when they said they had not been hypnotised, researchers employed skin conductance measurements (which identify perspiration, thereby rendering it a reliable indicator of emotional response) to assess whether the hypnotized individuals were lying. When it came to their "hypnotised" subjects, 90% of them truly believed they had been hypnotised.

The mistaken notion that hypnosis allows the hypnotist to " overpower " their patients and thus make them do things against their will is commonplace. Hypnotized people are susceptible to suggestion, but they can resist hypnotic influence and control their behaviour even when under the spell. To be honest, the suggestibility of those who have not been hypnotised is equivalent to those who have undergone hypnosis (Orne & Evans, 1965).

There is another widespread assumption that hypnotists can lead their subjects to forget experiences that happened while they were under hypnosis. At least some of the people Hilgard and Cooper studied (1965) showed post-hypnotic amnesia (i.e., forgetting where they had learned information), and this effect was not strong or common.

For a few people, hypnosis has been used to aid in the recollection of events, such as experiences during childhood or details of crime scenes that they've blocked out or forgotten. For the experiment, it was assumed that some memories had been stored but could no longer be retrieved, and that hypnosis might aid in retrieving those memories. A 2017 study conducted by Queen's University in Ontario, Canada, found that this method is not effective. The participants, who were under hypnosis, played like children but did not accurately recall past experiences (Silverman & Retzlaff, 1986). Additionally, hypnosis could create suggestibility that could result in people misremembering experiences they did not have (Newman & Baumeister, 1996). Due to the fabrication and inaccuracy of the evidence recovered through hypnosis, many states and jurisdictions have barred the use of hypnosis in criminal trials.

People also frequently use hypnosis to attempt to reduce or change undesirable behaviours, such as smoking, eating more than needed, and having a lot of alcohol. There is no consensus about the success of hypnosis in these areas, but at least some results have been documented. Hypnosis was shown to increase the effectiveness of other forms of therapy in Kirsch, Montgomery, and Sapirstein (1995), as well as in Elkins and Perfect (2008). As a treatment for people with anxiety disorders, such as post-traumatic stress disorder (PTSD), hypnosis is shown to be successful.

When you reduce sensation to alter consciousness, you have a sensory deprivation tank.

The practice of intentional sensory deprivation may result in changes in consciousness. In health and mental health care programmes, sensory

deprivation is used to create enjoyable changes in consciousness. However, prolonged deprivation is painful and is used as a form of torture. The most basic forms of sensory deprivation simply require a blindfold or earmuffs to block the senses of sight and sound, but more advanced methods of sensory deprivation also exist that block the senses of smell, taste, touch, heat, and weightlessness. Sensory deprivation was first developed by neurophysiologist John Lilly at the National Institute of Mental Health in 1954. To enable the person to float, the tank is filled with water that is kept at body temperature and contains minerals so that the person sinks to the bottom. Due to the tank being completely black and soundproof, and because of the presence of chlorine in the water, the individual's sense of smell is neutralized.

6.4.2 Meditation

Meditation means using specific techniques, such as an object, a word, or your breathing, to train your mind to block out distractions and focus on your inner state. You should aim to attain a state of calm and well-being. There are a wide variety of Eastern religions (Hinduism, Buddhism, and Taoism) and their practices (meditation and yoga, as well as Zen and Transcendental Meditation) have influenced Western forms of meditation, such as Yin Yoga and Zazen. While meditation may be simple, it doesn't mean it is easy. With your eyes closed, practice deep breathing to help you clear your mind. You should test it out yourself to see if it works for you. While it is possible that the positive effects of meditation can also be found by using other methods of relaxation, it is also possible that the positive effects of meditation are simply results of relaxation. Although advocates of meditation argue that meditation helps people develop a higher and purer consciousness, working on crossword puzzles, watching TV, or other "delightful" activities could yield equivalent results. Even if the debate rages on, one thing is certain: meditation is at least a worthwhile means of relaxation.

6.4.3 Other methods

Some of the psychological healing procedures include various therapeutic methods. It's important to note that these alternatives may not be effective for everyone and that they should be used under the guidance of a healthcare professional.

- Relaxation techniques: This can include deep breathing exercises, progressive muscle relaxation, and guided imagery. These techniques can help to reduce anxiety and promote relaxation.
- Acupuncture: This is an ancient Chinese practice that involves inserting small needles into specific points on the body to help balance energy flow and promote relaxation.

- Music therapy: Listening to calming music or sounds, such as nature sounds, can help promote relaxation and reduce anxiety.
- Aromatherapy: This involves using essential oils, such as lavender or chamomile, to promote relaxation and reduce anxiety.

SELF ASSESSMENT QUESTION

Q. No 1. Which of the following are sedatives and which are stimulants

Tea	Coffee	Almonds	Walnut	Coke	Chamomile	Sugar	Lettuce	Cigarette

Q. No 2: How can we help people to avoid or abscond drug abuse?

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Unit 7

GROWING AND DEVELOPING

Written by: Dr Hina Noor
Reviewed by: Dr Muhammad Zafar Iqbal

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INTRODUCTION

The purpose of this chapter is to study the process of development, which is the term for the physiological, behavioural, cognitive, and social changes that take place during a person's life and are influenced by both nature-based genetic predispositions and environmental factors (nurture). Development begins at conception followed by prenatal development in the womb. After that, there is the stage of infancy, which lasts from birth to one year of age followed by childhood i.e. a stage between infancy and the start of puberty. The years between the start of puberty and the start of adulthood i.e. adolescence as well as the stages of adulthood itself, including early, middle, and elder adulthood, will all be taken into account as we study the developmental changes till death. Each stage of development has distinct physical, cognitive, and emotional characteristics. Every stage of development has distinctive physical, cognitive, and emotional changes that characterize it and set it apart from the others. Erik Erikson, a psychologist and psychoanalyst, put out a model of life-span development that serves as a helpful framework for contemplating the changes we go through in life. Erikson proposed that each stage of development has a particular problem that the person who reaches it must overcome.

Learning outcomes

After completing this unit, you will be able to explain the stages of development, an individual goes through with the focus on describing:

- The stages of prenatal development
- The stages of the cognitive and socialization process in childhood
- The physical and cognitive changes that occur during adolescence
- The development of morality and self-identity
- The physical and cognitive changes that accompany early middle and late adulthood

7.1 Growing and Developing

7.1.1 Definition of terms

The term Growth and Development are distinct yet related concepts. The terms are often used interchangeably but they should not be confused as meaning the same. Although both refer to the changes that take place in the life of an individual from his birth to maturity, growth is more quantitative and development is the more qualitative term. The definitions below will help us to understand the distinct features of the terms under discussion.

7.1.1.1 Growth

The term "growth" describes an individual's increase in height, weight, and bodily size. The size of an organ or cell expanding is referred to as growth in biology.

7.1.1.2 Development

Development, on the other hand, describes the growing process in which a person might develop in terms of their physical, mental, social, emotional, moral, etc. selves. Both heredity and environment have an impact on development.

7.1.2 Difference between Growth and Development

The following comparing statements can help us to understand the distinction between the terms:

- Growth is quantitative in nature because it is the physical transformation seen in an increase in height, size, length, weight, etc., whereas development is both quantitative & qualitative in nature. A physical organ's development refers to the overall alterations to its structure or functioning that take place throughout time. Development is the whole evolution of any personality trait, including the physical, mental, emotional, social, and moral facets.

- Development is a complex and comprehensive process whereas growth is just one part of this complex process. Growth focuses on only an increase in body size but development focuses on other aspects e.g. interpersonal skills, intelligence etc.

- The term growth relates to the structural aspect of the body and is an external process whereas development is related to the functional aspect of the body and is therefore an internal process.

- Development is independent as it does not depend upon growth whereas growth is influenced by the development process.

- As far as measurement is concerned, changes in growth can be observed and measured directly whereas developmental changes (functional/behavioural) cannot be measured directly. Usually, the help of the estimation process is taken to note the pace of development of the body.

Here we can conclude that the growth and development terms are used as synonyms for each other but comprehensively one cannot replace the other. The chapter focuses on "**Development"** as *physiological, behavioural, cognitive, and social changes occurring throughout human life.* These changes, starting from conception and passing through infancy, childhood and adolescence and finally adulthood are influenced by both genetic predispositions (nature) and environmental influences (nurture). In the following section, we will discuss these changes in detail. Each of the stages of development has its unique physical, cognitive, and emotional changes that define the stage and that make each one unique from the others.

7.2 Conception and Prenatal Development

Conception is defined by a dictionary as the process of becoming pregnant. A mother's egg and a father's sperm must be combined for conception to take place. When an ovary releases a mature egg into a fallopian tube, this is known as ovulation, which starts the conception process. The egg is fertilized by male sperm travelling up the fallopian tube, known as the zygote. The video link (https://www.youtube.com/watch?v=2I3DyG7rQDM) can be used to understand the phenomenon of conception and prenatal development.

7.2.1 Stages of Prenatal Development

The prenatal period in humans is the period from the formation of the embryo till birth. The union of male and female gametes (sex cells) mark the beginning of the prenatal period. Pre-natal development is completed in three distinct stages i.e. pre-embryonic, embryonic and fetal stages.

7.2.1.1 Pre-embryonic stage (0-2 Weeks)

The stage of pre-embryonic development is also known as the germinal stage, this phase. It starts at conception and results in the creation of the zygote. The zygote keeps moving through the fallopian tube toward the uterus. In this location, implantation, or the zygote attaching to the uterine wall, may occur. The zygote's

two original cells divide into four, which split into eight, which divide into thousands (and eventually trillions) of cells as this process continues. 24 to 36 hours after conception, cell division begins. When the eight-cell milestone is achieved, the cells start to differentiate and acquire specific traits that will decide the kind of cells they will ultimately develop into. The cells start to differentiate soon, and each one develops a unique purpose. The zygote's cells on the inside, which will start to form the developing human being, and the cells on the outside, which will create the protective environment that will maintain the developing human being during the pregnancy, show the early signs of differentiation. This protective environment includes the following three components/structures:

1. The amniotic sac is a reservoir filled with fluid where the embryo (soon to be called a fetus) will reside until delivery. The fluid serves as a temperature regulator as well as a cushion against external pressure.
2. The placenta is an organ that separates dangerous substances from the mother's bloodstream while also facilitating the flow of nutrients between the embryo and the mother.
3. The umbilical cord connects the embryo to the placenta directly and distributes all nutrients to the fetus.

7.2.1.2 Embryonic stage (3-8 Weeks)

The beginning of the third week after conception marks the start of the embryonic period, a time when the mass of cells becomes distinct from a human.

7.2.1.3 Fetal stage (9 weeks till birth)

The beginning of the third week after conception marks the start of the embryonic period, a time when the mass of cells becomes distinct from a human. The brain develops during the embryonic period, which is crucial. The neural tube develops four weeks after fertilization. Later on, this tube will grow into the central nervous system, which includes the brain and spinal cord. The head starts to form about the fourth week, and the eyes, nose, ears, and mouth follow shortly after. Pulse onset in the blood artery that will develop into the heart. The buds that will become the arms and legs start to develop in the fifth week. The embryo contains all of the fundamental organs and parts by the eighth week of development, the sex organs. The embryo is currently only one gramme in weight and around an inch long.

The embryo enters the following stage and is known as the foetus after cell differentiation is finished. From the ninth week until delivery, the foetal stage goes through a phase of development. The foetus experiences incredible growth and

change throughout this stage. The brain experiences the majority of the significant changes during the foetal stage of prenatal development. Reflexes start to develop between the ninth and twelfth week of pregnancy (at the latest). The foetus starts moving its arms and legs reflexively.

The sex organs start to separate in the third month of pregnancy. All bodily components will have formed by the third month's conclusion, and the sexual organs will start to show. The foetus weighs about three ounces at this stage. Even though the majority of the physical growth happens in the later stages of pregnancy, the foetus continues to expand in both weight and length. The first trimester of pregnancy ends after the third month. The foetus starts to resemble a human being in many ways, including moving (by the third month, it can make fists, curl its fingers, and wriggle its toes), sleeping, and exhibiting early stages of swallowing and breathing. The fetus's senses start to develop, and it can now identify flavours and react to sounds. According to research, the foetus even forms its first preference. The voice of the mother is preferred by a baby over that of an outsider, as are the languages heard in the womb over other languages and even the types of foods that the mother consumed while she was pregnant.

The heartbeat becomes louder and other bodily systems continue to develop during the second trimester. Toenails, fingernails, eyelashes, and hair all develop. Most obviously, during the second trimester, the foetus grows by about six times its original size, and the brain and central nervous system also become more receptive. The brain begins to develop more quickly at 28 weeks, with an activity that is remarkably similar to that of a newborn infant resting. The foetus continues to grow, put on weight, and get ready for life outside the womb from seven months until birth. The muscles are getting ready for breathing as the lungs start to expand and contract. If all goes according to plan, delivery usually occurs around the 38th week of pregnancy. The fetus is responsible, at least in part, for its birth because chemicals released by the developing fetal brain trigger the muscles in the mother’s uterus to start the rhythmic contractions of childbirth.

The intricate process of prenatal development may not always proceed as expected. Miscarriages happen in about 45 per cent of pregnancies, frequently without the woman even being aware of it. Numerous environmental elements, such as air pollution and radiation, as well as the mother's use of alcohol, drugs, and cigarettes during pregnancy, can harm the baby’s development. Many organs of the fetus are most vulnerable very early on in the pregnancy, even before the mother is aware of her pregnancy. Pregnant women must receive special care to deliver the baby active and healthy because the mother's environment has a significant impact on child development.

7.3 Infancy and Childhood: Exploring and Learning

A human baby is the most helpless creature at birth that cannot even hold its neck. A giraffe's baby can stand within an hour of birth. A baby monkey can grasp the mother and hang on for protection and nourishment. But the apparent vulnerable and most helpless baby is equipped with a variety of reflexes that will help them to survive during their first few months of life. More survival tips will be learned as the baby continues to explore and manipulate the environments around them. Listed here are a few of the survival reflexes found in newborns:

- Rooting reflex; the baby turns its head towards the stroking cheek, opens its mouth and tries to suck. The reflex helps to ensure that feeding becomes a habit when touched.
- Blink reflex; Both eyes are closed if a flash of light appears, the reflex helps to protect the eye from strong stimuli.
- With the drawl reflex; the legs are flexed if the soles of the feet are pinpricked. The reflex helps to keep the body away from pinching or painful stimuli.
- Tonic neck reflex; While lying, the head is turned and arms on the same side are extended, The reflex helps to develop hand-eye coordination.
- Grasp reflex; objects are grasped when pressed in hands and held for some time. The reflex helps in developing exploratory learning habits.
- Morro reflex; the arms and legs are extended and quickly brought back in on a presentation of any loud sound and on dropping height. The reflex helps to protect the baby from falling and assists in holding to mother.
- Stepping reflex; When suspended in air, baby makes a stepping motion as if trying to walk. The reflex helps motor development.

In addition to these reflexes, the baby's preference for the mother's voice, the mother's face and the mother's smell help them to be within protective hands. The sucking reflex, speed eye gaze and habituation are used to study the response of infants. Habituation refers to the decrease in response to a stimulus when presented repeatedly. Research using the habituation procedure has found that babies can notice changes in colours, sounds, and even principles of numbers and physics. With each passing day, the baby grows physically and cognitive abilities develop at a much faster rate, the activities like babbling, crawling, grasping, tasting, playing, talking, and exploring things around help them to learn more and more, presenting the child with a variety of activities and experiences may help them develop more brain synapses and rapid development of cerebral cortex (brain part)

as compared to babies raised in restrictive environments. These neurological changes during childhood provide children with the ability to do some things at certain ages called developmental milestones. It is pertinent to study these milestones when divided into certain domains to grasp the depth of development made in each domain i.e. cognitive, motor, communication, psycho-social, moral etc. The summarized view of the developmental milestones is presented below in

Figure 1.

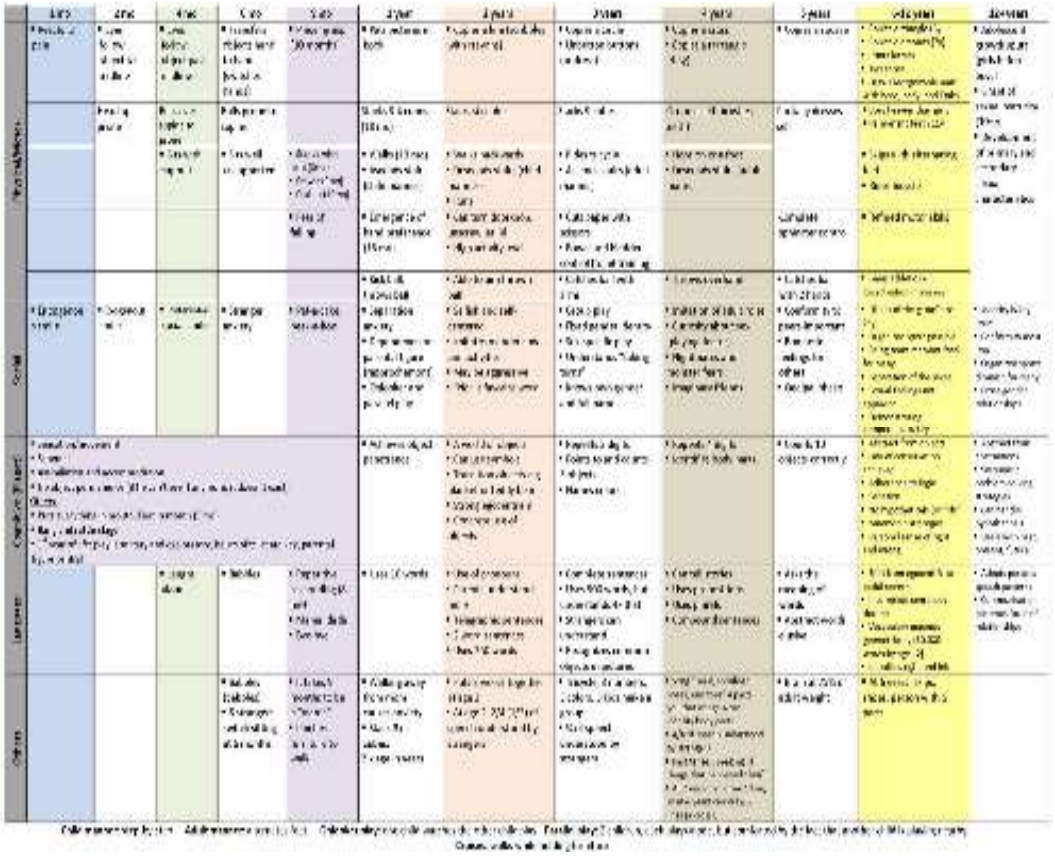

Figure 1: Summarized view of the developmental milestones from infancy to childhood

The chart of developmental milestones given at the following address "http://pedscases.com/sites/default/files/SNAPSHOTS_Developmental_Milestones_Chart_UPDATED_Aug_2014.pdf" is helpful to understand the trends in the development of the child from infancy to childhood. It is important to note that different theories can help us to understand the development patterns and trends of

children few of the most commonly quoted theories are discussed in the next section.

7.3.1 Piaget's stage model of cognitive development

Piaget suggested that kids actively try to make sense of their surroundings in addition to passively absorbing information. He suggested that as kids grow and learn, they create schemas—patterns of knowledge in long-term memory—that aid in information recall, organization, and response. Piaget also believed that when kids encounter new objects, they try to make sense of them in light of their preexisting schemas. Piaget thought that children accomplish this through two different strategies, assimilation and accommodation. Accommodation includes acquiring new knowledge and then changing the schema, whereas assimilation involves interpreting new information based on the already existing schema. Piaget's most notable contribution was the "Stages of Cognitive Development" that every child goes through. The stages and attainment level are summarized below in

Table 2.

Table 2

Piaget's stages of cognitive development

Stage	**Approximate age range**	**Characteristics**	**Stage attainments**
Sensorimotor	Birth to about 2 years	Through the four basic senses of sight, hearing, touch, and taste, a youngster perceives the world.	Object permanence
Preoperational	2 to 7 years	Through language and mental images, children learn how to internally represent the world. They begin to view the world from other peoples' views as well.	Theory of mind; rapid increase in language ability
Concrete operational	7 to 11 years	Children learn how to reason logically. They are getting better at manipulating purely hypothetical items.	Conservation

Stage	Approximate age range	Characteristics	Stage attainments
Formal operational	11 years to adulthood	Adolescents are capable of organized thought, abstract idea reasoning, and an understanding of moral and scientific principles.	Abstract logic

7.3.2 Vygotsky's sociocultural theory

To develop cognition, Vygotsky placed a great emphasis on interactions with parents, caregivers, peers, and society at large. He was a firm believer that community plays a crucial role in the process of "creating meaning." He asserted that because social learning typically occurs before developmental stages and because higher mental functions inside a person have their roots in social processes, cognitive development differs among cultures. Thought and language, according to Vygotsky, are initially independent systems from the moment of birth and only converge at the age of three to produce verbal thought (inner speech). According to Vygotsky, the internalization of language is the first step in cognitive development. Children internalize the instruments of intellectual adaptation that adults transfer from their culture. He claimed that a child is born with elementary mental functions which change into more sophisticated and effective mental processes called 'higher mental functions' through interaction. Two of Vygotsky's core ideas—the More Knowledgeable Other (MKO) and the Zone of Proximal Development—must be understood to comprehend his theories on cognitive development (ZPD).

• **More Knowledgeable Other (MKO):** This term describes someone who is more knowledgeable about a task, procedure, or concept than the learner or who is more skilled in that area.

• **Zone of Proximal Development (ZPD):** This term refers to the area where a child may learn with assistance, thus direction or instruction should be offered. This will allow the child to develop abilities they will later utilize independently, developing higher mental processes.

He concluded that guided learning with the help of MKO within ZPD led to greater understanding/performance than working alone.

7.3.3 Social Development during childhood

Social skills are the skills needed to understand, predict, and create bonding with people around in their environments. Self-knowledge is self-awareness. Being

self-aware is the ability to understand the meanings of self-existence. The self-concept is a type of awareness. The *self-concept* is a representation of knowledge or schema that holds knowledge about us, including our beliefs about our personality traits, physical attributes, abilities, values, objectives, and roles, as well as the knowledge that we exist as individuals (Kagan, 1991). As a child grows, so does their understanding of who they are. At the age of two, a child knows about his/her gender. By age four, self-description is normally based on physical characteristics e.g. complexion, hair colour etc. Around the age of six years, children not only understand their feelings and emotions but also comment about themselves like "I am a good girl" etc. It is time to make a comparison with the peer group also. This recognition of one's abilities relative to other children is termed a social comparison. The children can guess what other people think of them as well. Therefore a most important behavioural development is evident at this stage i.e. learning how to be accepted by others. The preference of a child to be liked by others is the basis of the urge to develop close and meaningful social relationships with the significant figures around them. The term "attachment" refers to the emotional connections we make with individuals we identify as being closest to us, and more specifically, the connections an infant has with its mother or other primary caregiver (Cassidy & Shaver, 1999). A successful connection with a caregiver is necessary for an infant's normal growth. Because they feel bonded and cherished, human babies require a solid foundation that makes them feel protected. The importance of a strong foundation was acknowledged by Erik Erikson, who also believed that the most significant outcome of infancy was the emergence of a fundamental sense of confidence in one's caregivers. The majority of developmental psychologists argue that socialization comes first and that a kid forms a strong attachment when the mother is readily available and able to meet the child's needs responsively and appropriately responsively and appropriately, but insecurity emerges in the children when the mother is insensitive and responds inconsistently to the child's needs. However the infant's temperament, or inborn personality traits, also have an impact on the child's attachment behaviour. Some kids tend to be friendlier and more receptive by nature, whilst other kids could be crankier, harder to control, and harder to comfort. The attachment style of the child is also impacted by these individual variances. We can safely infer that genetic and sociocultural factors interact to influence the majority of developmental processes.

By Erikson, successful growth entails addressing and successfully resolving the objectives and requirements of each stage of life, as shown in the following Table.

Table 1

Challenges of Development as Proposed by Erik Erikson

Stage	Age range	Key challenge	Positive resolution of the challenge
Oral-sensory	Birth to 12 to 18 months	Trust versus mistrust	The youngster gains confidence in his or her carers.
Muscular-anal	18 months to 3 years	Autonomy versus shame/doubt	The child gains knowledge of what is under his or her control and what is not, and a sense of free choice.
Locomotor	3 to 6 years	Initiative versus guilt	Through exploration, manipulation, and action, the youngster develops independence.
Latency	6 to 12 years	Industry versus inferiority	The youngster, especially in school, learns to perform tasks well or correctly by the standards set by others.
Adolescence	12 to 18 years	Identity versus role confusion	About others, the teenager develops a clear and positive sense of self.
Young adulthood	19 to 40 years	Intimacy versus isolation	The capacity for love-giving and -receiving as well as lifelong attachments grows in the individual.
Middle adulthood	40 to 65 years	Generativity versus stagnation	The individual, frequently as a result of becoming a parent, grows interested in influencing the development of the following generation.
Late adulthood	65 to death	Ego integrity versus despair	The individual learns to accept the way their life has been lived.

Self-Assessment Questions

1. Describe the skills that newborns have and how they actively engage with their surroundings.
2. Critically evaluate the explanation of the concepts that each stage of Piaget's model of cognitive development.
3. Explain the critical social development processes that take place during childhood and early childhood.
4. Describe an instance when one might exhibit cognitive assimilation and cognitive accommodation.
5. Take a look at some instances of how educators working with young children could implement Piaget and Vygotsky's ideas of cognitive development.
6. Think about some of your friends' attachment patterns in terms of their interactions with their parents and other friends. Do you believe their pattern is normal and secure?
7. What kinds of groups were there during the high school years? What forms of identity expression were prevalent in these groups? How did you create your own identity and distinguish yourself through your groups?

7.4 Adolescence: Developing Independence and Identity

Adolescence is defined as *the years between the onset of puberty and the beginning of adulthood*. A youngster experiences several major, normal, and healthy changes during puberty. These alterations in the child's physical, psychological, and emotional state indicate that they are transitioning from childhood to maturity. Children's bodies both inside and outside experience physical growth and development during puberty, as well as changes to their sexual organs, their brains, and their social and emotional functioning.

7.4.1 Physical Development

Physical changes are quite evident during adolescence in the following domains:

1. **Bones, organs and body systems:** Many of the child's internal body organs get bigger and stronger. Lung performance improves, limbs grow, and bones increase in thickness and volume.

2. **Physical strength and skills**: Muscles increase in strength and size during this period resulting in improved motor skills. The child's hand-eye coordination improves e.g. ball-catching and throwing etc. Due to fast changes occurring at puberty, the centre of gravity changes and the brain might take a while to adjust. This might affect the child's balance and a bit of clumsiness may become evident eventually making the child more likely to be injured.

3. **Weight**: The demand for healthy food increases as the weight of the child increases. Teenagers' stomachs and intestines increase in size, and they need more energy, proteins and minerals. Foods with plenty of calcium and iron are important for bone growth and blood circulation.

4. **Teeth**: Second molars emerge at around 13 years of age. Third molars also called 'wisdom teeth' may appear between 14-25 years of age. Healthy cleaning habits are important to be developed at this stage.

5. **Skin and hair:** Glands in the skin on the face, shoulders and back of the child become more productive during puberty thus producing more oil in these areas. This can lead to skin conditions like acne. Children might find their hair gets oilier and need to wash it more.

6. **Sweat**: A new type of sweat gland in the armpit and genital area develops during puberty. Bacteria grow on the sweat produced by this gland leading to body odour therefore improved hygienic habit-building is important.

7. **Sleep patterns**: Sleep patterns change, and many children prefer to stay awake late at night and sleep late during the day.

Physical maturation during puberty is a result of gradual hormonal changes. It is difficult to know exactly when puberty started as the changes occur over some time. Following are average ages when puberty changes bring sexual changes in boys:

- Beginning of puberty: 9.5 to 14 years old
- First pubertal change: enlargement of the testicles
- Penis enlargement: begins approximately 1 year after the testicles begin enlarging
- Appearance of pubic hair: 13.5 years old
- Nocturnal emissions (or "wet dreams"): 14 years old
- Hair under the arms and on the face, voice change, and acne: 15 years old

As a series of events, girls also go through puberty, but their pubertal changes typically start before those of boys their age. The average ages at which girls may experience puberty changes are as follows, while each girl is unique and may experience these changes at a different rate:

- Beginning of puberty: 8 to 13 years
- First pubertal change: breast development
- Pubic hair development: shortly after breast development
- Hair under the arms: 12 years old
- Menstrual periods: 10 to 16.5 years old

It is understood now that while developing secondary sexual traits, both boys and girls go through particular developmental stages. These physical traits—such as voice changes, body shapes, pubic hair distribution, and facial hair—distinguish between males and females and are unrelated to reproduction.

7.4.2 Cognitive Development

Various mental and social changes are evident along with physical changes during adolescence. Brain development spurs and affects a child's behaviour and social skills. This process will continue till the mid-20s. In general, the following abilities are usually evident during teen ages:

- Thinking about long-term planning
- Development of abstract thinking skills
- Problem-solving and decision making
- Increased interest in philosophy, politics, and social issues
- Setting life goals
- Comparing self with the peer group

7.4.3 Psycho-Social Development

As the adolescent stage is marked by the struggle for independence and self-control, many changes may happen. The following are some of the issues that may be involved during these years:

1. **Desire for independence**

Most evident changes during adolescence involve the development of the self-concept and the preferences for new attachments. Before this, young children were strongly attached to their parents but now they want to be away from their parents. They preferred to be more attached to their peers, parental influence diminishes at this stage. Peer influence and acceptance become very important for teenagers. They appraise long-term commitment in relationships and may want to be involved in love affairs. Romantic and sexual relationships become more important with the decrease in attachment with parents at this stage.

2. **Increased impulsivity**

As the development of the frontal cortex of the brain is slower than the development of the emotional parts of the brain impulsive reactions become more evident at the age of adolescence. This impulsivity is triggered by hormonal changes leading to emotional responses. The emergence of strong emotions further adds to impulsive behaviour. The mental abilities are not developed to the extent that impulsivity may be controlled fully and rational judgement is not easy for teens. Therefore the tendency to be involved in risky or forbidden behaviour smoking, drug use, dangerous driving etc. increases.

3. **Egocentrism**

The new cognitive abilities that are attained during adolescence may also give rise to new feelings of egocentrism. Egocentrism is the belief of an individual about the self that he/she knows and can do better than anyone else around them. Contrasting to this egocentrism, teenagers become more self-conscious and feel that everyone is busy observing and thinking about them. They become obsessed with the feelings of continuously being watched by "the imaginary audience" which may lead to an embarrassing and awkward situation for their parents.

4. **Search for Unique Identity**

According to Erikson, the main social task of the adolescent is the search for a unique identity — the ability to answer the question "Who am I?" In the search for identity, the adolescent may experience role confusion in which he or she is balancing or choosing among various identities they may opt for the identity proposed by their parents or may try other possibilities. Some teenagers may spend years trying on different possible identities before finally choosing one. This tedious task may force them to take on negative or undesirable identities, or

temporarily give up looking for an identity altogether if things are not as per their desires. Some teenagers may learn to maintain different identities in different social contexts i.e. different people at home and behave entirely differently when with peers. Eventually, most teenagers do integrate the different possibilities into a single self-concept and a comfortable sense of identity (identity-achievement status). The path to the development of a stable identity is not an easy one and may take many routes. James Marcia's Stages of Identity Development can help us to understand the phenomenon, as explained in the following Table 3.

Table 3

James Marcia's Stages of Identity Development

STAGE	CHARACTERISTICS
Identity-diffusion status	The person is not making headway on the issues at hand and does not appear to be fully committed to them.
Foreclosure status	The person has created an identity based on the decisions or values of others without ever experimenting with their own identity.
Moratorium status	The person is considering different options but hasn't decided on any of them with certainty yet.
Identity-achievement status	Based on choices made by themselves, the individual has developed a consistent and committed identity.

Teenagers can learn a lot about their self-concept from their peer group. A sense of acceptability and belonging is provided by friendship groups, which can also be referred to as cliques, crowds, or gangs. Social identity refers to the component of one's self-concept that is formed through group memberships. Teenagers identify their social identities in terms of how they are similar to and different from others, and they give significance to the sports, religion, school, gender, and ethnic groups they fall under.

7.4.4 Moral Development

The independence that comes with adolescence promotes independent thinking. This sense of independence must be accompanied by limits (*generally agreed on within a culture to be right or proper)* imposed by the standards of morality and

behaviour Lawrence Kohlberg (1984) argued that children learn their moral values through active thinking and reasoning and that moral development follows a series of stages i.e. the *pre-conventional level*, the *conventional level*, and the *post-conventional level*.

- **Pre-conventional morality (Young children)**

Pre-conventional morality is the first stage of moral development and lasts until approximately age 9. At this stage, children don't have a personal concept of morality and moral decisions are made as per the standards prescribed by the adults. The consequences of actions are the major source of following rules/standards. The stage is further divided into two stages:

> **Stage 1:** *Punishment and Obedience Orientation*: The child or person behaves well to avoid punishment. A person must have done something wrong if they are punished.
>
> **Stage 2:** *Individualism and Exchange*: At this stage, kids understand that other valid points of view aren't always dictated by the authorities. Distinct people have different perspectives of view.

- **Conventional morality (Older children, adolescents, most adults):**

Conventional morality is the second stage of moral development characterized by an acceptance of social rules for being right or wrong and internalization of the moral standards made by adults/authority. A social system stressing the responsibilities of relationships and perseverance of social order becomes desirable and ultimately influences our belief about right and wrong deeds.

> **Stage 3:** *Good Interpersonal Relationships*: The child or person acts morally upright to be regarded as a decent person by others. As a result, responses are related to other people's approval.
>
> **Stage 4:** *Maintaining Social Order*: As the kid or person gains awareness of broader social norms, judgments are made about upholding the law and abstaining from wrongdoing by following the rules.

- **Post-conventional morality (Older children, adolescents, most adults):**

Understanding universal ethical principles like the preservation of life at all costs, the importance of human dignity etc. is the major characteristic of the third stage of moral development. Individual judgment is based on self-chosen principles, and moral reasoning is based on individual rights and justice. Only 10-15% think through ethical principles for themselves as most people take their moral views from those around them.

Stage 5. *Social Contract and Individual Rights*: The individual realizes that while rules/laws must exist for the good there are times when they may become against the interest of particular individuals and breaking rules becomes inevitable. For example, the protection of life is more important than breaking the law against stealing.

Stage 6. *Universal Principles*: People at this stage develop their own set of moral guidelines which may or may not fit within the framework of existing law e.g. principles of human rights, justice, and equality. The person will be prepared to act to defend these principles even at the cost of going against the rest of society and readily accept the consequences of disapproval and or imprisonment. Kohlberg doubted few people reached this stage.

7.5 Early and Middle Adulthood: Building Effective Lives

Early adulthood and middle adulthood, which are roughly the ages of 25 to 45 and 45 to 65, respectively, are the two periods of our lives that are the longest. By overcoming two of Erik Erikson's life challenges—learning to love and be loved in a committed, long-term relationship and growing interested in influencing the development of a new generation, frequently by becoming parents—these are also the times when the majority of us contribute the most to society.

7.5.1 Physical and cognitive development

Early and middle adulthood are characterized physically by a slow, progressive reduction in physiological function. Through the mid-20s, muscular mass initially increases and then steadily declines. Similar to how muscle mass increases, skeletal mass decreases starting around the age of 30. The cardiac muscle eventually becomes weaker as a result of the gradual buildup of cholesterol in the arteries over the course of adulthood. The production of both male and female hormones also diminishes with age. There is also very clear evidence of a decrease in the speed of responding to tasks demanding increased intellectual concentration as well as physical strength. This slowdown of behaviour is a result of the brains of older adults having slower rates of electrical activity. The alterations in learning, memory, or other intellectual processes, however, do not match the loss in mental processing. Young people are better at organizing new information than older people, but older people are either equally good at or better at remembering and using this information as accumulated cultural knowledge.

7.5.2 Psycho-Social Changes in Early and Middle Adulthood

In contrast to childhood or adolescence, which are mostly supported by parents, the ability to establish a successful and autonomous life is the most distinguishing characteristic of adulthood. Even while the timing of the significant life events in early and middle adulthood varies, a broad pattern known as a social clock nevertheless frequently holds. The "appropriate time" for significant life events, such as leaving one's childhood home, getting married, and having children, is determined by culture. This is referred to as the "social clock." Adults must work for a living and start their own families. Young adults NOT following the *social clock* e.g. continuing living with their parents, not marrying or deciding not to have their children are considered deviant, and they may be stigmatized by society.

In **early adulthood**, an individual is concerned with developing the ability to share intimacy, seeking to form relationships and finding intimate love. Long-term relationships are formed, and often marriage and children result. Both the partners' physical and emotional health are better off after marriage. Married people report more life happiness than single people do, as well as having fewer health issues. Parenting also entails a significant and ongoing commitment that can put a great deal of stress on the parents. Children require a lot of time and money, which can lead to stress and lower marital happiness. This decline is especially true for women, who shoulder the majority of the responsibility for caring for the home and children while also working more often and pursuing jobs. Additionally, young adults must make career choices. Along with friends, political beliefs, residential location, child care, job stress, and many other facets of life, work/career choice also influences socioeconomic status. Along with income, other factors such as achievement, recognition, pleasure, security, and challenges are also crucial for job longevity and career choice.

During **middle childhood,** having a sincere concern for the welfare of future generations and making a difference in the world through job and family is a problem that faces middle-aged adults. The issue at this point is referred regarded as generativity vs. self-absorption by Erik Erikson. While it's not generally accepted that everyone has a midlife crisis, many people in their 40s and 50s realize that more than half of their lives have already passed. Some people may feel as though time is running out and that they must make quick, severe changes to accomplish their goals as a result of this realization, while others may choose to find fulfilment in the way their lives are going at the moment. Most middle-aged persons are not miserable, despite the difficulties of early and middle adulthood. As families have been formed, jobs have begun, and certain life goals have been partially attained, these years are frequently highly fulfilling.

Robert Havighurst summarizes the seven **major tasks** to be carried out in the middle years, as follows:

- Accepting and adjusting to physiological changes, such as menopause
- Reaching and maintaining satisfaction in one's occupation
- Adjusting to and possibly caring for ageing parents
- Helping teenage children become responsible adults
- Achieving adult social and civic responsibility
- relating to one's spouse as a person
- Developing leisure-time activities

7.6 Late Adulthood: Aging, Retiring, and Bereavement

Before entering adulthood late adulthood stage, most individuals are successful in developing secure attachments, having families and finding appropriate careers. Eventually, though, as people reach their 60s and beyond, the ageing process causes rapid changes in their physical, cognitive, and social capacities and demands, and life starts to come to a natural end. As a result, the final life stage, known as late adulthood, which starts in the 60s, emerges. Most people agree that late adulthood (old age) starts at the age of 65. reconcile the struggle between integrity and despair, Erik Erikson contends that it is crucial to find meaning and fulfilment in life right now rather than to grow resentful and dejected. A healthy life can be lived despite the changes brought on by age, which do not always impact everyone in the same manner. Researchers have found that the people who are best able to adjust well to changing situations early in life are also able to better adjust later in life. Perceptions of individuals about adults also matter. People who had more positive perceptions about ageing lived longer as compared to people who believed that the elderly are sick, vulnerable, and grumpy.

7.6.1 Physical changes

In their twenties, people normally reach their physical peak strength and endurance, after which they tend to slowly deteriorate. Several physiological changes, including some degree of brain atrophy and a slowing of neurological functions, may take place in later adulthood. Less effective respiratory and circulatory systems, as well as the potential for greater constipation due to alterations in the gastrointestinal tract. Decreased bone mass, particularly in women, causes bone density problems such as osteoporosis. Without consistent exercise, muscles

deteriorate. Skin dries out and loses flexibility. In both sexes, hair loss happens. All sensory faculties, including smell, taste, touch, hearing, and vision, are also less sensitive. There have been reports of slower neuronal and motor response times.

7.6.2 Cognitive Changes

Older adults do appear to process information more slowly; they may take longer to evaluate information and comprehend language, and they often take longer than younger ones to recall words they already know, even though they are perfectly capable of doing so once they are seen. Additionally, older folks have a harder time suppressing, controlling, and maintaining their focus on the discussion's subject. Slower processing and less precise executive control, however, are not always indicative of poorer memory or even lower intelligence. Perhaps one reason why older people move more slowly is that they just know more. Older persons have higher crystallised intelligence, or general understanding of the universe as reflected in semantic, lexical, and linguistic abilities. Therefore, adults typically score better than younger individuals on tests of history and geography, as well as on crossword puzzles, where this information is helpful. The elderly have the advantage of wisdom over young people who have fluid intelligence, which is the capacity to think and gather information quickly and abstractly. This superior knowledge is combined with a slower and more thorough processing style, as well as a more sophisticated understanding of how the world works. The difference between fluid and crystallised intelligence helps to explain why older people do not always do worse on activities that also require experience (i.e., crystallized intelligence), even though they have generally worse memory. For instance, a youthful chess player may think more swiftly, but a more seasoned chess player has more experience to draw from. Because they have more interpersonal experience than younger ones, older folks are also better at grasping the subtleties of social interactions.

Working memory loss is thought to be the main cause of older people performing less well in cognitive activities. Late-life intellectual development does not invariably lead to decreased capacity. While crystallised intelligence—the capacity to use accumulated knowledge to solve problems and make decisions—has been shown to slightly increase over the course of a person's lifetime, fluid intelligence—the capacity to recognise and use patterns and relationships to solve problems—does decline in later years. If elderly individuals receive corrective training, 40 to 60 per cent of their cognitive function declines could be halted.

One common misconception about older people is that they all have genera**liz**ed memory loss. However, new research shows that healthy older people only

experience specific forms of memory deficiencies, while other types of memory are either relatively unaffected or even get better with age. Cognitive deficiencies exhibited in older adults are typically caused by dementia. But only about 15% of adults over 65 experience these diseases. Alzheimer's disease, a fatal, ultimately progressive illness that starts with confusion and memory loss and ends with the inability to care for oneself, is the most common cause of dementia in the country.

7.6.2.1 Dementia and Alzheimer's Disease

Alzheimer's disease is a type of dementia that, over the course of years, results in a loss of emotions, cognition, and physical functioning and that is ultimately fatal. Dementia is defined as a progressive neurological disease that includes loss of cognitive abilities significant enough to interfere with daily behaviours. The probability of having Alzheimer's doubles almost every five years after age 65, and both dementia and Alzheimer's disease are most commonly seen in people 65 and older. After age 85, the risk rises to around 8% per year. The brain cells that make the neurotransmitter acetylcholine gradually lose their ability to function in both biological illnesses. The brain becomes less and less functional without this neurotransmitter since the neurons cannot connect.

While there is growing evidence that the environment also plays a role in dementia and Alzheimer's, these conditions are partially inherited. There is a higher likelihood of maintaining mental acuity in older adults who continue to keep their minds active by doing cognitive activities like reading, playing an instrument, attending lectures, or doing crossword puzzles, as well as those who maintain social connections with others and keep themselves physically fit. In other words, while physical illnesses can affect anyone at any time, the more people keep their brains active and the more they lead active, healthy lifestyles, the healthier their brains will remain.

7.6.3 Social Changes during Aging: Retiring Effectively

The 21st century's longer life expectancy means that seniors can anticipate spending around a quarter of their lives in retirement. Major life shifts like leaving one's job can result in anxiety, despair, and other negative changes in one's self-concept and sense of self. On the other hand, retirement may present a chance for a successful shift from workplace and career duties to stronger roles in the family and the community. People who have worked in physically taxing or monotonous employment may find respite in retirement, especially if they have alternative means of stimulation and identity expression. While forced to retire before they're ready and when they have enough money, people have been reported to be happier

in retirement. Chronic health problems such as arthritis, rheumatism, and hypertension increasingly interfere with the quality of life of most individuals as they age.

The following recommendations are made to make the retirement phase a positive one:

1. Maintain a part-time job after retiring to transition into retirement gradually.
2. Making financial arrangements for retirement is a wonderful idea, but it also makes sense to make plans for alternative careers or interests to fit into your post-employment life.
3. Retire together with someone – if you're still married, it's a good idea to retire along with your spouse so you may both continue working part-time and follow a joint retirement plan.
4. Have a happy marriage - People who have marital issues typically find retirement more difficult since they no longer have a home life to retreat to and are unable to take solace in their jobs. Working on their marriages can help couples retire much more easily. Take care of physical and financial health — a sound financial plan and good physical health can ensure a healthy, peaceful retirement.
5. Retire early from a stressful job — people who stay in stressful jobs due to fears like losing pensions or remaining jobless feel trapped and become a source of severe emotional setback. In such circumstances leaving a job early may make retirement a relief.
6. Retire "on time" — retiring too early or too late can create stress and feelings like underachievement in their goals.

It has been noted that people tend to be adaptable and usually adjust to their retired life, whether they follow these tips or not. A study found ten times as many widows as widowers. The reason behind the phenomenon can be the fact that women live 5 to 7 years longer than men and are usually married to men older than them, Widowhood is particularly stressful if the death of the spouse occurs early in life. In such cases, the support of friends, particularly other widows, can be very helpful.

Even though the body and mind are slowing, most older adults nevertheless maintain an active lifestyle, remain as happy as they were when younger — or are happier — and increasingly value their social connections with family and friends. Researchers found that people's memories of their lives became more positive with

age and that older adults tended to speak more positively about events in their lives, particularly their relationships with friends and family, than did younger adults.

Self-Assessment Questions

1 Compare your actions, beliefs, and perspectives on work and marriage with those of your parents and grandparents. How are they distinct from one another?

2 Create a timeline representing your own desired or intended social clock. What elements do you believe will increase or decrease the likelihood that you can adhere to the schedule?

3 How do people in your culture feel about getting older? What myths exist about the elderly? Do any other, more advantageous ways for people in your community to learn how to think about ageing?

4 What advice would you give your parents regarding how to best maintain healthy physical and cognitive performance into late age in light of the material you have read in this chapter?

Conclusions

Due to the numerous fundamental genetic and environmental elements that can have a favourable or negative impact on a person's entire life, human development from conception to adulthood includes a crucial and significant period. The human being, who completes the first stage of development in the womb, is influenced by a variety of factors, including a) parental origin (father and mother), genetic inheritance, and nutrition of the mother before and during pregnancy; b) incidents that affected the mother and, as a result, the child as it developed in the womb, at birth (delivery), during the perinatal period, and in the early years of life. Throughout the preschool, school, and adolescent years, when a good diet and a balanced lifestyle are crucial and depend on direction from important individuals around, development is still fragile.

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Unit 8

LEARNING

Written by: Dr Sidra Rizwan
Reviewed by: Dr Muhammad Zafar

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INTRODUCTION

Any educational endeavour is a learning situation; the task of educational psychology is to study the learner in that situation. The first learning situation outside the family a child (or a learner) encounters is the school, which is again teaching-learning condition oriented. The teacher's duty in this setting is to apply the general propositions received from psychology and apply them in the classroom. But not one single strategy employed so far has yielded any fruitful result. A more practical-oriented strategy is required synthesize the learner, the teacher the instructional techniques and the educational managers on the one hand and produce qualified students to meet the demands of the day, on the other.

The world we live in today is shaped to a considerable degree by the decisions people make individually and collectively. Any decision-making needs to possess some knowledge and use them in solving problems. In other words, the kind of perceiving, thinking and evaluating that goes into problem-solving has to be considered. Historically, possession of knowledge and its utilization are learned during the developmental years of children through interacting with parents, employers, religious and political leaders as well as teachers in the classroom. The interaction with the teachers is no less important in the process of our lifelong learning even if the exposure to school is brief and transitory (this is stated considering the number of dropouts at the school level in our country). It has an impact on their lives, nevertheless.

The world of tomorrow will be shaped not only by what today's children are learning from their teachers but also by the 'way' they are learning it, for it is the way knowledge is presented that determines how children will learn to solve problems. Thus according to Lindgren (1980), the 'how' of teaching includes not only teaching methods, but also teachers' attitudes and values, and a full range of teachers' classroom behaviours.

Learning outcomes

After completing this unit the students will be able to:

1. Elicit the details of various learning theories
2. Elaborate how behaviorists approach learning
3. Apply learning theories in real life scenarios

8.1 Learning

Learning is a process that is often not under our control and is wrapped up in the environments we inhabit and the relationships we make. It involves encountering signals from the senses; attending to them; looking for connections and meanings; and framing them so that we may act.

Over the last thirty years or so, 'learning' has become one of the most used words in the field of education. Adult education became lifelong learning; students became learners, teachers' facilitators of learning; schools are now learning environments; learning outcomes are carefully monitored. This *verification* of the language and practice of education (Biesta 2009, 2018) is in part due to the rise of individualizing neo-liberal policies. Developments in learning theory have also contributed.

Yet, for all the talk of 'learning', there has been little questioning about what it is, and what it entails. As Hower, Holmes & and Moors (2013) noted, 'Questions about learning are addressed in virtually all areas of psychology. It is therefore surprising to see that researchers are rarely explicit about what they mean by the term'.

There has been a similar situation in the field of education. It is almost as if 'learning' is something that is unproblematic and can be taken for granted. Get the instructional regime right, the message seems to be, and learning (as measured by tests and assessment regimes) will follow.

The reality is that learning, as Lynda Kelly (2002) put it, 'is a very individual, complex, and, to some degree, an indescribable process: something we just do, without ever thinking too much about it'. It is also a complex social activity. Perhaps the most striking result of recent research on learning in childhood and adolescence is that very little comes through conscious and deliberate teaching (Gopnik, 2016). It comes from participation in life.

Learning is an adaptive function by which our nervous system changes about stimuli in the environment, thus changing our behavioural responses and permitting us to function in our environment. The process occurs initially in our nervous system in response to environmental stimuli. Neural pathways can be strengthened, pruned, activated, or rerouted, all of which cause changes in our behavioural responses.

Instincts and reflexes are innate behaviours that occur naturally and do not involve learning. In contrast, learning is a change in behaviour or knowledge that results from experience. The field of behavioural psychology focuses largely on measurable behaviours that are learned, rather than trying to understand internal states such as emotions and attitudes.

8.2 Learning by Association: Classical Conditioning

Learning is a process of drawing connections between what is already known or understood and new information. Thus, prior knowledge is important to the learning process. People make connections and draw conclusions based on a sense of what they already know and have experienced. Learning can be viewed, in part, as a matter of encoding and storing information in memory, processing, categorizing and clustering material, and later retrieving this information to be applied at the appropriate times and situations. For learning to occur, facts, concepts and ideas must also be stored, connected to other facts, concepts, and ideas, and built upon. Knowing in advance what the big ideas are and how they relate to each other conceptually helps learners to make sense of information and to remember and use it more flexibly.

Classical conditioning (Pavlov, 1897) is a form of learning whereby a conditioned stimulus (CS) becomes associated with an unrelated unconditioned stimulus (US) to produce a behavioural response known as a conditioned response (CR). The conditioned response is the learned response to the previously neutral stimulus. The unconditioned stimulus is usually a biologically significant stimulus such as food or pain that elicits an unconditioned response (UR) from the start. The conditioned stimulus is usually neutral and produces no particular response at first, but after conditioning it elicits the conditioned response.

Extinction is the decrease in the conditioned response when the unconditioned stimulus is no longer presented with the conditioned stimulus. When presented with the conditioned stimulus alone, the individual would show a weaker and weaker response, and finally no response. In classical conditioning terms, there is a gradual weakening and disappearance of the conditioned response. Related to this, spontaneous recovery refers to the return of a previously extinguished conditioned response following a rest period. Research has found that with repeated extinction/recovery cycles, the conditioned response tends to be less intense with each period of recovery.

8.2.1 Pavlov's Famous Study

Ivan Pavlov (1849–1936) was a Russian scientist whose work with dogs has been influential in understanding how learning occurs. Through his research, he

established the theory of classical conditioning. The best-known of Pavlov's experiments involves the study of the salivation of dogs. Pavlov was originally studying the saliva of dogs as it related to digestion, but as he conducted his research, he noticed that the dogs would begin to salivate every time he entered the room—even if he had no food. The dogs were associating his entrance into the room with being fed. This led Pavlov to design a series of experiments in which he used various sound objects, such as a buzzer, to condition the salivation response in dogs.

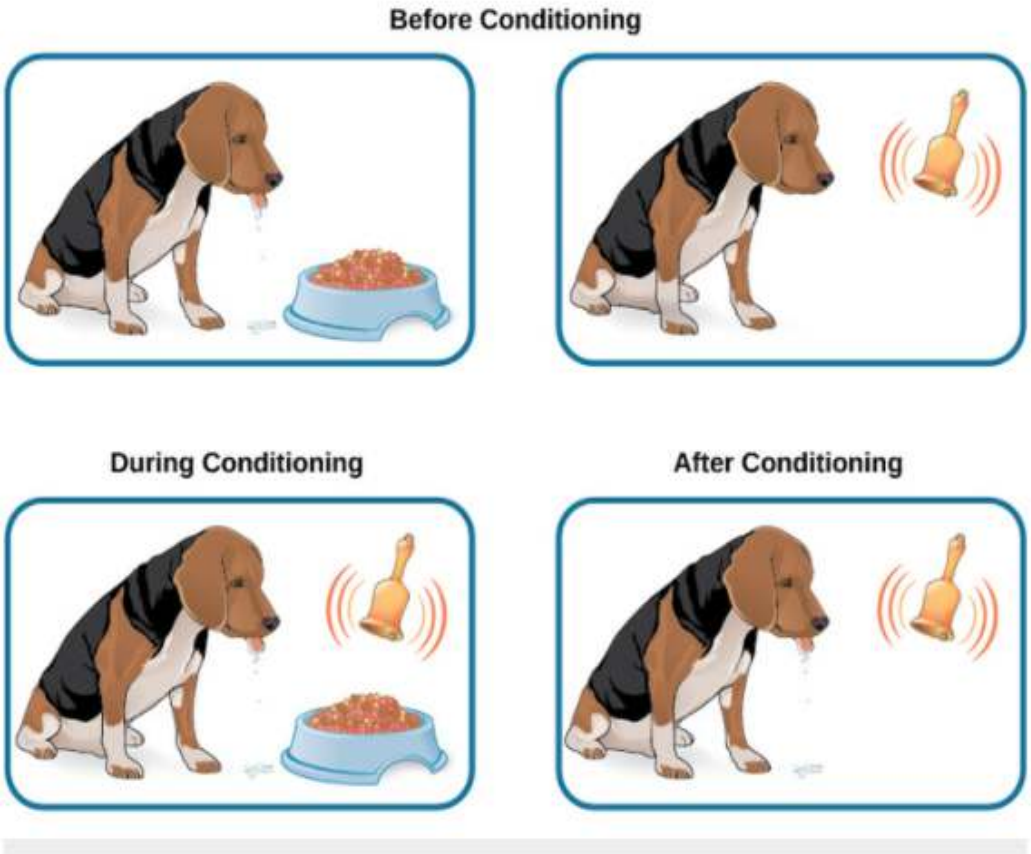
He started by sounding a buzzer each time food was given to the dogs and found that the dogs would start salivating immediately after hearing the buzzer—even before seeing the food. After some time, Pavlov began sounding the buzzer without giving any food at all and found that the dogs continued to salivate at the sound of the buzzer even in the absence of food. They had learned to associate the sound of the buzzer with being fed.

If we look at Pavlov's experiment, we can identify the four factors of classical conditioning at work:

The unconditioned response was the dogs' natural salivation in response to seeing or smelling their food.

- The *unconditioned stimulus* was the sight or smell of the food itself.
- The *conditioned stimulus* was the ringing of the bell, which previously had no association with food.
- The *conditioned response,* therefore, was the salivation of the dogs in response to the ringing of the bell, even when no food was present.

Pavlov had successfully associated an unconditioned response (natural salivation in response to food) with a conditioned stimulus (a buzzer), eventually creating a conditioned response (salivation in response to a buzzer). With these results, Pavlov established his theory of classical conditioning.



Classical conditioning: Before conditioning, an unconditioned stimulus (food) produces an unconditioned response (salivation), and a neutral stimulus (bell) does not have an effect. During conditioning, the unconditioned stimulus (food) is presented repeatedly just after the presentation of the neutral stimulus (bell). After conditioning, the neutral stimulus alone produces a conditioned response (salivation), thus becoming a conditioned stimulus.

8.2.2 Behaviorism and Other Research

Pavlov's research contributed to other studies and theories in behaviourism, which is an approach to psychology interested in observable behaviours rather than the inner workings of the mind. The philosopher Bertrand Russell argued that Pavlov's work was an important contribution to the philosophy of mind. Pavlov's research also contributed to Hans Eysench's personality theory of introversion and extroversion. Eysench built upon Pavlov's research on dogs, hypothesizing that the

differences in arousal that the dogs displayed were due to inborn genetic differences. Eysench then extended the research to human personality traits.

Pavlov's research further led to the development of important behaviour-therapy techniques, such as flooding and desensitizing, for individuals who struggle with fear and anxiety. *Desensitizing* is a kind of reverse conditioning in which an individual is repeatedly exposed to the thing that is causing anxiety. *Flooding* is similar in that it exposes an individual to the thing causing the anxiety, but it does so in a more intense and prolonged way.

8.2.3 Classical Conditioning in Everyday Life

Classical conditioning is used not only in therapeutic interventions but in everyday life as well. Advertising executives, for example, are adept at applying the principles of associative learning. Think about the car commercials you have seen on television: many of them feature an attractive model. By associating the model with the car being advertised, you come to see the car as being desirable (Cialdini, 2008). You may be asking yourself, does this advertising technique work? According to Cialdini (2008), men who viewed a car commercial that included an attractive model later rated the car as being faster, more appealing, and better designed than men who viewed an advertisement for the same car *without* the model.

SELF ASSESSMENT QUESTIONS

Give a small description of the following:

Learning	
Stimulus	
Response	
Conditioning	
Extinction	

8.3 Changing Behavior: Operant Conditioning

8.3.1 Reinforcement and Punishment

Reinforcement and punishment are principles of operant conditioning that increase or decrease the likelihood of a behaviour.

Reinforcement and punishment are principles that are used in operant conditioning. *Reinforcement* means you are increasing a behaviour: it is any consequence or outcome that increases the likelihood of a particular behavioural response (and that therefore *reinforces* the behavior). The strengthening effect on the behaviour can manifest in multiple ways, including higher frequency, longer duration, greater magnitude, and short latency of response. *Punishment* means you are decreasing a behavior: it is any consequence or outcome that decreases the likelihood of a behavioural response.

Extinction, in operant conditioning, refers to when a reinforced behaviour is extinguished entirely. This occurs at some point after reinforcement stops; the speed at which this happens depends on the reinforcement schedule, which is discussed in more detail in another section.

8.3.2 Positive and Negative Reinforcement and Punishment

Both reinforcement and punishment can be positive or negative. In operant conditioning, positive and negative do not mean good and bad. Instead, *positive* means you are adding something and *negative* means you are taking something away. All of these methods can manipulate the behaviour of a subject, but each works unique.

	Reinforcement (Increase / maintain behavior)	**Punishment** (Decrease behavior)
Positive (add stimulus)	**Add** pleasant stimulus *to* **Increase / maintain** behavior	**Add** aversive stimulus *to* **Decrease** behavior
Negative (remove stimulus)	**Remove** aversive stimulus *to* **Increase / maintain** behavior	**Remove** pleasant stimulus *to* **Decrease** behavior

8.3.3 Operant conditioning:

In the context of operant conditioning, whether you are reinforcing or punishing a behaviour, "positive" always means you are *adding* a stimulus (not necessarily a good one), and "negative" always means you are *removing* a stimulus (not necessarily a bad one. See the blue text and yellow text above, which represent positive and negative, respectively. Similarly, reinforcement always means you are *increasing* (or maintaining) the level of behavior, and punishment always means you are *decreasing* the level of behavior. See the green and red backgrounds above, which represent reinforcement and punishment, respectively.

- Positive reinforcers add a wanted or pleasant stimulus to increase or maintain the frequency of a behaviour. For example, a child cleans her room and is rewarded with a cookie.
- Negative reinforcers remove an aversive or unpleasant stimulus to increase or maintain the frequency of a behaviour. For example, a child cleans her room and is rewarded by not having to wash the dishes that night.
- Positive punishments add an aversive stimulus to decrease a behaviour or response. For example, a child refuses to clean her room so her parents make her wash the dishes for a week.
- Negative punishments remove a pleasant stimulus to decrease a behaviour or response. For example, a child refuses to clean her room and so her parents refuse to let her play with her friend that afternoon.

- **Primary and Secondary Reinforcers**

The stimulus used to reinforce a certain behaviour can be either primary or secondary. A primary reinforcer, also called an *unconditioned reinforcer,* is a stimulus that has innate reinforcing qualities. These kinds of reinforcers are not learned. Water, food, sleep, shelter, sex, touch, and pleasure are all examples of primary reinforcers: organisms do not lose their drive for these things. Some primary reinforcers, such as drugs and alcohol, merely mimic the effects of other reinforcers. For most people, jumping into a cool lake on a very hot day would be reinforcing and the cool lake would be innately reinforcing—the water would cool the person off (a physical need), as well as provide pleasure.

A secondary reinforcer also called a *conditioned reinforcer,* has no inherent value and only has reinforcing qualities when linked or paired with a primary reinforcer. Before pairing, the secondary reinforcer has no meaningful effect on a subject. Money is one of the best examples of a secondary reinforcer: it is only worth something because you can use it to buy other things—either thing that satisfies basic needs (food, water, shelter—all primary reinforcers) or other secondary reinforcers.

- **Schedules of Reinforcement**

Reinforcement schedules determine how and when a behaviour will be followed by a reinforcer.

A schedule of reinforcement is a tactic used in operant conditioning that influences how an operant response is learned and maintained. Each type of schedule imposes a rule or program that attempts to determine how and when a desired behaviour occurs. Behaviours are encouraged through the use of reinforcers, discouraged through the use of punishments, and rendered extinct by the complete removal of a stimulus. Schedules vary from simple ratio- and interval-based schedules to more complicated compound schedules that combine one or more simple strategies to manipulate behaviour.

8.3.4 Continuous vs. Intermittent Schedules

Continuous schedules reward a behaviour after every performance of the desired behaviour. This reinforcement schedule is the quickest way to teach someone a behaviour, and it is especially effective in teaching a new behaviour. Simple *intermittent* (sometimes referred to as partial) schedules, on the other hand, only reward the behaviour after certain ratios or intervals of responses.

- **Types of Intermittent Schedules**

There are several different types of intermittent reinforcement schedules. These schedules are described as either fixed or variable and as either interval or ratio.

- **Fixed vs. Variable, Ratio vs. Interval**

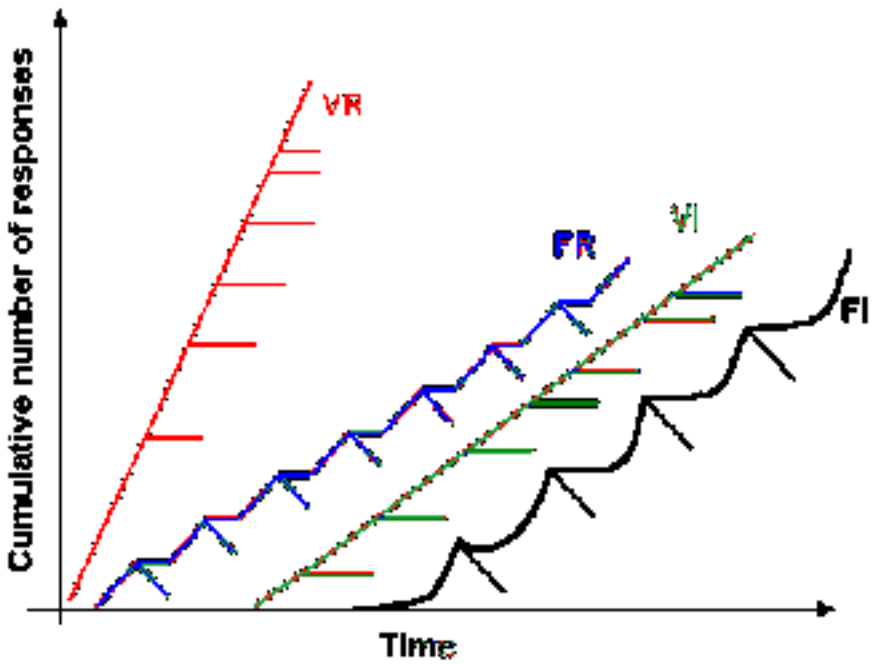
Fixed refers to when the number of responses between reinforcements, or the amount of time between reinforcements, is set and unchanging. *Variable* refers to when the number of responses or amount of time between reinforcements varies or changes. *Interval* means the schedule is based on the time between reinforcements, and *ratio* means the schedule is based on the number of responses between

reinforcements. Simple intermittent schedules are a combination of these terms, creating the following four types of schedules:

- A *fixed-interval schedule* is when behaviour is rewarded after a set amount of time. This type of schedule exists in payment systems when someone is paid hourly: no matter how much work that person does in one hour (behaviour), they will be paid the same amount (reinforcement).
- With a *variable-interval schedule,* the subject gets the reinforcement based on varying and unpredictable amounts of time. People who like to fish experience this type of reinforcement schedule: on average, in the same location, you are likely to catch about the same number of fish in a given period. However, you do not know exactly when those catches will occur (reinforcement) within the period spent fishing (behaviour).
- With a *fixed-ratio schedule,* there are a set number of responses that must occur before the behaviour is rewarded. This can be seen in payment for work such as fruit picking: pickers are paid a certain amount (reinforcement) based on the amount they pick (behaviour), which encourages them to pick faster to make more money. In another example, Carla earns a commission for every pair of glasses she sells at an eyeglass store. The quality of what Carla sells does not matter because her commission is not based on quality; it's only based on the number of pairs sold. This distinction in the quality of performance can help determine which reinforcement method is most appropriate for a particular situation: fixed ratios are better suited to optimize the *quantity* of output, whereas a fixed interval can lead to a higher *quality* of output.
- In a *variable-ratio schedule,* the number of responses needed for a reward varies. This is the most powerful type of intermittent reinforcement schedule. In humans, this type of schedule is used by casinos to attract gamblers: a slot machine pays out an average win ratio—say five to one—but does not guarantee that every fifth bet (behaviour) will be rewarded (reinforcement) with a win.

All of these schedules have different advantages. In general, ratio schedules consistently elicit higher response rates than interval schedules because of their predictability. For example, if you are a factory worker who gets paid per item that you manufacture, you will be motivated to manufacture these items quickly and consistently. Variable schedules are categorically less-predictable so they tend to resist extinction and encourage continued behaviour. Both gamblers and fishermen alike can understand the feeling that one more pull on the slot-machine lever, or one more hour on the lake, will change their luck and elicit their respective rewards.

Thus, they continue to gamble and fish, regardless of previously unsuccessful feedback.



Simple reinforcement-schedule responses: The four reinforcement schedules yield different response patterns. The variable-ratio schedule is unpredictable and yields high and steady response rates, with little if any pause after reinforcement (e.g., gambling). A fixed-ratio schedule is predictable and produces a high response rate, with a short pause after reinforcement (e.g., eyeglass sales). The variable-interval schedule is unpredictable and produces a moderate, steady response rate (e.g., fishing). The fixed-interval schedule yields a scallop-shaped response pattern, reflecting a significant pause after reinforcement (e.g., hourly employment).

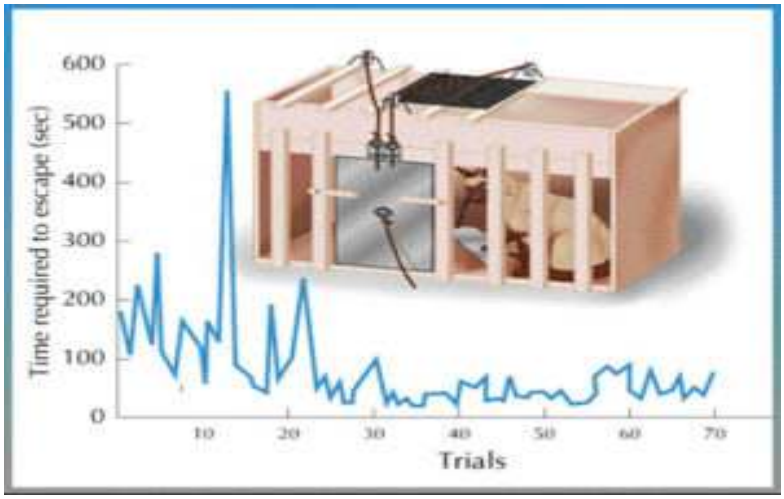
Extinction of a reinforced behaviour occurs at some point after reinforcement stops, and the speed at which this happens depends on the reinforcement schedule. Among the reinforcement schedules, variable-ratio is the most resistant to extinction, while fixed-interval is the easiest to extinguish.

8.4 Thorndike's Law of Effect

Operant conditioning is a theory of learning that focuses on changes in an individual's observable behaviours. In operant conditioning, new or continued behaviours are impacted by new or continued consequences. Research regarding

this principle of learning first began in the late 19th century with Edward L. Thorndike, who established the *law of effect*.

Thorndike's most famous work involved cats trying to navigate through various puzzle boxes. In this experiment, he placed hungry cats into homemade boxes and recorded the time it took for them to perform the necessary actions to escape and receive their food reward. Thorndike discovered that with successive trials, cats would learn from previous behaviour, limit ineffective actions, and escape from the box more quickly. He observed that the cats seemed to learn, from an intricate trial and error process, which actions should be continued and which actions should be abandoned; a well-practised cat could quickly remember and reuse actions that were successful in escaping to the food reward.

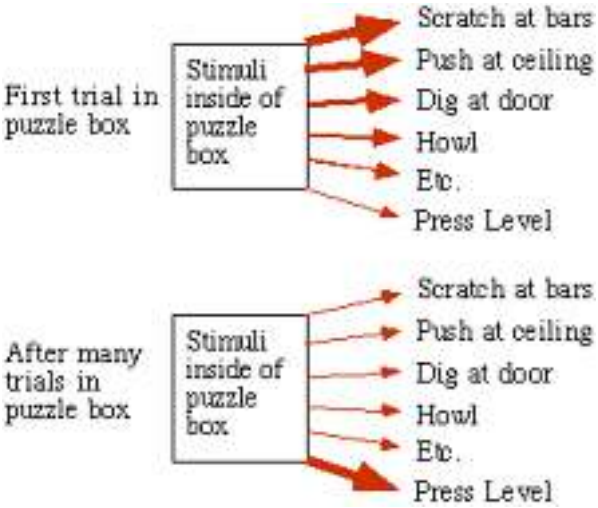


Thorndike's puzzle box: This image shows an example of Thorndike's puzzle box alongside a graph demonstrating the learning of a cat within the box. As the number of trials increased, the cats were able to escape more quickly by learning.

The Law of Effect

Thorndike realized not only that stimuli and responses were associated, but also that behaviour could be modified by consequences. He used these findings to publish his now famous "law of effect" theory. According to the law of effect, behaviours that are followed by consequences that are satisfying to the organism are more likely to be repeated, and behaviours that are followed by unpleasant consequences are less likely to be repeated. Essentially, if an organism does something that brings about a desired result, the organism is more likely to do it

again. If an organism does something that does not bring about the desired result, the organism is less likely to do it again.



Law of effect: Initially, cats displayed a variety of behaviours inside the box. Over successive trials, actions that helped escape the box and receive the food reward were replicated and repeated at a higher rate.

Thorndike's law of effect now informs much of what we know about operant conditioning and behaviourism. According to this law, behaviours are modified by their consequences, and this basic stimulus-response relationship can be learned by the operant person or animal. Once the association between behavior and consequences is established, the response is reinforced, and the association holds the sole responsibility for the occurrence of that behavior. Thorndike posited that learning was merely a change in behaviour as a result of a consequence and that if an action brought a reward, it was stamped into the mind and available for recall later.

From a young age, we learn which actions are beneficial and which are detrimental through a trial and error-process. For example, a young child is playing with her friend on the playground and playfully pushes her friend off the swing set. Her friend falls to the ground begins to cry, and then refuses to play with her for the rest of the day. The child's actions (pushing her friend) are informed by their consequences (her friend refusing to play with her), and she learns not to repeat that action if she wants to continue playing with her friend.

The law of effect has been expanded to various forms of behaviour modification. Because the law of effect is a key component of behaviourism, it does not include any reference to unobservable or internal states; instead, it relies solely on what can be observed in human behaviour. While this theory does not account for the entirety of human behaviour, it has been applied to nearly every sector of human life, particularly in education and psychology.

Operant conditioning is a theory of behaviourism that focuses on changes in an individual's observable behaviours. In operant conditioning, new or continued behaviours are impacted by new or continued consequences. Research regarding this principle of learning was first conducted by Edward L. Thorndike in the late 1800s and then brought to popularity by B. F. Skinner in the mid-1900s. Much of this research informs current practices in human behaviour and interaction.

8.5 Skinner's Theories of Operant Conditioning

Almost half a century after Thorndike's first publication of the principles of operant conditioning and the law of effect, Skinner attempted to prove an extension to this theory—that all behaviours are in some way a result of operant conditioning. Skinner theorized that if a behaviour is followed by reinforcement, that behaviour is more likely to be repeated, but if it is followed by some sort of aversive stimuli or punishment, it is less likely to be repeated. He also believed that this learned association could end, or become extinct if the reinforcement or punishment was removed.

Skinner's most famous research studies were simple reinforcement experiments conducted on lab rats and domestic pigeons, which demonstrated the most basic principles of operant conditioning. He conducted most of his research in a special cumulative recorder, now referred to as a "Skinner box," which was used to analyze the behavioural responses of his test subjects. In these boxes, he would present his subjects with positive reinforcement, negative reinforcement, or aversive stimuli in various timing intervals (or "schedules") that were designed to produce or inhibit specific target behaviours.

In his first work with rats, Skinner would place the rats in a Skinner box with a lever attached to a feeding tube. Whenever a rat pressed the lever, food would be released. After experiencing multiple trials, the rats learned the association between the lever and food and began to spend more of their time in the box procuring food than performing any other action. It was through this early work that Skinner started to understand the effects of behavioural contingencies on actions. He discovered that the rate of response—as well as changes in response features—depended on what occurred *after* the behaviour was performed, not before. Skinner named these actions *operant behaviours* because they *operated* on the environment to produce an outcome. The process by which one could arrange the contingencies of reinforcement responsible for producing a certain behaviour then came to be called *operant conditioning.*

To prove his idea that behaviourism was responsible for all actions, he later created a "superstitious pigeon." He fed the pigeon at continuous intervals (every 15 seconds) and observed the pigeon's behaviour. He found that the pigeon's actions would change depending on what it had been doing in the moments before the food was dispensed, even though those actions had nothing to do with the dispensing of food. In this way, he discerned that the pigeon had fabricated a causal relationship between its actions and the presentation of reward. It was this development of "superstition" that led Skinner to believe all behaviour could be explained as a learned reaction to specific consequences.

In his operant conditioning experiments, Skinner often used an approach called *shaping*. Instead of rewarding only the target, or desired, behaviour, the process of shaping involves the reinforcement of successive approximations of the target behaviour. Behavioural approximations are behaviours that, over time, grow increasingly closer to the actual desired response.

Skinner believed that all behaviour is predetermined by past and present events in the objective world. He did not include room in his research for ideas such as free will or individual choice; instead, he posited that all behaviour could be explained using learned, physical aspects of the world, including life history and evolution. His work remains extremely influential in the fields of psychology, behaviourism, and education.

In his operant conditioning experiments, Skinner often used an approach called shaping. Instead of rewarding only the target, or desired, behavior, the process of shaping involves the reinforcement of successive approximations of the target behavior. The method requires that the subject perform behaviours that at first merely resemble the target behavior; through reinforcement, these behaviours are gradually changed, or *shaped*, to encourage the performance of the target behavior itself. Shaping is useful because it is often unlikely that an organism will display anything but the simplest of behaviours spontaneously. It is a very useful tool for training animals, such as dogs, to perform difficult tasks.

8.5 Learning by Insight and Observation

Learning by insight means sudden grasping of the solution, a flash of understanding, without any process of trial and error. All discoveries and inventions have taken place through insight. Of course, the discoverer possessed a complete knowledge of the whole situation in peace-meal.

This theory is also called Gestalt Theory of Learning an explanation of Gestalt School of Psychology. The word Gestalt in the German language means 'whole', 'total pattern' or 'configuration'. This school believes that the whole is more important than the parts. So learning also takes place as a whole. In this respect, Kohlar performed several experiments on monkeys and arrived at the result that the highest type of learning is through insight.

Gestalt psychologists dismiss the theory of 'trial and error', hit and miss' strive and succeed'. Another modern psychologist E.C. Tolman also rejects the trial and error theory and approaches the insight theory.

According to Tolman, in all learning some intelligence is at work. It is the learner who actively participates in the act of getting new experiences. He organizes his perceptions and observations and gives meaning to them. It is his whole mind that perceives, constructs and reconstructs experience. In his 'purposive theory of learning', he explains the behaviour of rats in teaching the goal through many trials as a result of 'insight' or 'making a cognitive map' of the maze.

8.5.1 Experiments of Theory of Learning by Insight:

Experiment with the box:

Kohlar kept a monkey (named Sultan) hungry for some time, and then shut him in a large cage. He hung bananas from the ceiling, and kept a box on the floor of the cage, fast beneath. The monkey could not reach the banana. Another box was put in a corner of the cage.

But Sultan could not get the idea of placing one box on the other and thus reaching the banana. Ultimately Kohlar gave a demonstration of putting one box on the other. Sultan could now learn the whole situation. He used his intelligence and insight to put the two boxes one upon the other, stand on these and then reach the bananas.

Experiment with two sticks

In another experiment, Kohler kept two sticks in the cage. One end of the shorter sticks could be fitted in the one end of the longer sticks, to make them longer. The monkey did not get the idea of forming the two sticks through trial and error. When Kohlar gave a hint by putting his finger in the whole of the bigger stick, the monkey viewed the whole situation and performed the right task by understanding the insight.

8.5.2 Criterion or Essentials of Learning by Insight:

- **Comprehension as a whole:**

Learning by insight requires full comprehension of the situation as a whole.

- **Clear goal:**

The goal must be quite clear, to begin with.

- **Power of generalisation:**

The learner must possess the power of generalisation along with those of differentiation.

- **Suddenness of solution:**

The suddenness of the solution is the hallmark of learning by insight i.e., the solution flashes suddenly to the learner. No lengthy reasoning is involved.

- **New forms of objects:**

As a result of insight into the problem or situation, objects appear in new forms and patterns.

- **Transfer:**

Transfer of learning occurs as a result of insight. The principles learnt in one situation are applied to the other situation.

- **Behaviour change:**

Insight changes our behaviour to the extent that we have learnt through insight.

8.5.3 Laws of Insight Formation:

- **Capacities:**

Insight depends upon the capacity of the organism. Individuals offer in their capacities. The more developed the individual; the more will be the capacity to develop insight.

- **Previous experience:**

Insight depends upon relevant previous experience and maturation. Some practice, trial and error and maturation up to the level are essential before insight develops. A child of five years cannot develop mathematical insight since he has not done sufficient practice in it.

- **Experimental arrangement:**

The development of insight depends upon experimental arrangement.

- **Fumbling and searching:**

Insight follows a period of fumbling and searching.

- **Readily repeated:**

Insightful solutions can be readily repeated.

- **Use in new situations:**

Insight once achieved can be used in new situations.

- **Wholesome experience:**

Experience of insight is always wholesome. The whole is just not equal to its parts.

8.6 Observational Learning

The Observational Learning process was propounded by Albert Bandura in his Social Learning Theory, which focused on learning by imitation or observing people's behaviour. For observational learning to take place effectively, four important elements will be essential: Motivation, Attention, Memory and Motor Skills.

According to Albert Bandura, learning cannot simply be based merely on associations or reinforcements which he has mentioned in his writings in his book **Social Learning Theory** which was published in 1977. Instead, his focus was on learning based on observation, which he has proven through his well-known Bobo Doll experiment. He reckoned that children keenly observe their surroundings and the behaviour of people around them, particularly their caregivers, teachers and siblings and try to imitate those behaviours in day-to-day lives. He also tried proving through his experiment that children can easily imitate negative behaviours or actions.

Another important principle of Bandura's Social Learning Theory was that learning something by way of observation, need not necessarily mean that it would lead to a change in the behaviour. This behavioural change is entirely influenced by the felt need or motivation of a person to endorse and adopt a behavioural change.

Key Steps Involved in Observational Learning

- **Attention**: Attention is very important for learning to take place effectively by following observational techniques. A novel concept or a unique idea is expected to attract attention far more strongly than those which are routine or mundane in nature.
- **Retention**: It is the ability to store the learned information and recall it later, which is equally affected by some factors.

- **Reproduction**: It involves practising or emulating the learnt behaviour, which will further lead to the advancement of the skill.
- **Motivation**: Motivation to imitate the learnt behaviour of a model depends a lot on reinforcement and punishment. For example, an office-goer may be motivated to report to the office on time by seeing his colleague being rewarded for his punctuality and timeliness.

8.7 Using the Principles of Learning to Understand Everyday Behavior

The principles of learning are some of the most general and most powerful in all of psychology. It would be fair to say that these principles account for more behaviour using fewer principles than any other set of psychological theories. The principles of learning are applied in numerous ways in everyday settings. For example, operant conditioning has been used to motivate employees, to improve athletic performance, to increase the functioning of those suffering from developmental disabilities, and to help parents successfully toilet-train their children.

8.7.1 Using Classical Conditioning

Classical conditioning has long been and continues to be, an effective tool in marketing and advertising. The general idea is to create an advertisement that has positive features such that the ad creates enjoyment in the person exposed to it. The enjoyable ad serves as the unconditioned stimulus (US), and the enjoyment is the unconditioned response (UR). Because the product being advertised is mentioned in the ad, it becomes associated with the US, and then becomes the conditioned stimulus (CS). In the end, if everything has gone well, seeing the product online or in the store will then create a positive response in the buyer, leading him or her to be more likely to purchase the product.

A similar strategy is used by corporations that sponsor teams or events. For instance, if people enjoy watching a university basketball team playing basketball, and if that team is sponsored by a product, such as Pepsi, then people may end up experiencing positive feelings when they view a can of Pepsi. Of course, the sponsor wants to sponsor only good teams and good athletes because these create more pleasurable responses.

8.7.2 Reinforcement in Social Dilemmas

The basic principles of reinforcement, reward, and punishment have been used to help understand a variety of human behaviours. The general idea is that, as predicted by principles of operant learning and the laws of effect, people act in ways that maximize their *outcomes*, where **outcomes** are defined as *the presence of reinforcers and the absence of punishers*.

A **social dilemma** such as the commons dilemma is *a situation in which the behaviour that creates the most positive outcomes for the individual may in the long term lead to negative consequences for the group as a whole*. The dilemmas are arranged in such a way that it is easy to be selfish because the personally beneficial choice (such as using water during a water shortage or driving to work alone in one's car) produces reinforcements for the individual. Furthermore, social dilemmas tend to work on a type of time delay.

8.7.3 Transfer of Training or Learning

Transfer of training or learning occurs when learning of one set of materials influences the learning of another set of materials later. For example, driving a new car, the movements and responses in driving a new car will have similarities and differences when compared to movements and responses in driving the old car. The individual has to adapt his old habits and learn new ones.

According to Crow and Crow, transfer of learning means "the carryover of habits, thinking, feeling, working, knowledge and skills from one learning area to another learning area".

According to Bigger transfer of learning occurs when a person's learning in one situation influences his learning and performance in other situations.

Forms of Transfer:

Positive transfer:

When learning in one situation facilitates the learner in another situation, we call it a positive transfer. For example, learning to ride a bicycle helps to ride a bike, etc.

Negative transfer:

When learning in one situation hinders learning in a new situation it is called negative transfer. For example, some companies like to take fresh employees to avoid the process of negative learning due to earlier experience. Her earlier experience acts as a hindrance.

Zero transfer:

When learning in one situation does not influence learning in another situation is said to be zero learning. For example, learning to write Urdu won't influence our learning to write in English.

Positive and negative:

Sometimes, previous learning may partly help as well as partly hinder the performance in a new situation. This is called positive as well as negative.

SELF ASSESSMENT QUESTIONS

Q Choose the correct option for each statement.

i). Which of the following is an example of a reflex that occurs at some point in the development of a human being?
   a) child riding a bike
   b) teen socializing
   c) infant sucking on a nipple
   d) toddler walking

ii). Learning is best defined as a relatively permanent change in behaviour that _____.
   a) is innate
   b) occurs as a result of experience
   c) is found only in humans
   d) occurs by observing others

iii). Two forms of associative learning are _____ and _____.
   a) classical conditioning; operant conditioning
   b) classical conditioning; Pavlovian conditioning
   c) operant conditioning; observational learning
   d) operant conditioning; learning conditioning

iv). In _____ the stimulus or experience occurs before the behaviour and then gets paired with the behaviour.
   a) associative learning
   b) Observational learning
   c) operant conditioning
   d) Classical conditioning

8.8 Theories of Learning

Major theories of learning that have been proposed by psychologists and educational theorists in the past three to four centuries helped us to learn how human beings learn. Here is the chronological sequence of how the set of theories emerged and made it possible for researchers to know more about learning:

Behaviorism is the oldest theory, associated with psychologists such as B.F. Skinner, and Thorndike focused on observable behaviours and external stimuli about responses in overt behaviour. It suggests that learning occurs through the reinforcement and punishment of specific behaviours.

Cognitivism was proposed by psychologists such as Jean Piaget and Benjamin Bloom. Cognitive theories emphasize the role of mental processes in learning. They

propose that learning involves the active construction of knowledge through processes like attention, memory, and problem-solving.

Constructivism derived from cognitive theories and the founder was John Dewey who was an educational psychologist. Mainly constructivism suggests that learners actively build their understanding of the world by integrating new knowledge with existing knowledge. Learning is viewed as a personal, active, and social process.
Social Learning Theory was proposed by Albert Bandura, and Lev Vygotsky who theorized that cognitive development occurs in collaboration with others. this theory emphasizes the role of observational learning and modeling. It suggests that individuals learn by observing others' behaviour, as well as the consequences of social interaction.

Humanistic theories, such as those developed by Carl Rogers and Abraham Maslow, focused more on the individual's innate drive for self-actualization and personal growth. They emphasize the importance of a supportive and nurturing learning environment that values the learner's needs and goals.

Experiential Learning Theory was proposed by David Kolb, This theory suggests that learning is a cyclical process involving concrete experiences, reflective observation, abstract conceptualization, and active experimentation. It emphasizes the importance of hands-on, real-world experiences in the learning process.

Multiple Intelligences Theory was then developed by Howard Gardner, this theory challenges the traditional notion of intelligence and suggests that individuals possess different types of intelligence, such as linguistic, logical-mathematical, spatial, musical, bodily-kinesthetic, interpersonal, intrapersonal, and naturalistic.

Connectivism is known to be the theory of twenty-first century. It was first introduced in 2005 by two theorists, George Siemens and Stephen D., Siemens. Connectivism is based on the theory that we learn when we make connections, or “links,” between various "nodes" of information, and we continue to make and maintain connections to form knowledge. Thus it establishes how human beings learn together by connecting in physical or virtual spaces to give meanings to reality through diverse points of view.

These are just a few examples of the major theories of learning. Each theory offers a unique perspective on how learning occurs and provides insights into effective teaching and instructional strategies. It's important to note that learning is a complex process, and different theories can complement each other in understanding various aspects of learning.

Self Assessment

Fill in the blanks

- Learning is an _____ function by which our nervous system changes about stimuli in the environment, thus changing our _____responses and permitting us to function in our environment.
- Instincts and reflexes are _____behaviors they occur naturally and do not involve learning. In contrast, learning is a change in behavior or knowledge that results from experience.
- The field of behavioural psychology focuses largely on _____behaviours that are learned, rather than trying to understand internal states such as emotions and attitudes.
- Classical conditioning is a process by which we learn to _____ events, or stimuli, that frequently happen together; as a result of this, we learn to _____ events.
- Ivan Pavlov conducted a famous study involving _____in which he trained (or conditioned) the dogs to associate the sound of a bell with the presence of a piece of meat.
- ” Reinforcement ” refers to any consequence that _____the likelihood of a particular behavioral response;
- ” punishment ” refers to a consequence that _____the likelihood of this response.Both reinforcement and punishment can be positive or negative.
- In operant conditioning, positive means you are _____ something and negative means you are _____ something away.
- Operant conditioning is the learning process by which behaviours are reinforced or punished, thus _____ or _____ a response.
- Edward Thorndike coined the term “law of _____” in which behaviours that are followed by consequences that are _____ to the organism are more likely to be repeated, and behaviours that are followed by _____consequences are less likely to be repeated.
- B. F. Skinner researched operant conditioning by conducting experiments with rats in what he called a “Skinner box.” He differentiated between _____ and _____reinforcement and also explored the concept of extinction.
- Albert Bandura noticed that children often learn through _____ adults, and he tested his theory using his famous _____ experiment.

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Unit 9

REMEMBERING
AND
JUDGING

Writer: Dr Afshan Huma
Reviewer: Dr Muhammad Zafar Iqbal

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INTRODUCTION

Humans use their previous knowledge and experiences to understand and interpret their environment. This process is called cognitive mapping. Our brains store information about the relationships between objects and events in the world, as well as our past experiences and emotions associated with those experiences. When we encounter new information, we use our prior knowledge to make sense of it and fit it into our existing understanding of the world. For example, if you have previously learned that a certain type of flower is associated with a specific smell, you will use that information to help you recognize that flower again in the future. Similarly, if you have experienced the sound of thunder before, you will be able to identify it and associate it with a coming storm. In this way, previous knowledge helps us to form expectations and predictions about our environment, which can inform our behaviour and decision-making. However, it is important to note that previous knowledge can also lead to biases and errors in perception, particularly when the current situation does not match our prior experiences. Remembering past experiences involves the formation and storage of memories in the brain. The process of memory formation is complex and involves multiple brain regions and systems.

Learning about memory and judgment is important because these are fundamental cognitive processes that play a critical role in many aspects of daily life. Understanding how memories are formed and retrieved can help individuals improve their memory skills, which can have a positive impact on academic performance and overall quality of life. Teaching about judgment and decision-making can help individuals understand how their past experiences and prior knowledge influence their perceptions and beliefs, and how these can lead to biases and errors in judgment. This knowledge can help individuals critically evaluate information and make more informed and rational decisions.

Learning outcomes

After completion of this unit, you will be able to

1. Elaborate on how human beings store knowledge and recognize new scenarios

2. Explain the process of information stored in memory and retrieval for future use

9.1 Remembering and Judging

Memory plays an important role in the cognition of new experiences by allowing us to use our past experiences and knowledge to make sense of new information. When we encounter new information, our brains automatically search for similar past experiences stored in memory. This allows us to compare and contrast the new information with what we already know and helps us to make sense of the new experience. Psychologists have been working on cognitive psychology and dedicated their work to examining how people think. It attempts to explain how and why we think the way we do by studying how our memory system functions, how creativity affects our problem-solving, the impact of language on our development, the different types of intelligence, and the fundamental errors we make in decision-making. Cognitive psychologists also focus on how we organize thoughts and information gathered from our environments into meaningful categories. The three subjects of this chapter are memory, defined as the ability to store and retrieve information over time, cognition, defined as the processes of acquiring and using knowledge, and problem-solving, defined as the strategies used to find a solution. It is useful to consider memory and cognition in the same chapter because they work together to help us interpret and understand our environments. The next chapter will examine intelligence and language (Stangor and Walinga, 2014).

Memory also helps us to form expectations and predictions about new experiences. When we encounter a new situation, we use our past experiences and knowledge to anticipate what might happen next. For example, if you see a traffic light turn red, you expect that the cars will stop and pedestrians will have a chance to cross the street. Memory and cognition represent the two major interests of cognitive psychologists. The cognitive approach became the most important school of psychology during the 1960s, and the field of psychology has remained in large part cognitive since that time. The cognitive school was greatly influenced by the development of the electronic computer, and although the differences between computers and the human mind are vast, cognitive psychologists have used the computer as a model for understanding the workings of the mind.

Although cognitive psychology began in earnest at about the same time that the electronic computer was first being developed, and although cognitive psychologists have frequently used the computer as a model for understanding how the brain operates, research in cognitive neuroscience has revealed many important differences between brains and computers (Lally and Valentine-French, 2018). Our memories allow us to do relatively simple things, such as remembering where we put our keys or the name of the current governor of Illinois. Memories also allow us to form complex memories, such as how to ride a bicycle or write a

computer program. Moreover, our memories define us as individuals as they are our experiences, our relationships, our successes, and our failures. Without our memories, we would not have a life, but our memories are far from perfect (Schacter, 1996). At least for some things, our memory is very good (Bahrick, 2000). Once we learn a face, we can recognize that face many years later. We know the lyrics of many songs by heart, and we can give definitions for tens of thousands of words, but memory is not like a video. When people are asked to remember an event, they use current knowledge to put together a story that makes sense. This story is assembled using their original perceptions and memory traces, emotions, beliefs, and other experiences from life.

Thus memory can help us to learn from new experiences. When we encounter a new situation, we compare it to our past experiences and knowledge to see what we can learn. For example, if you make a mistake in a new job, you can use your past experiences to learn from the mistake and avoid making the same mistake in the future. Overall, memory plays a critical role in the cognition of new experiences by allowing us to use our past experiences and knowledge to make sense of new information, form expectations and predictions, and learn from new experiences.

9.2 Memory- Types and Stages

9.2.1 Types

Explicit memory, also known as declarative memory, is the conscious, intentional recollection of past experiences, facts, and events. This type of memory is usually accessed through deliberate effort and involves the ability to retrieve specific details about a past event or piece of information. Examples of explicit memory include remembering your first day of school or reciting the capitals of different countries. Explicit memory is assessed using measures in which the individual being tested must consciously attempt to remember the information. A recall memory test is a measure of explicit memory that involves bringing from memory information that has previously been remembered. Explicit memory includes episodic and semantic memories.

a. **Episodic memory** refers to the firsthand experiences that we have had (e.g., recollections of our high school passing out day or of the fantastic dinner we had last year).

b. **Semantic memory** refers to our knowledge of facts and concepts about the world (e.g., that the absolute value of −90 is greater than the absolute value of 9 and that one definition of the word “affect” is “the experience of feeling or emotion”).

Implicit memory, also known as non-declarative memory, is the unconscious, unintentional recollection of past experiences, facts, and events. This type of memory is usually accessed automatically and without conscious awareness. Examples of implicit memory include riding a bike, typing on a keyboard, or playing a musical instrument. Further types of implicit memory are:

a. Procedural memory refers to our often unexplainable knowledge of how to do things. When we walk from one place to another, speak to another person in English, dial a cell phone, or play a video game, we are using procedural memory.

b. Classical conditioning effects, in which we learn, often without effort or awareness, to associate neutral stimuli (such as a sound or a light) with another stimulus (such as food), which creates a naturally occurring response, such as enjoyment or salivation.

c. Priming, or changes in behaviour as a result of experiences that have happened frequently or recently. Priming refers both to the activation of knowledge (e.g., we can prime the concept of kindness by presenting people with words related to kindness) and to the influence of that activation on behaviour (people who are primed with the concept of kindness may act more kindly)

Implicit memory is often revealed through the performance of a task, rather than through explicit recall. For example, if you learned to ride a bike as a child, you may not be able to explicitly describe how you do it, but you can still ride a bike without consciously thinking about it. Implicit memory is thought to be primarily processed by different brain regions than explicit memory (Krupa, Thompson, & Thompson, 1993). While the hippocampus handles explicit memory, the cerebellum and the amygdala concentrate on implicit and emotional memories, respectively. Research shows that the cerebellum is more active when we are learning associations and in priming tasks, and animals and humans with damage to the cerebellum have more difficulty in classical conditioning studies (Woodruff-Pak, Goldenberg, Downey-Lamb, Boyko, & Lemieux, 2000). The storage of many of our most important emotional memories, particularly those related to fear, is initiated and controlled by the amygdala (Sigurdsson, Doyère, Cain, & LeDoux, 2007). Explicit memory involves conscious, intentional recollection of past experiences, while implicit memory involves unconscious, unintentional recollection of past experiences. Explicit memory is accessed through deliberate effort and involves the ability to retrieve specific details about a past event, while implicit memory is often revealed through performance on a task, rather than through explicit recall.

9.2.2 Stages

- **Sensory memory** refers to the initial stage of memory processing that involves the immediate, temporary storage of information received from our senses. Sensory memory is typically very short-lived, lasting only a few seconds at most. Sensory memory is a memory buffer that lasts only very briefly and then, unless it is attended to and passed on for more processing, is forgotten. The purpose of sensory memory is to give the brain some time to process the incoming sensations and to allow us to see the world as an unbroken stream of events rather than as individual pieces. The two main examples of sensory memory are iconic memory, which is visual sensory memory, and echoic memory, which is auditory sensory memory.

- **Short-term memory**, also known as working memory, refers to the stage of memory processing that involves the temporary storage and manipulation of information. Short-term memory is responsible for holding onto information that we need to actively work with or use in some way. Examples of short-term memory include remembering a phone number that we just looked up or following a set of directions. STM is the place where small amounts of information can be temporarily kept for more than a few seconds but usually for less than one minute (Baddeley, Vallar, & Shallice, 1990). Information in short-term memory is not stored permanently but rather becomes available for us to process, and the processes that we use to make sense of, modify, interpret, and store information in STM are known as working memory.

- **Long-term memory** refers to the stage of memory processing that involves the relatively permanent storage and retrieval of information. Long-term memory can be further divided into two main types: explicit (declarative) memory and implicit (non-declarative) memory. Explicit memory refers to the conscious recollection of facts, events, and experiences, while implicit memory refers to the unconscious influence of past experiences on current behaviour or thought. The capacity of long-term memory is large, and there is no known limit to what we can remember (Wang, Liu, & Wang, 2003). Although we may forget at least some information after we learn it, other things will stay with us forever. In the next section, we will discuss the principles of long-term memory.

In summary, sensory memory is the initial stage of memory processing that involves the immediate, temporary storage of information received from our senses, short-term memory is the temporary storage and manipulation of information and

long-term memory is the relatively permanent storage and retrieval of information, which can be further divided into explicit and implicit memory.

9.2.3 Processes

Here is a brief overview of the main stages of memory formation that to begin during the sensation and perception of environmental stimuli:

- **Encoding** is the process of taking in information and transforming it into a form that can be stored in the brain. The information is encoded in a specific way depending on the type of information, such as visual, auditory, or semantic.
- **Consolidation** After encoding, the memory is consolidated over time by being transferred from short-term memory to long-term memory. This process involves the strengthening of connections between neurons in the brain, which makes the memory more stable and less susceptible to decay.
- **Storage** is done in various regions of the brain, including the hippocampus and the cortex. Different types of memories are stored in different regions, and the strength of the connections between neurons determines how easily the memory can be retrieved.
- **Retrieval** is to dig out from storage. When we want to recall a memory, the memory is retrieved from storage and becomes active again in our brain. This process is influenced by factors such as the strength of the memory, the cues that are present at the time of retrieval, and the emotional state you are in.

SELF ASSESSMENT EXERCISE

Fill in the blanks

a. Memory refers to the ability to _____ and _____ information over time.

b. _____memory refers to experiences that can be intentionally and consciously remembered, and it is measured using recall, recognition, and relearning.

c. Explicit memory includes _____ and _____ memories.

d. _____memory refers to the influence of experience on behaviour, even if the individual is not aware of those influences.

e. The three types of implicit memory are _____memory, _____, and _____

f. Information processing begins in _____memory, moves to _____ memory, and eventually moves to _____ memory.

9.3 How We Remember: Cues to Improving Memory

The process of remembering past experiences involves the interaction of multiple brain regions and systems, and the formation and retrieval of memories are influenced by various factors such as attention, emotions, and prior knowledge. Remembering our childhood involves the retrieval of memories that were formed during that period. Childhood memories can be influenced by several factors, such as the age of the individual at the time the memory was formed, the emotional significance of the event, and the frequency of recall.

One theory of memory formation suggests that memories are formed and stored in different brain regions depending on the type of information being remembered. For example, episodic memories, which are memories of specific events, are thought to be stored in the hippocampus, a region of the brain that is important for spatial navigation and memory.

Some memories can be particularly vivid because they are often associated with strong emotions and sensory experiences. Additionally, early experiences can have a significant impact on later development, and recalling childhood memories can provide insight into one's personality and life experiences.

However, it's important to note that memories can be fallible, and older memories in particular may be subject to distortion and reconstruction over time. For example, memories may be influenced by subsequent experiences or may be altered by suggestions or social cues. Remembering involves the retrieval of memories that were formed during that period, and can be influenced by a range of factors including emotional significance, sensory experiences, and subsequent experiences.

Cognitive strategies to enhance memory include Maintenance rehearsal; Association and Chunking :

- **Maintenance rehearsal:** This is the simple repetition of information to maintain it in short-term memory. It involves rehearsing information repeatedly to keep it fresh in your mind for a brief period. This technique is

useful for retaining information that only needs to be recalled for a short period, such as phone numbers or addresses.

- **Association:** This strategy involves linking new information to pre-existing knowledge or memories. The idea is to create a connection between new information and something you already know, making it easier to remember. For example, you might associate a new name with a famous person or a particular event.

- **Chunking:** This strategy involves grouping information into smaller, more manageable units, or chunks. By breaking down larger pieces of information into smaller, more meaningful parts, it becomes easier to remember. This technique is often used to remember lists of items, such as a phone number, by breaking them into smaller groups of numbers.

- **Context-dependent learning**: This occurs when the environment or context in which information is learned is the same as the environment or context in which it is recalled. For example, if you study for an exam in a particular room, you may have better recall if you take the exam in the same room.

- **State-dependent learning:** It is the internal state of the person while learning and recalling information is the same. For example, if you study for an exam while in a particular emotional state, you may have better recall if you are in the same emotional state during the exam.

- **Primacy effect:** This occurs when people are more likely to remember items that are presented first in a list. This is thought to be because the initial items have more time to be processed and encoded into long-term memory.

- **Recency effect:** This occurs when people are more likely to remember items that are presented last in a list. This is thought to be because these items are still in working memory when the recall task is initiated.

Overall, these effects suggest that the context in which information is learned, as well as the order in which it is presented, can have a significant impact on long-term memory. By understanding these effects, we can develop more effective strategies for learning and remembering information. For example, by studying in an environment similar to the one where we will be recalling information, we may be able to improve our memory retention.

Some more techniques to improve memory are given below:

9.4 Accuracy and Inaccuracy in Memory and Cognition

It is hard to directly assess accuracy and inaccuracy in memory and cognition. The study of human memory is a complex and multi-disciplinary field that draws on a range of methods and techniques from psychology, neuroscience, and other related fields.

Technique	Description	Useful example
Use elaborative encoding.	Material is better remembered if it is processed more fully.	Think, for instance, "Proactive interference is like retroactive interference but it occurs in a forward manner."
Make use of the self-reference effect.	Material is better remembered if it is linked to thoughts about the self.	Think, for instance, "I remember a time when I knew the answer to an exam question but couldn't quite get it to come to mind. This was an example of the tip-of-the-tongue phenomenon."
Be aware of the forgetting curve.	Information that we have learned drops off rapidly with time.	Review the material that you have already studied right before the exam to increase the likelihood it will remain in memory.
Make use of the spacing effect.	Information is learned better when it is studied in shorter periods spaced over time.	Study a little bit every day; do not cram at the last minute.
Rely on overlearning.	We can continue to learn even after we think we know the information perfectly.	Keep studying, even if you think you already have it down.
Use context-dependent retrieval.	We have better retrieval when it occurs in the same situation in which we learned the material.	If possible, study under conditions similar to the conditions in which you will take the exam.
Use state-dependent retrieval.	We have better retrieval when we are in the same psychological state as we were when we learned the material.	Many possibilities, but don't study under the influence of drugs or alcohol, unless you plan to use them on the day of the exam (which is not recommended).

9.4.1 Methods of assessing memory:

Here are some common methods used to study human memory:

- **Behavioral experiments:** These experiments involve testing participants' ability to remember specific information, such as lists of words or images. The researcher then collects data on their performance to determine how well they remember the information.
- **Brain imaging techniques:** Brain imaging techniques such as fMRI, PET, and EEG can provide insights into the neural processes that underlie memory formation and retrieval. These techniques allow researchers to identify which areas of the brain are activated during memory tasks and how these areas communicate with each other.

- **Case studies:** Researchers can also study the memory performance of individuals with specific conditions or injuries that affect memory. For example, studies of patients with amnesia have provided valuable insights into the role of different brain structures in memory.

- **Computer modelling:** Computer modelling involves creating simulations of the memory process to test theoretical models and hypotheses about how memory works.

- **Longitudinal studies:** Longitudinal studies involve tracking participants' memory performance over time to identify patterns and changes in memory performance as people age.

These are just a few examples of the methods used to study human memory and cognition. By using a range of techniques and approaches, researchers can gain a better understanding of the complex mechanisms that underlie human memory. Assessing memory accuracy can be a complex task, as memory can be influenced by various factors, such as attention, motivation, emotion, and cognitive biases. However, several methods and techniques can be used to evaluate the accuracy of memory, including:

- **Recall and recognition tests:** These are commonly available tests used to assess memory accuracy. Recall tests require individuals to reproduce information from memory, while recognition tests require them to identify previously learned information from a set of options.

- **Source monitoring:** This involves assessing the accuracy of the source of the memory, such as determining whether the memory was experienced directly or learned from another source.

- **Confidence ratings**: Individuals can rate their confidence in the accuracy of their memories, which can be used to evaluate the likelihood of errors or inaccuracies.

- **Memory consistency**: Assessing the consistency of memory over time can provide insights into its accuracy. For example, if a person's memory of an event remains consistent over several retellings, it may be more likely to be accurate.

- **Corroborating evidence**: Gathering additional information or evidence that supports or contradicts a person's memory can help to determine its accuracy.

9.4.2 Memory Loss and Dysfunctional Cognitions

Amnesia is a condition characterized by a loss of memory, either temporarily or permanently. It can be caused by various factors, such as head trauma, stroke, certain medications, alcohol or drug abuse, or severe emotional trauma. Depending on the cause and severity, amnesia can be temporary or permanent. There are several different forms of memory loss:

a. **Retrograde amnesia**: Inability to remember events that occurred before the onset of amnesia. It usually affects recently stored memories, not memories from years ago.

b. **Anterograde amnesia**: Inability to form new memories after the onset of amnesia. It is a type of memory loss that occurs when you can't form new memories. In the most extreme cases, this means you permanently lose the ability to learn or retain any new information.

c. **Transient global amnesia**: Temporary memory loss usually lasting a few hours, commonly seen in older adults. It is an episode of confusion that comes on suddenly in a person who is otherwise alert. This confused state isn't caused by a more common neurological condition, such as epilepsy or stroke.

d. **Dissociative amnesia**: Loss of memory due to a traumatic event or stress. You can't recall information about yourself or events and people in your life, especially from a traumatic time. Dissociative amnesia can be specific to events in a certain time, such as intense combat, or more rarely, can involve complete loss of memory about yourself.

Dementia is a broad term used to describe a group of disorders that affect cognitive functioning, including memory, thinking, and behaviour. It is caused by damage to brain cells and can be progressive, meaning it gets worse over time. Dementia can also affect other aspects of cognitive function beyond memory, such as language and judgment. The most common types of dementia include:

a. **Vascular dementia**: A decline in cognitive function due to reduced blood flow to the brain. It refers to changes to memory, thinking, and behaviour resulting from conditions that affect the blood vessels in the brain. Cognition and brain function can be significantly affected by the size, location, and number of vascular changes.

b. **Frontotemporal dementia**: A type of dementia that affects the front and side areas of the brain, leading to changes in personality and behaviour, as well as memory loss.

c. **Korsakoff's syndrome**: A form of memory loss caused by severe vitamin B1 deficiency, often seen in people with alcoholism.

d. **Alzheimer's disease**: A progressive neurodegenerative disorder mostly in older age that affects memory, thinking, and behaviour.

e. **Parkinson's disease dementia**: A decline in cognitive functions seen in people with Parkinson's disease in advanced years.

Understanding the symptoms and causes of amnesia and dementia can help with early detection and diagnosis. Early diagnosis can lead to earlier interventions, such as medication, lifestyle changes, and therapies, which can slow the progression of these conditions. Knowing about amnesia and dementia can help people who care for those with these conditions to provide better care and support. It can also help them understand the changes in behaviour and cognitive function that may occur, and how to communicate effectively with those affected. Some forms of amnesia and dementia may be preventable. Knowing about the risk factors and causes can help individuals make lifestyle changes that can reduce their risk, such as eating a healthy diet, exercising regularly, and engaging in mentally stimulating activities.

SELF ASSESSMENT EXERCISE

Talk to people with the following issues and write down the issues they face in routine life

a. Retrograde amnesia:	
b. Anterograde amnesia:	
c. Transient global amnesia:	
d. Dissociative amnesia:	
e. Vascular dementia	
f. Frontotemporal dementia	
g. G.Korsakoff's syndrome	
h. Alzheimer's disease	
i. Parkinson's disease dementia	

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EPPSL Department,
Faculty of Education
Allama Iqbal Open University, Islamabad